Dubai Metro, UAE
Jebel Ali Depot

Structural Design Calculation
(Enhanced Design)
Building 10 – Waste Stock Yard

May 2007

ATKINS
WS Atkins & Partners Overseas
Supported by Meinhardt Infrastructure

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Revision, Review and Approval/Authorization Records

A3	Issued for Approval (ENHANCED DESIGN)	CK Lam May 07	P.Shepherd-Smith May 07	B. Maney May 07
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Distribution (if insufficient space, please use separate paper)

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2	Atkins (Office Copy)
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4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

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1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

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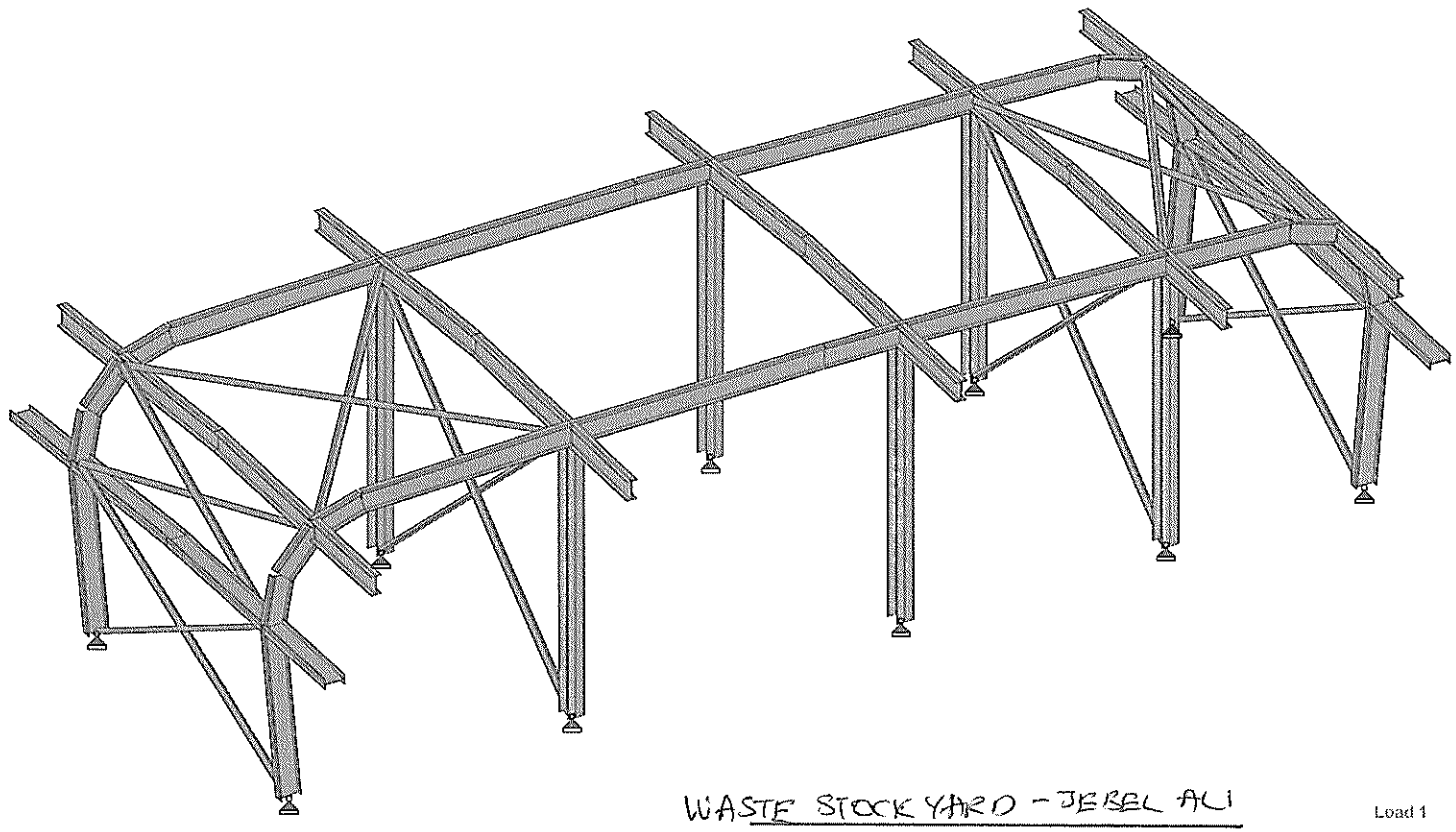


Site:	Dubai Metro Project - Jebel Ali	Date:	15-May-07	Job No.:	68010.01
	Waste Stock Yard				
	Loadings	Designed by:	Rudy	Sheet No.:	

Roof Loadings :

DL

W_{DL}:

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	0.8 kPa	x 1 =	0.8 kN/m

LL

0.6 kPa	x 1 =	0.6 kN/m

P_{LL}: A/C equipment S/W
= 5 kN (point load at mid-span of bay 2, 4)

Notional horizontal load, Ph
(Load Case of 1.4DL + 1.6LL)

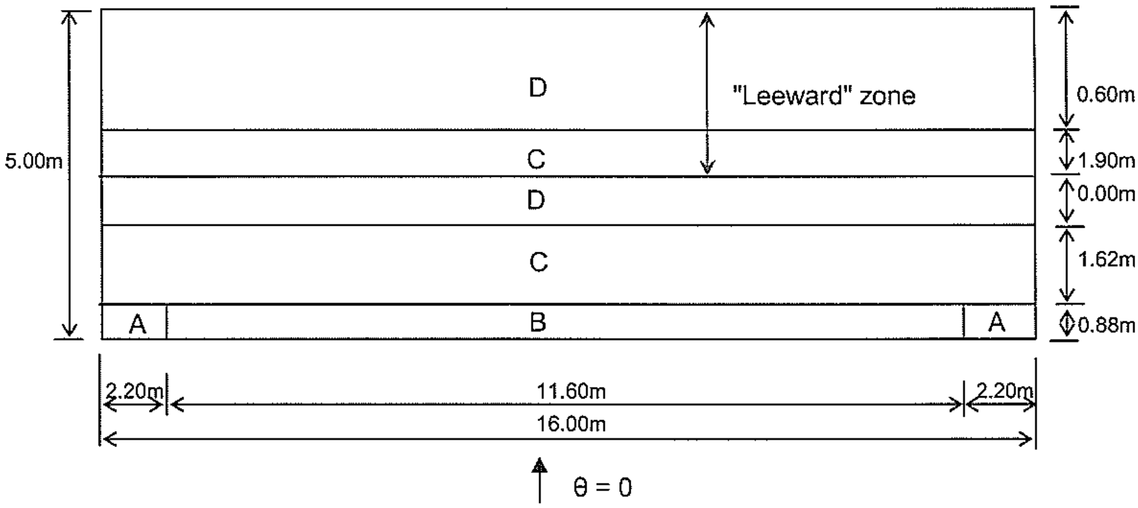
Tributary load area for the roof	Total Dead Load	Total Live Load
= (5+2*1.8)*(16+2*1.1)	125.216 kN	98.912 kN
= 156.52 m2		(Including A/C equipment)

Tributary load area for the wall	Total Dead Load	Total Live Load
= 4.4*2*(5+16)	92 kN	
= 184.8 m2	(excluding the M&E Finishes)	

Site:	Dubai Metro Project - Jebel Ali	Date:	15-May-07	Job No.:	68010.01
	Waste Stock Yard				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)

PLAN



Design as flat roof (with sharp eaves)

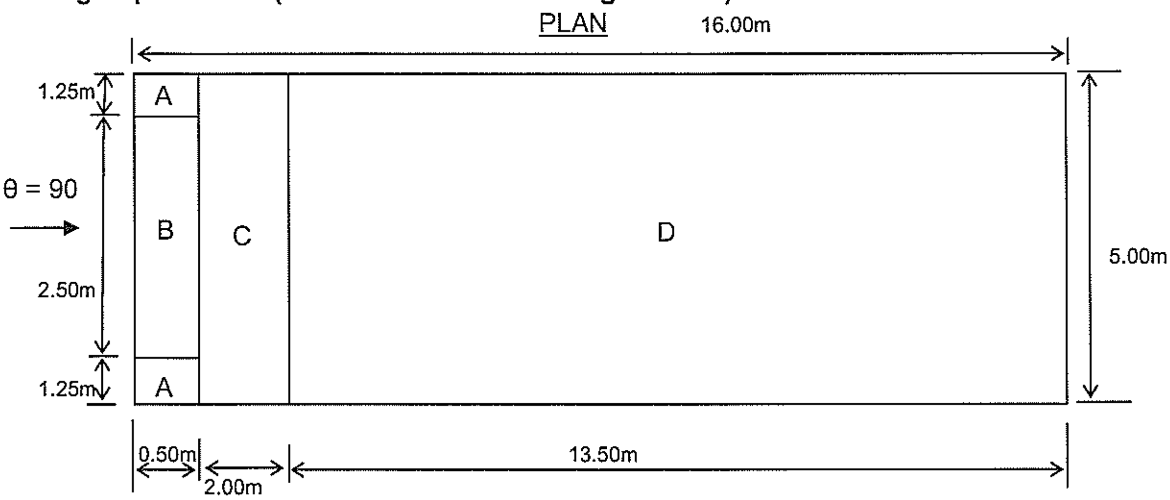
B = 16 m
H = 4.4 m
b = smaller of (B, 2H)
= 2 x 4.4
= 8.8 m

Zone	Windward		Leeward		
	Area (m^2)	Cpe	Area	Cpe1	Cpe2
A	3.9	-2			
B	10.2	-1.4			
C	25.9	-0.7	30.4	-0.7	-0.7
D	0.0	-0.2	9.6	-0.2	0.2
Total	40.0		40.0		
Average		-1.00		-0.58	-0.48

Calculation/Sketch **Meinhardt Infrastructure Pte Ltd**
Consulting Engineers, Planners & Managers

Site:	Dubai Metro Project - Jebel Ali	Date:	15-May-07	Job No.:	68010.01
	Waste Stock Yard				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)



Design as flat roof (with sharp eaves)

B = 5 m
H = 4.4 m
b = smaller of (B, 2H)
= 1 x 5
= 5 m

For maximum uplift, assume zone C coefficient => Cpe = -0.7
For maximum downward force, assume zone D coefficient => Cpe = 0.2

Wind Load Computation

D = 5.00m m
H = 4.4 m
D/H = 1.14

Elements	Effective Wind Speed Ve m/s	Dynamic Wind Pressure q kN/m^2	External Pressure Coefficient, Cpe				
			θ = 0 (normal to eave)			θ = 90 (normal to gable end)	
			Windward	Leeward			
Roof	45	1.24	-1.00	-0.58	-0.48	-0.7 (Zone C)	0.2 (Zone D)
Wall	45	1.24	0.84	-0.5	-0.5	-0.8 (Zone B)	-0.8 (Zone B)

Calculation/Sketch

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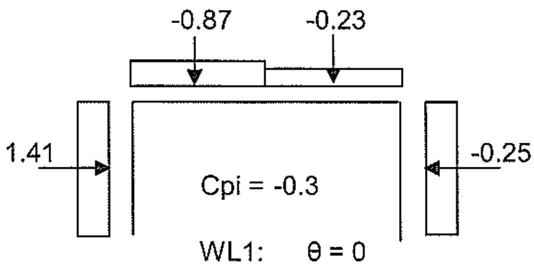
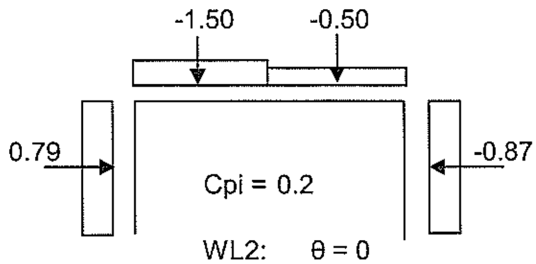
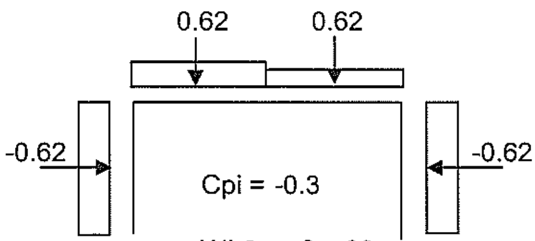
Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



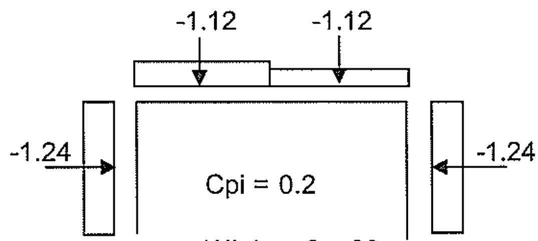
WL1: θ = 0
Internal suction
Roof Cpe for Leeward = -0.48



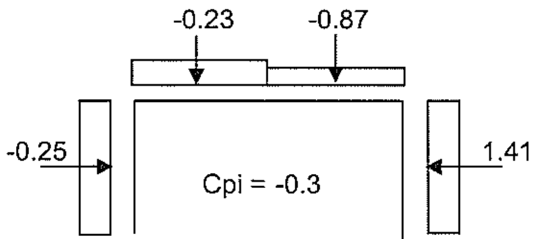
WL2: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20



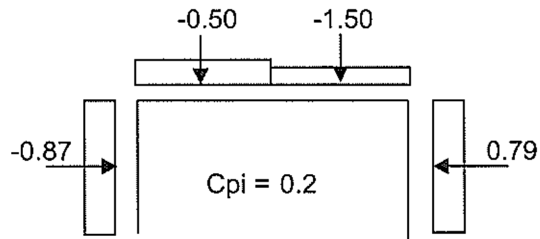
WL3: θ = 90
Internal suction
Roof Cpe = 0.20



WL4: θ = 90
Internal pressure
Roof Cpe = -0.70



WL5: θ = 0
Internal suction
Roof Cpe for Leeward = -0.48
(Reverse wind direction)



WL6: θ = 0
Internal pressure
Roof Cpe for Leeward = -0.20
(Reverse wind direction)

Site:	Dubai Metro Project - Jebel Ali	Date:	15-May-07	Job No.:	68010.01
	Waste Stock Yard				
	Earthquake Loading	Designed by:	Rudy	Sheet No.:	

Esitmation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Tanle- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

4) **Base Shear** $V_{base\text{-}S} = (C_V \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient C_V = 0.32 (Table 16-R assuming soil profile type S_D)
Importance Factor I = 1.00 (Table 16-K)
Numeric. Coef: Strength & Ductility R = 4.50 (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t\,(h_n)^{3/4}$ Sec.
Numerical Coefficient C_t = 0.0853 (Steel frame t.b.c)

5) **Limiting Values of Base shear** $V_{base\text{-}S\text{-}Max} = (2.5 \times C_a \times I \times W) / R$ = 32.64
$V_{base\text{-}S\text{-}Min} = 0.11 \times C_a \times I \times W$ = 6.46

Design Earthquake Loads

(Section 1630.6)

$E_x = \rho \times \text{Min.}\,(E_h \text{ or } V_{base\text{-}S\text{-}Max}) + E_V$
$E_{x\text{-}max} = \Omega \times \text{Min.}\,(E_h \text{ or } V_{base\text{-}S\text{-}Max})$
$V_{base\text{-}S} = E_h$ = Earthquake load in the element due to the base shear, V, as set out above (for non-structural components refer to section 1632)

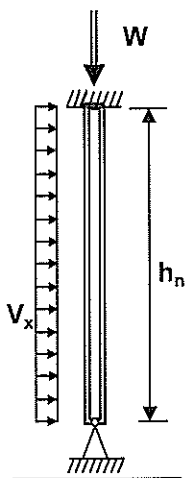
Vert. component of EQ ground motion $E_v = 0.5\,C_a\,I\,D$ (Table 16-K)
Seismic Coefficient C_a = 0.22 (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor Ω = 2.80 (Table 16-N)
Dead Load on the element $W = D$ = Given below
$\rho = 2 - 6.1 / (r_{max} \sqrt{A_B})$ = 1.25 (conservative) $(1.0 \le \rho \le 1.5)$
r_{max} = The maximum element-storey shear ratio
A_B = Ground floor area of the structure

Distribution of Horizontal Forces:

Distribution of Horizontal Force on the Column V_x = $\mathbf{E_x / h_n}$

Col	Gr. Load, W, kN	Ht h_n, m	Time, T sec.	$V_{base\text{-}S}$ kN	$V_{base\text{-}S}$ kN	E_v kN	E_x kN	$E_{x\text{-}max}$ kN	V_x kN/m	$V_{x\text{-}max}$ kN/m
C1	267	4.40	0.26	73.3	32.64	29.3779	70.2	91.4	16.0	**20.8**

Load per column = 4.4*16.0/10
= 7.02 kN



Distribution of Horozontal Forces on the Coulumn

Calculation/Sketch

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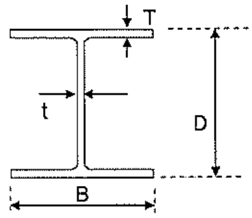
Consulting Engineers, Planners & Managers

Project:	**C856 - PPJ Station: TERS**	Job No.:	068024.01
Subject:	**Design of Waler**	**700x300 x 182 kg/m --- S2, S3**	
Description	Section F2-F2 South	Designed by Michelle	Checked by: Ravi — Sheet No.

Design of Steel Beam with Axial force and Bending Moment



Load per meter, w	= 1279.0 kN/m	**No. of walers**	= 2.0	
Span L_1	= 2.3 m	**Clear span (for Shear) L_2**	= 2.3 m	

1. Design Forcese

ULS Bending Moment $M_{X\text{-}max}$	= 473.6	kN-m	$w\,L_1^2\,/\,10$	Axial Force P_{max}	= 790 kN
ULS Shear Force V_{max}	= 1029.6	kN	$w\,L_2\,/\,2$	Load factor, γ_f	= 1.4
ULS Bending Moment $M_{Y\text{-}max}$	= 0.0	kN/m		Ultimate strut load	= 1107 kN

2. Individual Member Properties

Section size	174	700x300	**182 kg/m**	E =	2.05E+05	N/mm^2
Beam depth, D	=	700	mm	Depth between fillets, d =	616	mm
Beam width, B	=	300	mm	* Plastic modulus, Sx =	6.34E+06	mm^3
Flange thickness, T	=	24	mm	* Plastic modulus, Sy =	1.11E+06	mm^3
Web thickness, t	=	13	mm	[*calculated based on contribution from flanges & web]		
Sectional area, Ag	=	23200	mm^2	Radius of gyration, rx =	292	mm
Moment of inertia, Ix	=	1.98E+09	mm^4	Radius of gyration, ry =	68.3	mm
Moment of inertia, Iy	=	1.08E+08	mm^4	A,eff =	23200	mm^2
Section Modulus, Zx	=	5.64E+06	mm^3	Zx,eff =	4.70E+06	mm^3
Section Modulus, Zy	=	7.21E+05	mm^3	Zy,eff =	7.20E+05	mm^3

3. Material Properties

Steel grade	=	43				
Design strength, py	=	235	N/mm^2	Elastic modulus =	205	kN/mm^2

4. Section Classification — **Section classified as** Plastic

Epsilon = [275/py]^0.5 = 1.082

Outstand element of compression flange, b/T				Web subject to compression throughout, d/t			
Plastic	-	9.7		Plastic	-	60.9	r_1 = 0.42
Compact	-	10.8		Compact	-	66.4	
Semi-compact	-	16.2		Semi-compac	-	100.63	r_2 = 0.14
Actual b/T	=	7.23	***Flange is plastic.***	Actual d/t	=	17.40	***Web is plastic.***

5. Design Capacity Check

5.1 Local/Section Capacity Check

Shear Check

Shear capacity of beam , P_v = $0.6p_yA_v$ = = 1283 kN **> Design Value, OK**

Since d / t is not more than 63 ε → Shear buckling resistance check is not required *Clause 4.2.3*

Bending Moment Check *Clause 4.2.5*

Ag	=	23200	mm^2			
Sxx	=	6.34E+06	mm^3	Syy =	1.11E+06	mm^3

Moment Capacity of beam, M_{cx} **Min(PySx, 1.2PyZx)** (about major axis)

beam section = = **1489.4 kNm > Design Value, OK** Mcy = 261 kNm

5.2 Lateral Torsional Buckling Check *Clause 4.3*

$(A,eff/Ag)^{0.5}$	=	1.00			
Effective length, Lxx	=	= 2.3 m	Effective length, Lyy	= 2.3 m	
Slenderness Ratio: λ_x	=	7.9	Slenderness Ratio:	λ = 33.67	
Design Lamda, λ	=	33.7			

Member Buckling Strength

Equivalent Slenderness Ratio, $\lambda_{LT} = u\,v\,\lambda\,\sqrt{\beta_W}$: **29.46**

β_w	=	1.00		Torsional index, x =	31.4
Buckling parameter, u	=	0.887	For Equal Flanges	λ/x =	1.1
Slenderness factor, v	=	0.99		Slenderness factor	$v = 1/((1+0.05(\lambda/x)^2)^{0.25}$

Bending strength, $p_b = \dfrac{p_E\, p_y}{\phi_{LT} \sqrt{(\phi_{LT}^{\,2} - p_E\, p_Y)}}$ = **235.00** N/mm^2

Euler strength p_E

$p_E = (\pi^2 E / \lambda_{LT}^{\,2})$ = 2332.0 — Limiting Slenderness Ratio λ_{L0} = 37.12 $\lambda_{L0} = 0.4\sqrt{(\pi^2 E / p_y)} =$

α_{LT} = 7.00 B.2.2

$\phi_{LT} = \dfrac{p_y + (\eta_{LT}+1) p_E}{2}$ = 1284 — $\eta_{LT} = \alpha_{LT}(\lambda_{LT} - \lambda_{L0})/1000$ - = 0.05 It shall be >= 0.0

= 0.00

Buckling Resis. Momt. Mb = = **1489.43** kNm **> Design Value, OK**

Ratio of ULS BM / Capacity = 0.32

6. Design Check: Compression Members with Moments

6.1 Local Capacity Check / Cross-Section Capacity of Laced Strut -- Member Buckling Resistance -- Clauase - 4.8.3.2

$\dfrac{F_c}{A\, p_y} + \dfrac{M_x}{M_{cx}} + \dfrac{M_y}{M_{cy}}$ = 0.20 + 0.32 + 0.00 = 0.52 *< 1, ok* [Load case 1]

6.2 Member Compressive Strength and Member Buckling Resistance of Strut -- Clauase - 4.8.3.3

$(A,eff/Ag)^{0.5}$ = 1.00

Major axis			Minor axis		
Slenderness Ratio:	λ_x =	7.9	Slenderness Ratio λ_y =		33.7
Compressive Strength,	$p_{cx} = \dfrac{p_E\, p_y}{\phi + \sqrt{(\phi^2 - p_E\, p_Y)}} =$	**235 MPa**	*Compressive Strength,*	$p_{cx} = \dfrac{p_E\, p_y}{\phi + \sqrt{(\phi^2 - p_E\, p_Y)}} =$	**215 MPa**
Euler strength	$p_{EX} = (\pi^2 E / \lambda_x^{\,2}) =$	32611	Euler strength	$p_{EY} = (\pi^2 E / \lambda_y^{\,2}) =$	1784
	$\phi = \dfrac{p_y + (\eta+1) p_{Ex}}{2} =$	16423		$\phi = \dfrac{p_y + (\eta+1) p_{Ey}}{2} =$	1084
Perry Factor	$\eta = \alpha\ (\lambda_x - \lambda_0)/1000$ =	-0.04	Perry Factor	$\eta = \alpha\ (\lambda_y - \lambda_0)/1000$ =	0.08
Perry factor should not be less than 0.0		0.00	>= 0.0		0.08 >= 0.0

Limiting Slenderness Ratio $\lambda_0 = 0.2\sqrt{(\pi^2 E / p_y)} =$ = 18.6

Robertson Constant α = 3.50 (Major Axis)

Robertson Constant α = 5.50 (Minor axis)

Comp. resistance, P_{cx} = 5452 kN — Comp. resistance, P_{cy} = 4981 kN

Comp. resistance, P_c = **4981** kN

m_x = 1 m_y = 1 m_{LT} = 1

6.3 Simplified Method --- Clause 4.8.3.3.1

$\dfrac{F_c}{P_c} + \dfrac{m_x M_x}{p_y Z_x} + \dfrac{m_y M_y}{p_y Z_y} =$

For Major Axis In-plane Buckling :
0.22 + 0.36 + 0.00 = **0.58** ***< 1, ok*** [Load case 1]

$\dfrac{F_c}{P_c y} + \dfrac{m_{LT} M_{LT}}{M_b} + \dfrac{m_y M_y}{p_y Z_y} =$

For Lateral-Torsional Buckling :
0.22 + 0.32 + 0.00 = **0.54** ***< 1, ok*** [Load case 1]

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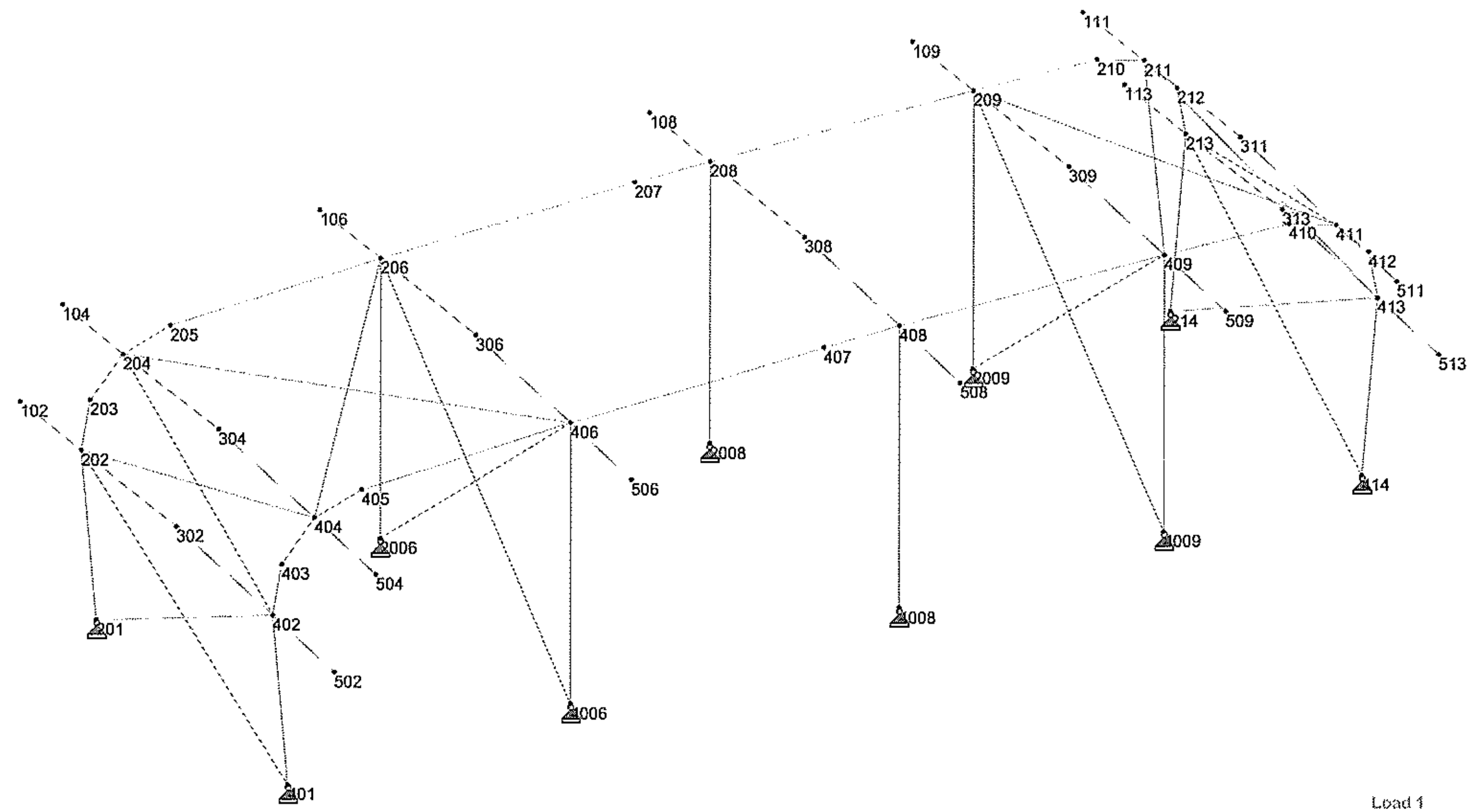

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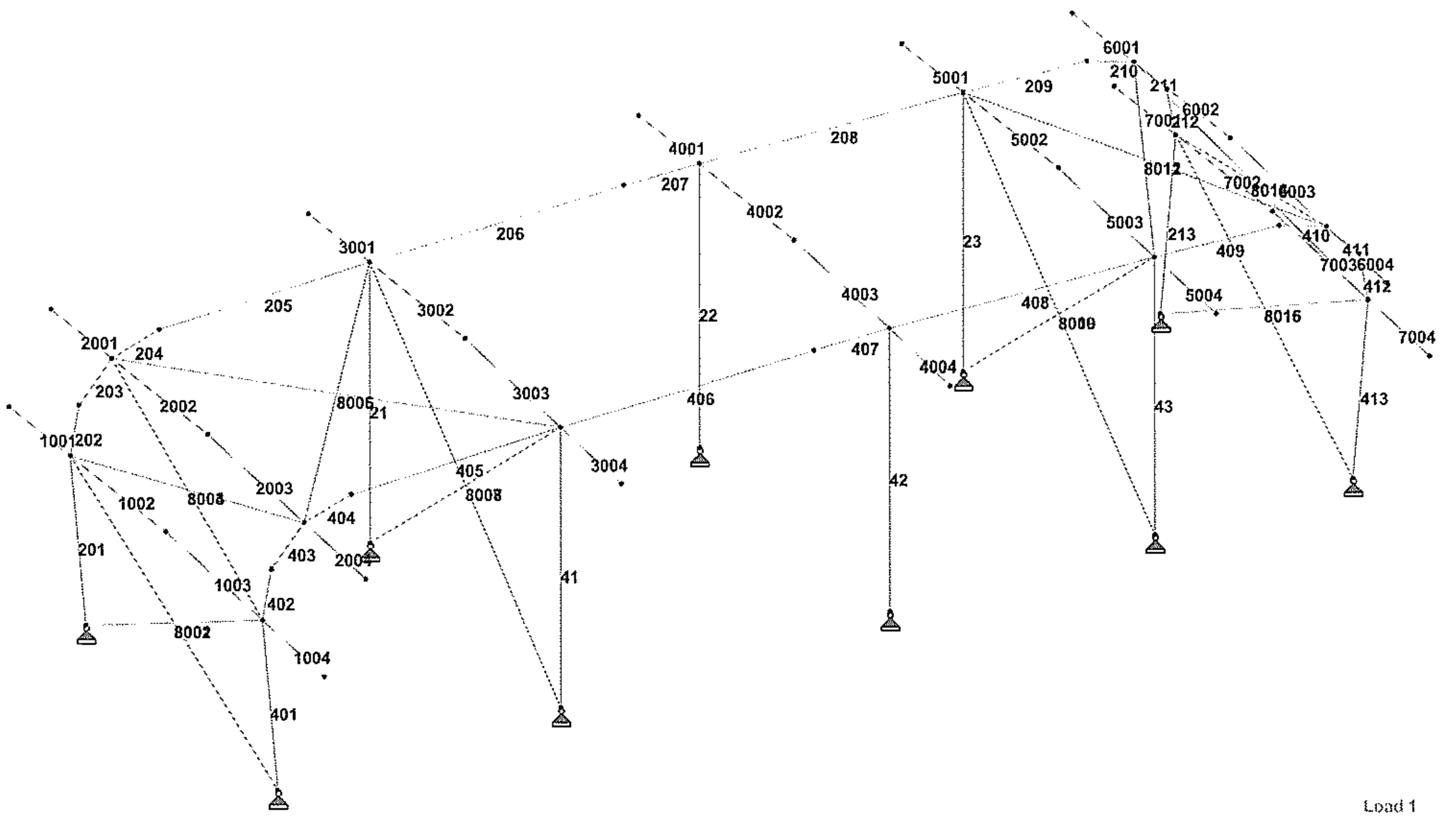

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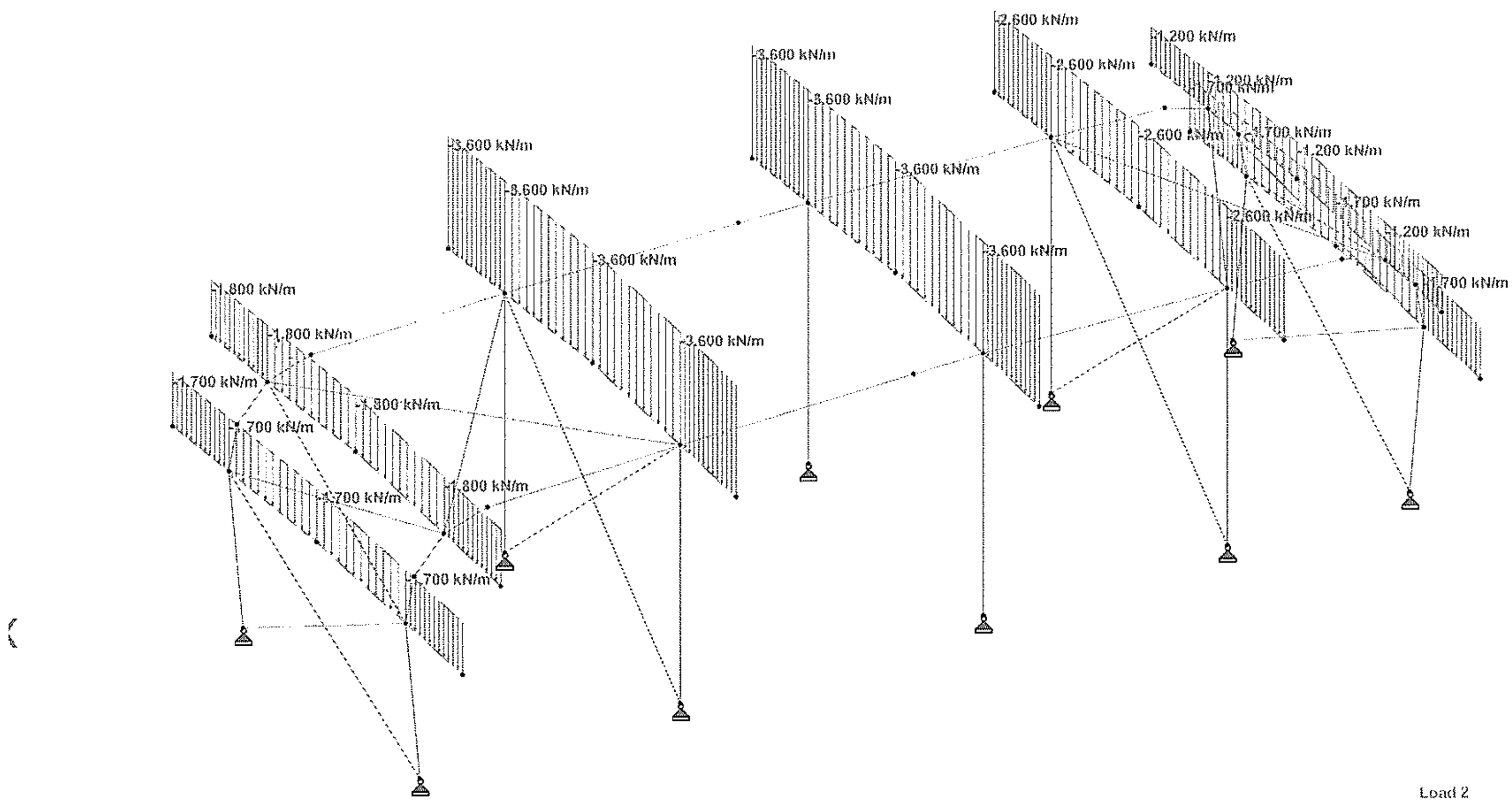

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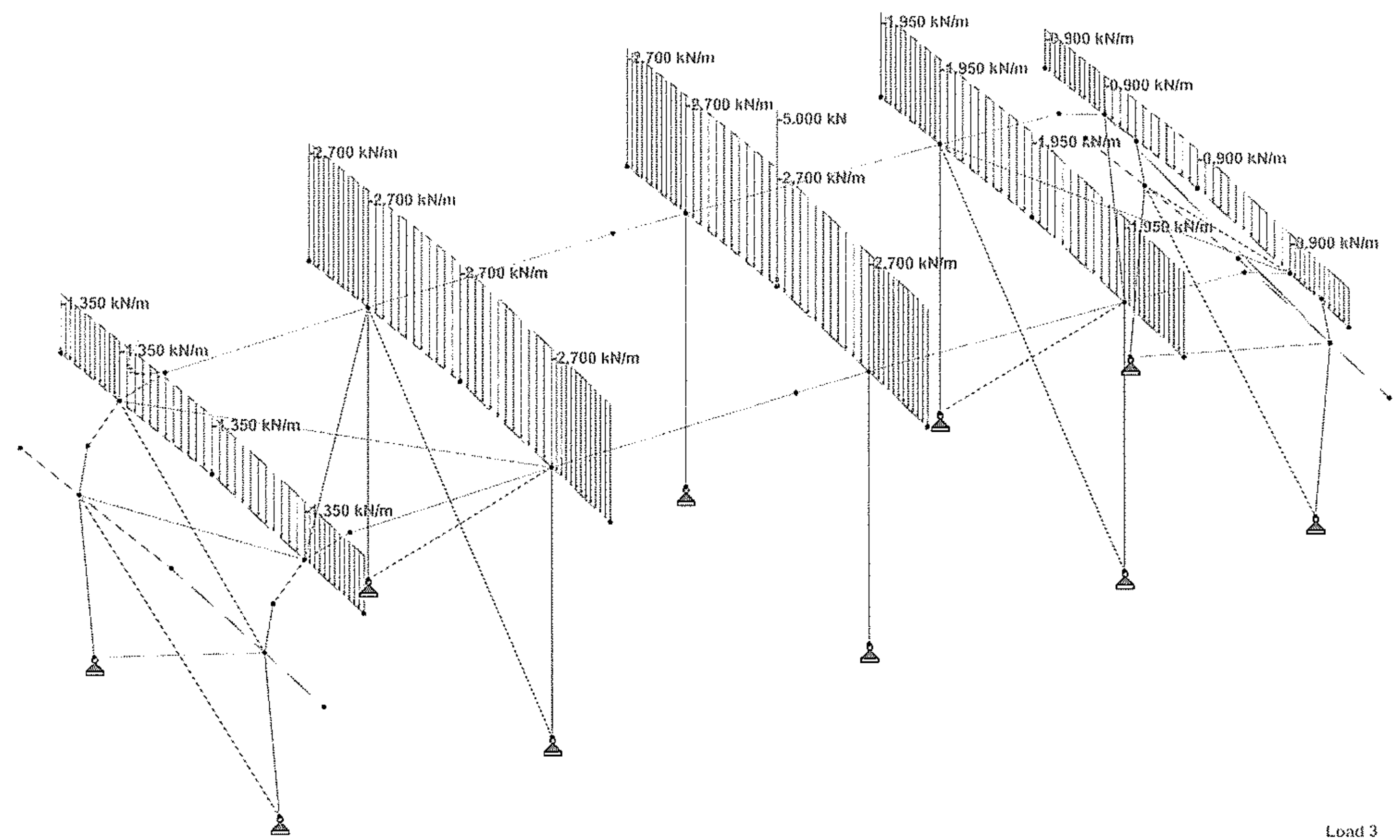

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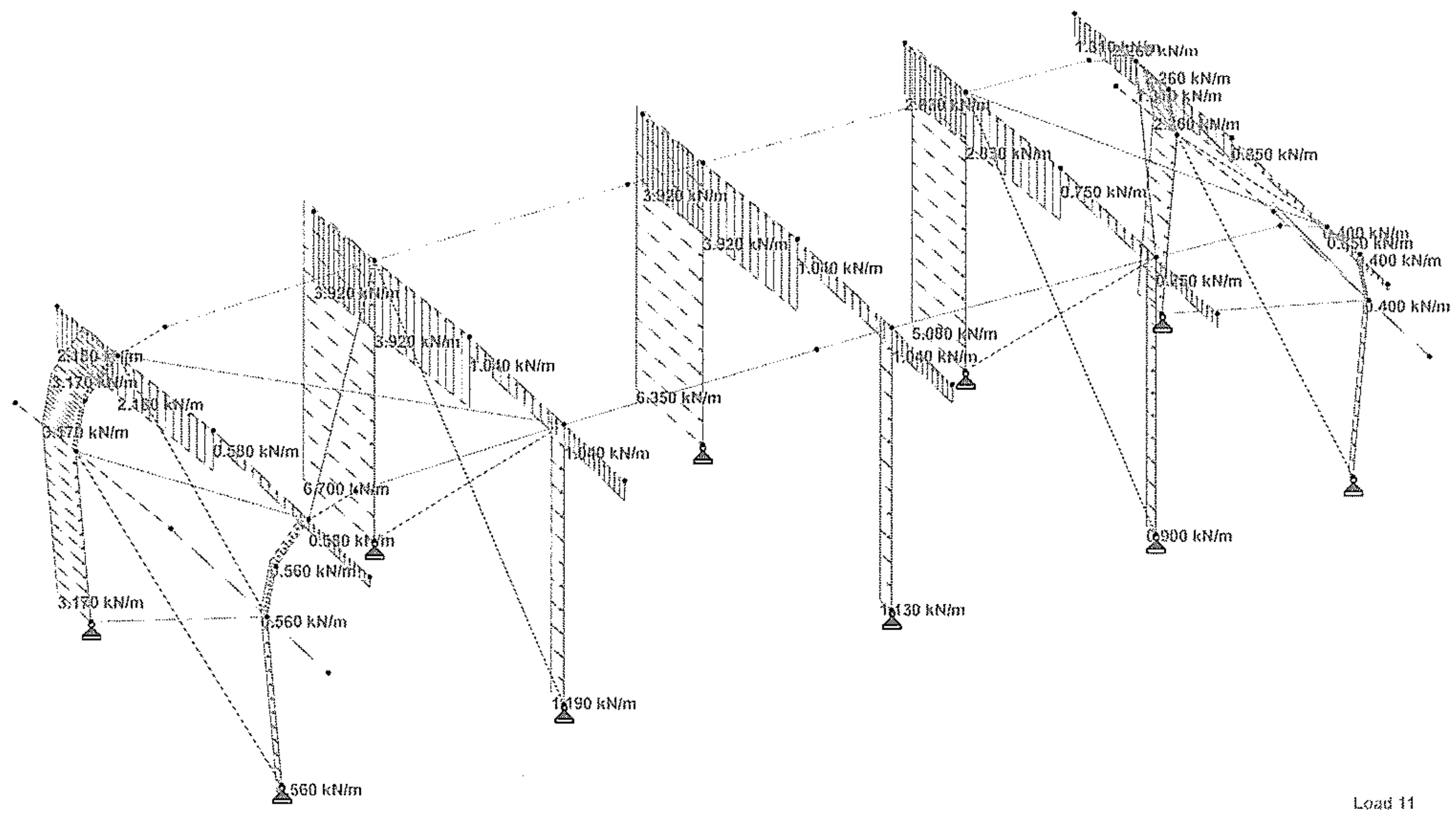

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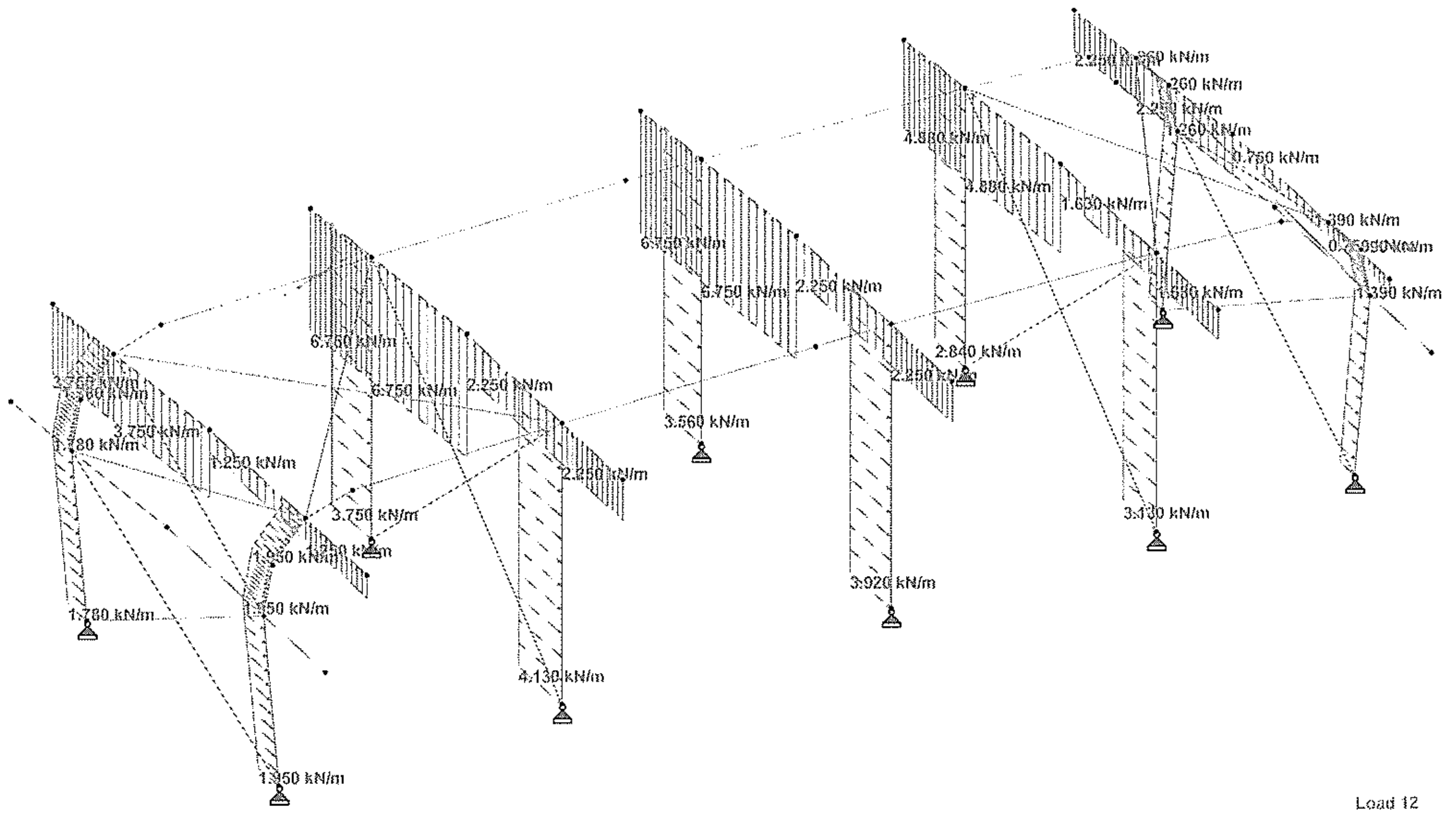

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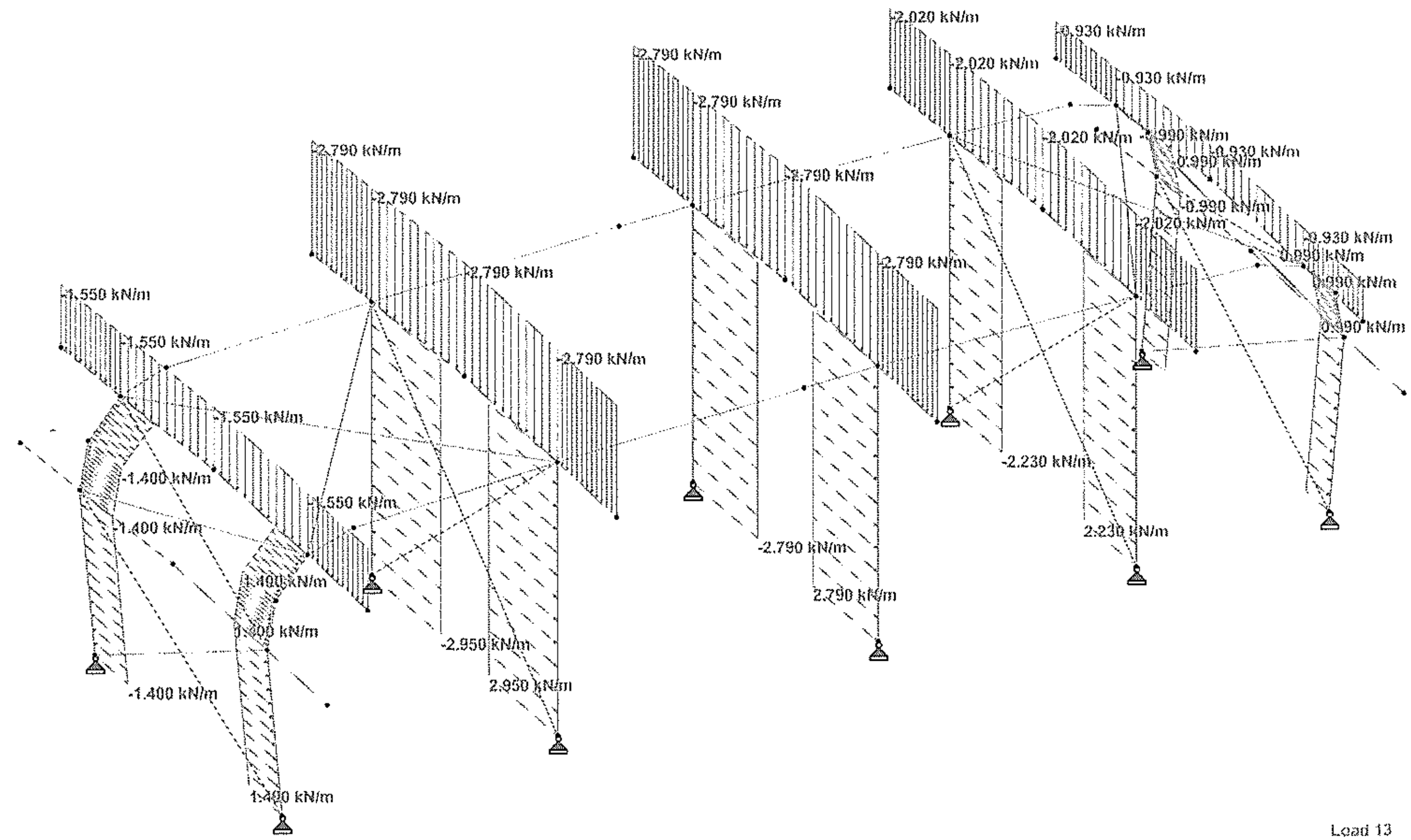

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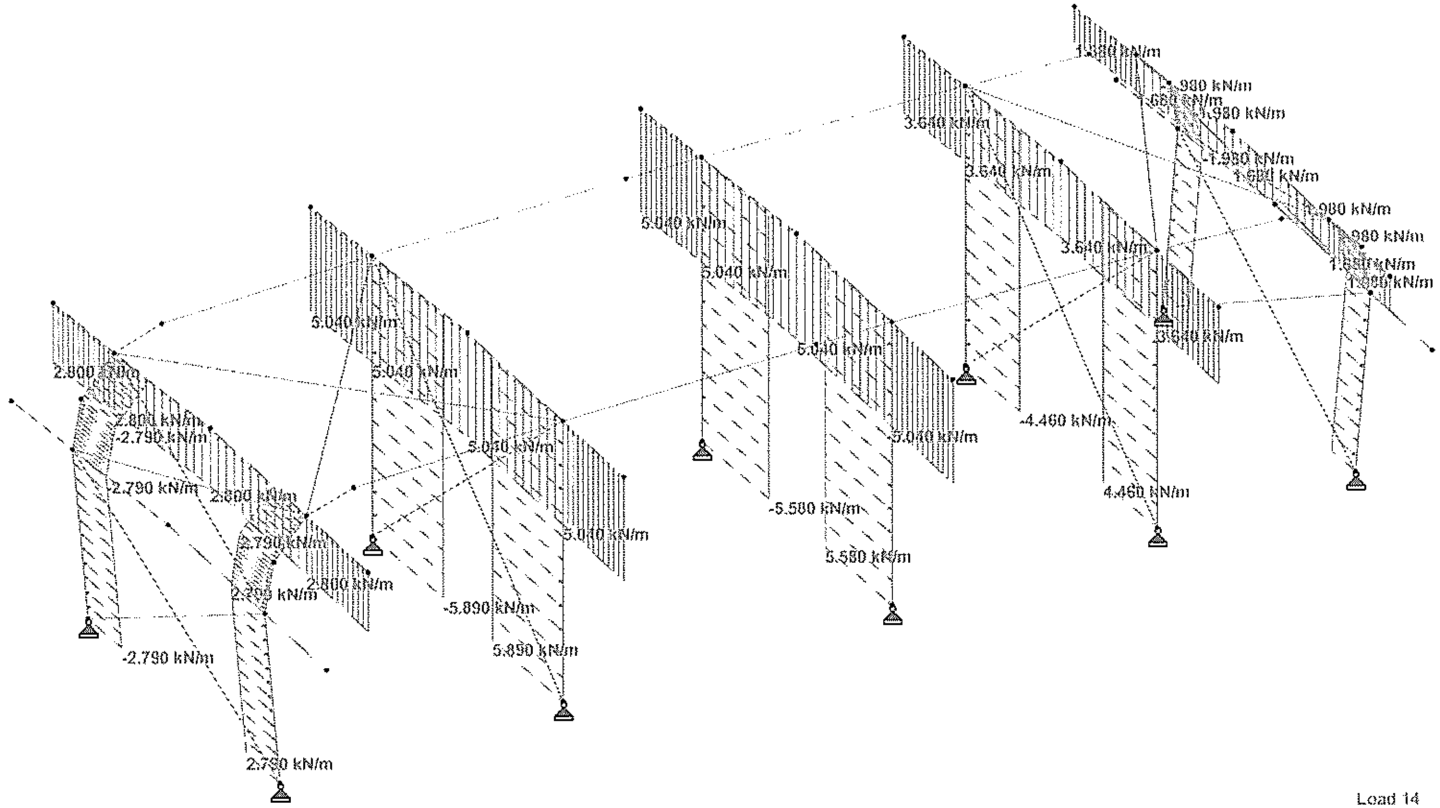

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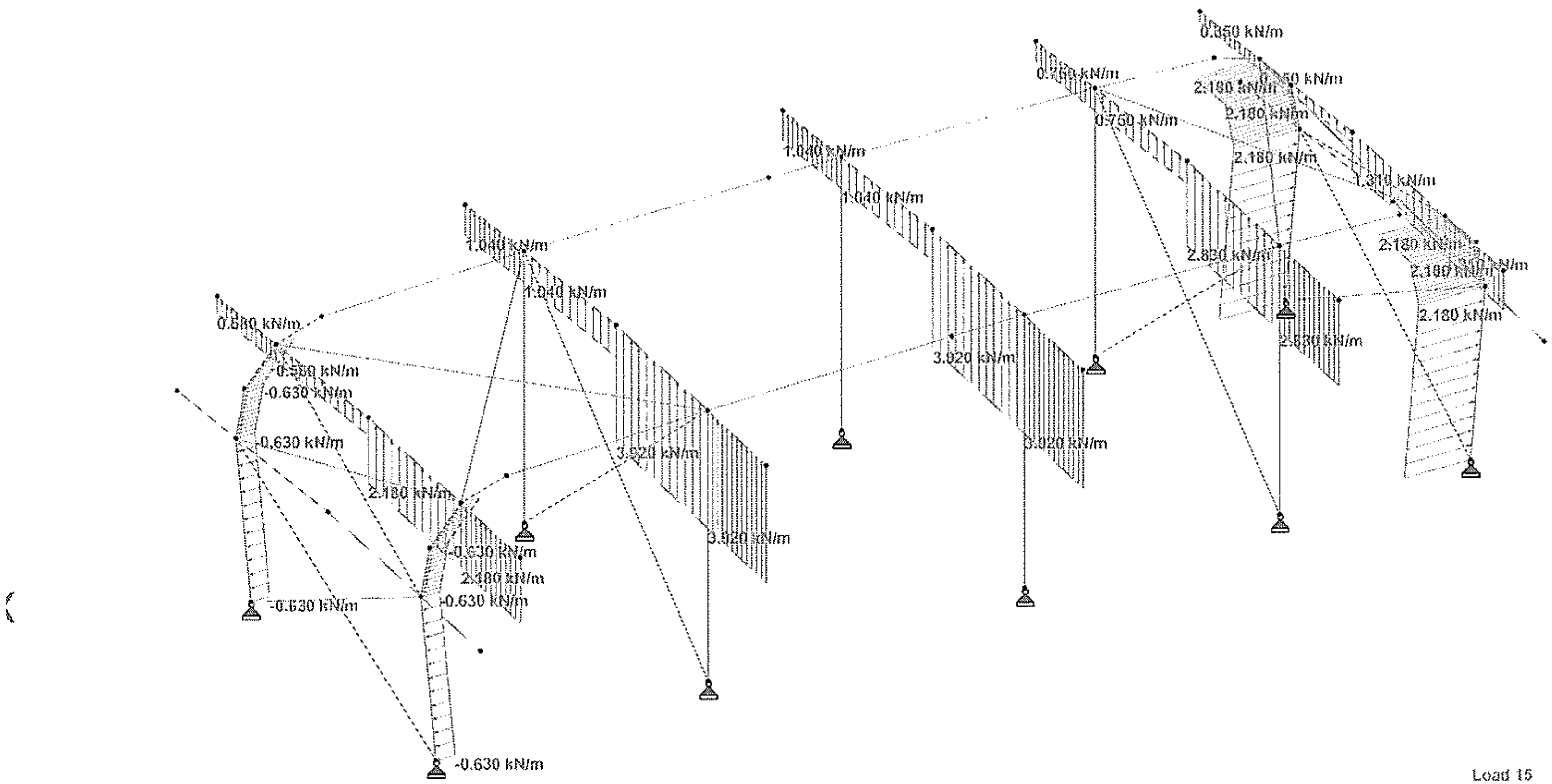

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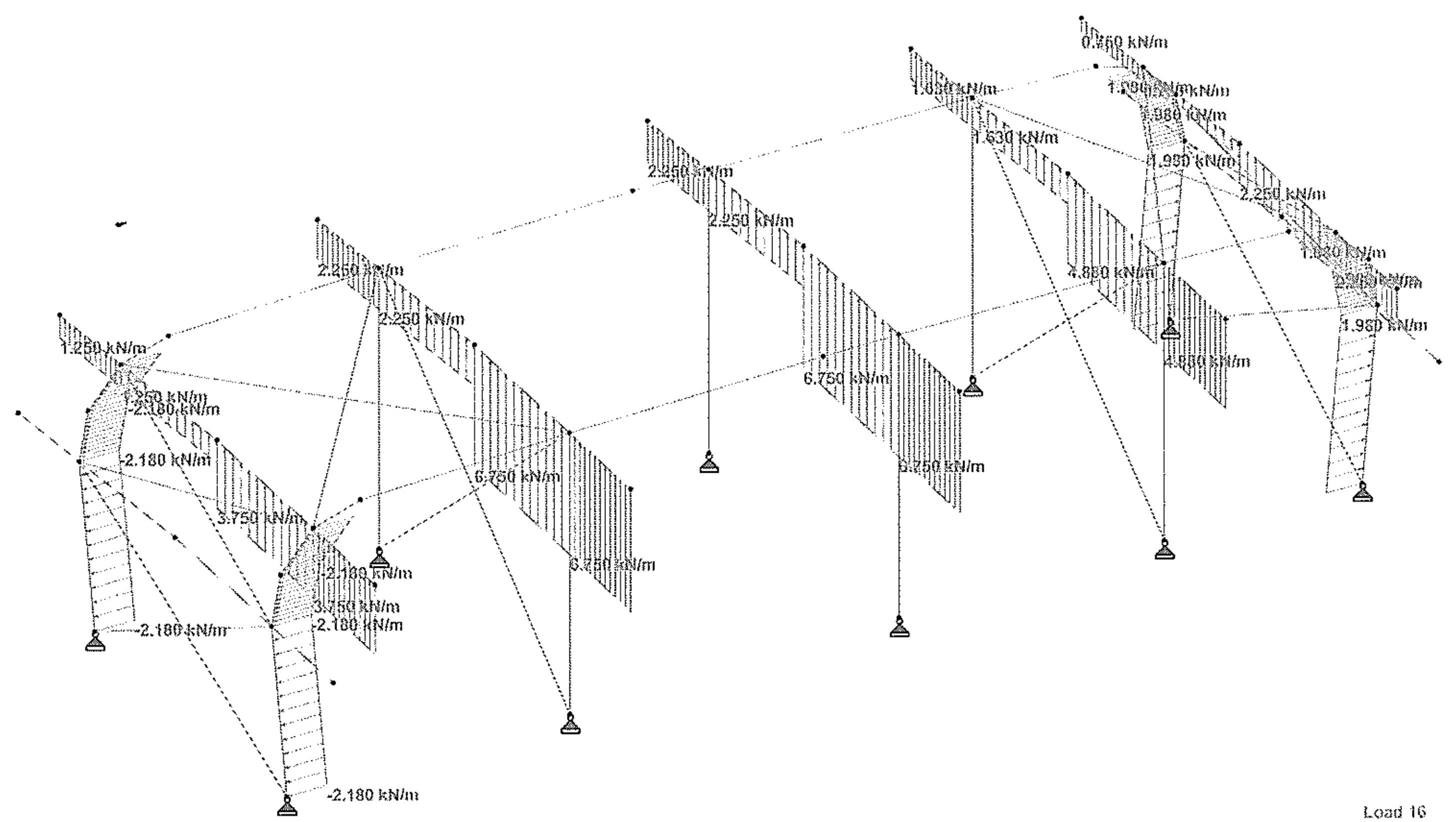

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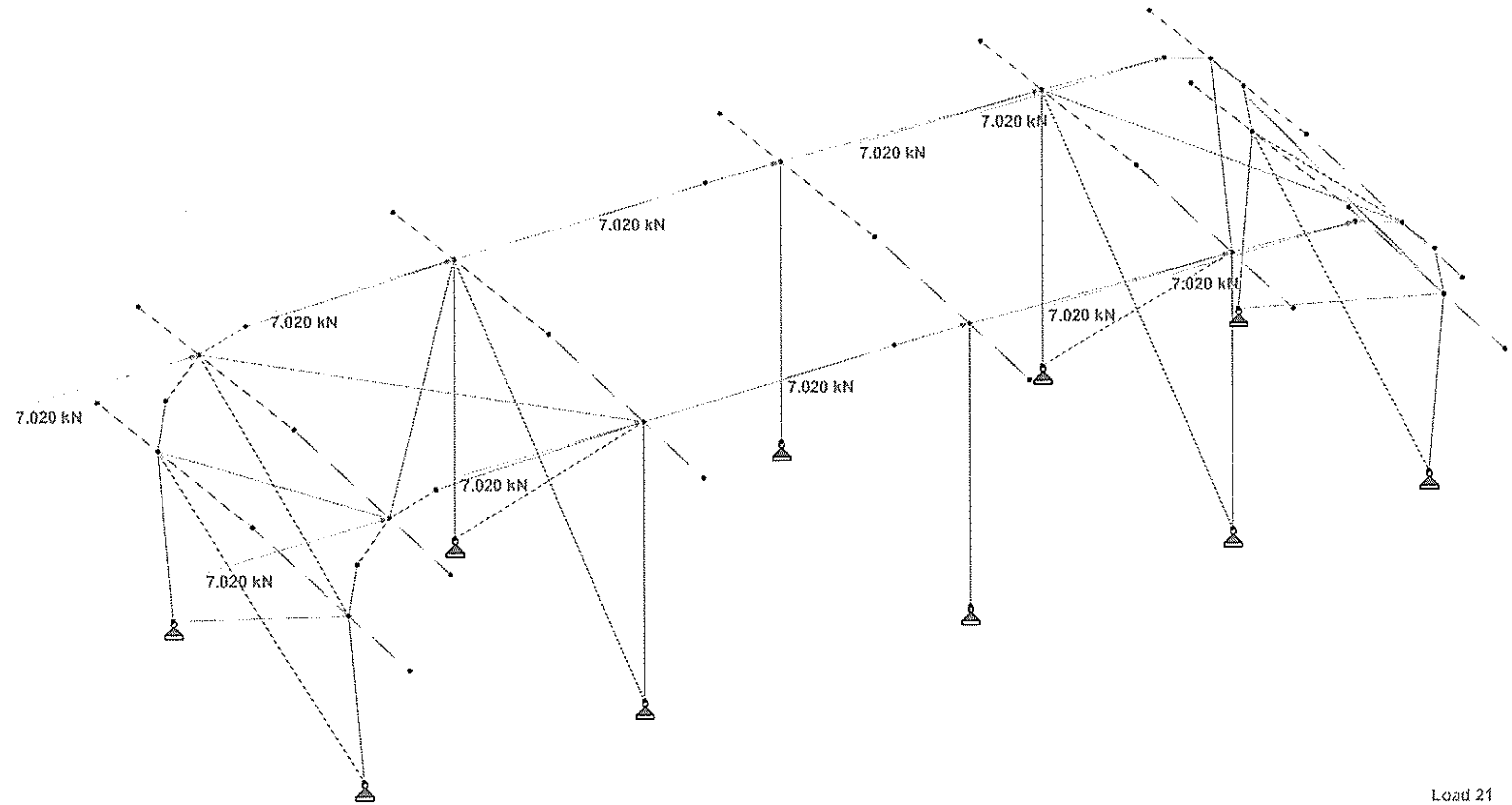
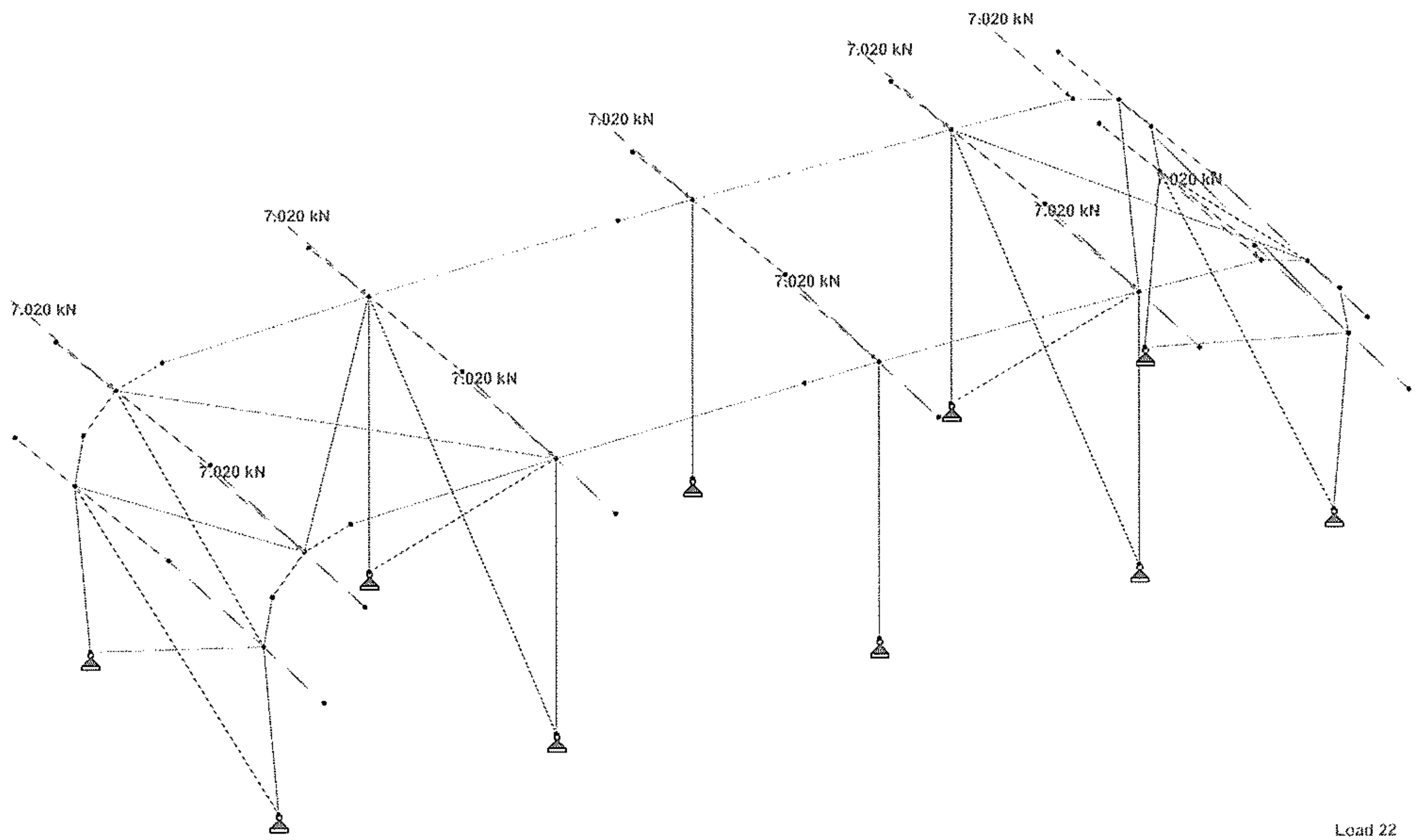
	By	Date MAY 2007	Chd	
Client	File Waste Stock Yard Rev1 -	Date/Time 15-May-2007 17:01		



Software licensed to MIPL	Job No	Sheet No 1	Rev
	Part		
Job Title	Ref		
	By	Date MAY 2007	Chd
Client	File Waste Stock Yard Rev1 -	Date/Time 15-May-2007 17:01	



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          *******************************************************
          *                                                     *
          *           STAAD.Pro                                 *
          *           Version  2006     Bld 1005.US             *
          *           Proprietary Program of                    *
          *           Research Engineers,  Intl.                *
          *           Date=    MAY 15, 2007                     *
          *           Time=     17: 1:56                        *
          *                                                     *
          *      USER ID: MIPL                                  *
          *******************************************************

     1. STAAD SPACE
INPUT FILE: Waste Stock Yard Rev1 - Jebel Ali.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE MAY 2007
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 102 -8.351 2.581 -4.100; 104 -7.722 3.871 -4.100
     9. 106 -3.850 4.198 -4.100; 108  1.150 4.239 -4.100
    10. 109  5.150 4.159 -4.100; 111  7.722 3.871 -4.100
    11. 113  8.351 2.581 -4.100
    13. 201 -8.125  0.000 -2.500; 202 -8.356  2.639 -2.500
    14. 203 -8.220  3.372 -2.500; 204 -7.726  3.931 -2.500
    15. 205 -7.015  4.155 -2.500; 206 -3.850  4.258 -2.500
    16. 207  0.000  4.306 -2.500; 208  1.150  4.300 -2.500
    17. 209  5.150  4.220 -2.500; 210  7.015  4.155 -2.500
    18. 211  7.726  3.931 -2.500; 212  8.220  3.372 -2.500
    19. 213  8.356  2.639 -2.500; 214  8.125  0.000 -2.500
    21. 302 -8.364  2.729 0.000; 304 -7.732  4.025 0.000
    22. 306 -3.850  4.352 0.000; 308  1.150  4.395 0.000
    23. 309  5.150  4.315 0.000; 311  7.732  4.025 0.000
    24. 313  8.364  2.729 0.000
    26. 401 -8.125  0.000 2.500; 402 -8.356  2.639 2.500
    27. 403 -8.220  3.372 2.500; 404 -7.726  3.931 2.500
    28. 405 -7.015  4.155 2.500; 406 -3.850  4.258 2.500
    29. 407  0.000  4.306 2.500; 408  1.150  4.300 2.500
    30. 409  5.150  4.220 2.500; 410  7.015  4.155 2.500
    31. 411  7.726  3.931 2.500; 412  8.220  3.372 2.500
    32. 413  8.356  2.639 2.500; 414  8.125  0.000 2.500
    34. 502 -8.351 2.581 4.100; 504 -7.722 3.871 4.100
    35. 506 -3.850 4.198 4.100; 508  1.150 4.239 4.100
    36. 509  5.150 4.159 4.100; 511  7.722 3.871 4.100
    37. 513  8.351 2.581 4.100
    39. 2006 -3.850 0.000 -2.500; 2008 1.150 0.000 -2.500
    40. 2009  5.150 0.000 -2.500
    42. 4006 -3.850 0.000 2.500; 4008 1.150 0.000 2.500
    43. 4009  5.150 0.000 2.500
    45. MEMBER INCIDENCES
    46. 21 2006 206; 22 2008 208; 23 2009 209
    47. 41 4006 406; 42 4008 408; 43 4009 409
```



```
    48. *
    49. 201 201 202 213; 401 401 402 413
    50. *
    51. 1001 102 202; 1002 202 302; 1003 302 402; 1004 402 502
    52. 2001 104 204; 2002 204 304; 2003 304 404; 2004 404 504
    53. 3001 106 206; 3002 206 306; 3003 306 406; 3004 406 506
    54. 4001 108 208; 4002 208 308; 4003 308 408; 4004 408 508
    55. 5001 109 209; 5002 209 309; 5003 309 409; 5004 409 509
    56. 6001 111 211; 6002 211 311; 6003 311 411; 6004 411 511
    57. 7001 113 213; 7002 213 313; 7003 313 413; 7004 413 513
    58. *
    59. 8001 201 402; 8002 401 202; 8003 202 404; 8004 402 204
    60. 8005 204 406; 8006 206 404; 8007 206 4006; 8008 406 2006
    61. 8009 209 4009; 8010 409 2009; 8011 209 411; 8012 211 409
    62. 8013 211 413; 8014 411 213; 8015 213 414; 8016 214 413
    64. DEFINE MATERIAL START
    65. ISOTROPIC STEEL
    66. E 2.05E+008
    67. POISSON 0.3
    68. DENSITY 76.8195
    69. ALPHA 1.2E-005
    70. DAMP 0.03
    71. END DEFINE MATERIAL
    73. MEMBER PROPERTY BRITISH
    74. 21 TO 23 41 TO 43 TAPERED 0.25 0.006 0.25 0.25 0.008 0.25 0.008
```



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    75. 201 TO 213 401 TO 413 2001 TO 2004 -
    76. 3001 TO 3004 4001 TO 4004 5001 TO 5004 -
    77. 6001 TO 6004 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
    78. 1001 TO 1004 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
    79. 7001 TO 7004 TAPERED 0.3 0.008 0.3 0.15 0.008 0.15 0.008
    80. 8001 TO 8016 TABLE ST UA90X90X6
    81. CONSTANTS
    82. BETA 90 MEMB 21 TO 23 41 TO 43 1001 TO 1004 7001 TO 7004
    83. MATERIAL STEEL ALL
    84. MEMBER TENSION
    85. 8001 TO 8016
    86. SUPPORTS
    87. 201 214 401 414 2006 2008 2009 4006 4008 4009 PINNED
    88. LOAD 1 SELFWEIGHT
    89. SELFWEIGHT Y -1
    90. LOAD 2 SDL
    91. MEMBER LOAD
    92. 1001 TO 1004 7001 TO 7004 UNI GY -1.70
    93. 2001 TO 2004 UNI GY -1.80
    94. 3001 TO 3004 4001 TO 4004 UNI GY -3.60
    95. 5001 TO 5004 UNI GY -2.60
    96. 6001 TO 6004 UNI GY -1.20
    97. LOAD 3 LL
    98. MEMBER LOAD
    99. 2001 TO 2004 UNI GY -1.35
   100. 3001 TO 3004 4001 TO 4004 UNI GY -2.70
   101. 5001 TO 5004 UNI GY -1.95
   102. 6001 TO 6004 UNI GY -0.90
   103. JOINT LOAD
   104. 308 FY -5
   106. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
```



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   107. MEMBER LOAD
   108. 2001 2002 UNI GY 2.18
   109. 2003 2004 UNI GY 0.58
   110. 3001 3002 4001 4002 UNI GY 3.92
   111. 3003 3004 4003 4004 UNI GY 1.04
   112. 5001 5002 UNI GY 2.83
   113. 5003 5004 UNI GY 0.75
   114. 6001 6002 UNI GY 1.31
   115. 6003 6004 UNI GY 0.35
   116. 21 UNI GZ 6.70
   117. 22 UNI GZ 6.35
   118. 23 UNI GZ 5.08
   119. 201 TO 203 UNI GZ 3.17
   120. 211 TO 213 UNI GZ 2.26
   121. 41 UNI GZ 1.19
   122. 42 UNI GZ 1.13
   123. 43 UNI GZ 0.90
   124. 401 TO 403 UNI GZ 0.56
   125. 411 TO 413 UNI GZ 0.40
   127. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
   128. MEMBER LOAD
   129. 2001 2002 UNI GY 3.75
   130. 2003 2004 UNI GY 1.25
   131. 3001 3002 4001 4002 UNI GY 6.75
   132. 3003 3004 4003 4004 UNI GY 2.25
   133. 5001 5002 UNI GY 4.88
   134. 5003 5004 UNI GY 1.63
   135. 6001 6002 UNI GY 2.25
   136. 6003 6004 UNI GY 0.75
   137. 21 UNI GZ 3.75
   138. 22 UNI GZ 3.56
   139. 23 UNI GZ 2.84
   140. 201 TO 203 UNI GZ 1.78
   141. 211 TO 213 UNI GZ 1.26
   142. 41 UNI GZ 4.13
   143. 42 UNI GZ 3.92
   144. 43 UNI GZ 3.13
   145. 401 TO 403 UNI GZ 1.95
   146. 411 TO 413 UNI GZ 1.39
   148. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)
   149. MEMBER LOAD
   150. 2001 TO 2004 UNI GY -1.55
   151. 3001 TO 3004 4001 TO 4004 UNI GY -2.79
   152. 5001 TO 5004 UNI GY -2.02
   153. 6001 TO 6004 UNI GY -0.93
   154. 21 UNI GZ -2.95
   155. 22 UNI GZ -2.79
   156. 23 UNI GZ -2.23
   157. 201 TO 203 UNI GZ -1.40
   158. 211 TO 213 UNI GZ -0.99
   159. 41 UNI GZ 2.95
   160. 42 UNI GZ 2.79
   161. 43 UNI GZ 2.23
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162. 401 TO 403 UNI GZ 1.40
163. 411 TO 413 UNI GZ 0.99
165. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
   STAAD SPACE                                          -- PAGE NO.    4

166. MEMBER LOAD
167. 2001 TO 2004 UNI GY 2.8
168. 3001 TO 3004 4001 TO 4004 UNI GY 5.04
169. 5001 TO 5004 UNI GY 3.64
170. 6001 TO 6004 UNI GY 1.68
171. 21 UNI GZ -5.89
172. 22 UNI GZ -5.58
173. 23 UNI GZ -4.46
174. 201 TO 203 UNI GZ -2.79
175. 211 TO 213 UNI GZ -1.98
176. 41 UNI GZ 5.89
177. 42 UNI GZ 5.58
178. 43 UNI GZ 4.46
179. 401 TO 403 UNI GZ 2.79
180. 411 TO 413 UNI GZ 1.98
182. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0)
183. MEMBER LOAD
184. 2001 2002 UNI GY 0.58
185. 2003 2004 UNI GY 2.18
186. 3001 3002 4001 4002 UNI GY 1.04
187. 3003 3004 4003 4004 UNI GY 3.92
188. 5001 5002 UNI GY 0.75
189. 5003 5004 UNI GY 2.83
190. 6001 6002 UNI GY 0.35
191. 6003 6004 UNI GY 1.31
192. 201 TO 203 401 TO 403 UNI GX -0.63
193. 211 TO 213 411 TO 413 UNI GX 2.18
195. LOAD 16 WL6 (INTERNAL PRESSURE, WIND ANGLE 0)
196. MEMBER LOAD
197. 2001 2002 UNI GY 1.25
198. 2003 2004 UNI GY 3.75
199. 3001 3002 4001 4002 UNI GY 2.25
200. 3003 3004 4003 4004 UNI GY 6.75
201. 5001 5002 UNI GY 1.63
202. 5003 5004 UNI GY 4.88
203. 6001 6002 UNI GY 0.75
204. 6003 6004 UNI GY 2.25
205. 201 TO 203 401 TO 403 UNI GX -2.18
206. 211 TO 213 411 TO 413 UNI GX 1.98
208. LOAD 21 EQ1 (EARTHQUAKE LOAD)
209. JOINT LOAD
210. 204 206 208 209 210 404 406 408 409 410 FX 7.02
211. LOAD 22 EQ1 (EARTHQUAKE LOAD)
212. JOINT LOAD
213. 204 206 208 209 210 404 406 408 409 410 FZ 7.02
215. LOAD COMB 1001 1.4DL + 1.6LL
216. 1 1.4 2 1.4 3 1.6
218. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
219. 1 1.2 2 1.2 3 1.2 11 1.2
220. LOAD COMB 1111 1.4DL + 1.4WL1
221. 1 1.4 2 1.4 11 1.4
222. LOAD COMB 1211 1.0DL + 1.4WL1
223. 1 1.0 2 1.0 11 1.4
224. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
225. 1 1.2 2 1.2 3 1.2 12 1.2
226. LOAD COMB 1112 1.4DL + 1.4WL2
   STAAD SPACE                                          -- PAGE NO.    5

227. 1 1.4 2 1.4 12 1.4
228. LOAD COMB 1212 1.0DL + 1.4WL2
229. 1 1.0 2 1.0 12 1.4
230. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
231. 1 1.2 2 1.2 3 1.2 13 1.2
232. LOAD COMB 1113 1.4DL + 1.4WL3
233. 1 1.4 2 1.4 13 1.4
234. LOAD COMB 1213 1.0DL + 1.4WL3
235. 1 1.0 2 1.0 13 1.4
236. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
237. 1 1.2 2 1.2 3 1.2 14 1.2
238. LOAD COMB 1114 1.4DL + 1.4WL4
239. 1 1.4 2 1.4 14 1.4
240. LOAD COMB 1214 1.0DL + 1.4WL4
241. 1 1.0 2 1.0 14 1.4
242. LOAD COMB 1015 1.2DL + 1.2LL + 1.2WL5
243. 1 1.2 2 1.2 3 1.2 15 1.2
244. LOAD COMB 1115 1.4DL + 1.4WL5
245. 1 1.4 2 1.4 15 1.4
246. LOAD COMB 1215 1.0DL + 1.4WL5
247. 1 1.0 2 1.0 15 1.4
248. LOAD COMB 1016 1.2DL + 1.2LL + 1.2WL6
249. 1 1.2 2 1.2 3 1.2 16 1.2
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250. LOAD COMB 1116 1.4DL + 1.4WL6
251. 1 1.4 2 1.4 16 1.4
252. LOAD COMB 1216 1.0DL + 1.4WL6
253. 1 1.0 2 1.0 16 1.4
256. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
257. 1 1.2 2 1.2 3 0.5 21 1.0
258. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
259. 1 1.2 2 1.2 3 0.5 22 1.0
260. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ1
261. 1 1.2 2 1.2 3 0.5 21 -1.0
262. LOAD COMB 1122 1.2DL + 0.5LL - 1.0EQ2
263. 1 1.2 2 1.2 3 0.5 22 -1.0
264. LOAD COMB 1221 0.9DL + 1.0EQ1
265. 1 0.9 2 0.9 21 1.0
266. LOAD COMB 1222 0.9DL + 1.0EQ2
267. 1 0.9 2 0.9 22 1.0
268. LOAD COMB 1321 0.9DL - 1.0EQ1
269. 1 0.9 2 0.9 21 -1.0
270. LOAD COMB 1322 0.9DL - 1.0EQ2
271. 1 0.9 2 0.9 22 -1.0
272. *SLS
273. LOAD COMB 2001 1.0DL + 1.0LL
274. 1 1.0 2 1.0 3 1.0
275. LOAD COMB 2002 1.0DL + 0.5LL
276. 1 1.0 2 1.0 3 0.5
277. LOAD COMB 2011 DL + LL + WL1
278. 1 1.0 2 1.0 3 1.0 11 1.0
279. LOAD COMB 2012 DL + LL + WL2
280. 1 1.0 2 1.0 3 1.0 12 1.0
281. LOAD COMB 2013 DL + LL + WL3
282. 1 1.0 2 1.0 3 1.0 13 1.0
283. LOAD COMB 2014 DL + LL + WL4
284. 1 1.0 2 1.0 3 1.0 14 1.0
   STAAD SPACE                                          -- PAGE NO.    6

285. LOAD COMB 2015 DL + LL + WL5
286. 1 1.0 2 1.0 3 1.0 15 1.0
287. LOAD COMB 2016 DL + LL + WL6
288. 1 1.0 2 1.0 3 1.0 16 1.0
290. PERFORM ANALYSIS

          P R O B L E M   S T A T I S T I C S
          -----------------------------------

   NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =     55/    76/    10
   ORIGINAL/FINAL BAND-WIDTH=    40/     9/     60 DOF
   TOTAL PRIMARY LOAD CASES =    11, TOTAL DEGREES OF FREEDOM =     300
   SIZE OF STIFFNESS MATRIX =      18 DOUBLE  KILO-WORDS
   REQRD/AVAIL. DISK SPACE  =     12.4/ 402640.4 MB

 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=    1


 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=    2


 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=    3


 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=   11


 **START ITERATION NO.          2

 **NOTE-Tension/Compression converged after  2 iterations, Case=   12


 **NOTE-Tension/Compression converged after  1 iterations, Case=   13
```

```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=   14


**START ITERATION NO.          2
**START ITERATION NO.          3
    STAAD SPACE                                         -- PAGE NO.    7

**START ITERATION NO.          4
**START ITERATION NO.          5

**NOTE-Tension/Compression converged after  5 iterations, Case=   15


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=   16


**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=   21


**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after  3 iterations, Case=   22


 291. LOAD LIST 1001 1011 TO 1016 1021 1022 1111 TO 1116 1121 1122 -
 292. 1211 TO 1216 1221 1222 1321 1322
 293. PRINT SUPPORT REACTION ALL
 SUPPORT  REACTION ALL
    STAAD SPACE                                         -- PAGE NO.    8
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SUPPORT REACTIONS -UNIT KN   METE   STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
201	1001	-0.46	34.25	-3.47	0.00	0.00	0.00
	1011	0.02	9.61	-24.04	0.00	0.00	0.00
	1111	-0.13	4.20	-28.01	0.00	0.00	0.00
	1211	0.06	-3.30	-27.03	0.00	0.00	0.00
	1012	-0.14	2.89	-22.27	0.00	0.00	0.00
	1112	-0.31	-3.64	-25.95	0.00	0.00	0.00
	1212	-0.13	-11.14	-24.97	0.00	0.00	0.00
	1013	-0.28	35.40	-1.69	0.00	0.00	0.00
	1113	-0.48	34.28	-1.94	0.00	0.00	0.00
	1213	-0.30	26.79	-0.96	0.00	0.00	0.00
	1014	-0.71	16.03	-1.81	0.00	0.00	0.00
	1114	-0.97	11.68	-2.08	0.00	0.00	0.00
	1214	-0.79	4.19	-1.10	0.00	0.00	0.00
	1015	-0.49	23.04	-3.33	0.00	0.00	0.00
	1115	-0.72	19.86	-3.85	0.00	0.00	0.00
	1215	-0.54	12.37	-2.87	0.00	0.00	0.00
	1016	4.10	20.73	-3.55	0.00	0.00	0.00
	1116	4.63	17.18	-4.10	0.00	0.00	0.00
	1216	4.81	9.68	-3.12	0.00	0.00	0.00
	1021	-10.53	11.55	-3.07	0.00	0.00	0.00
	1022	0.07	14.36	-17.29	0.00	0.00	0.00
	1121	9.55	38.44	-2.84	0.00	0.00	0.00
	1122	-1.05	35.63	11.38	0.00	0.00	0.00
	1221	-10.45	3.43	-2.32	0.00	0.00	0.00
	1222	0.15	6.23	-16.54	0.00	0.00	0.00
	1321	9.64	30.31	-2.09	0.00	0.00	0.00
	1322	-0.96	27.51	12.13	0.00	0.00	0.00
214	1001	0.66	27.65	-3.81	0.00	0.00	0.00
	1011	0.19	10.46	-19.11	0.00	0.00	0.00
	1111	0.32	7.75	-22.28	0.00	0.00	0.00
	1211	0.10	1.31	-21.20	0.00	0.00	0.00
	1012	0.28	6.61	-17.84	0.00	0.00	0.00

	1112	0.42	3.26	-20.79	0.00	0.00	0.00
	1212	0.20	-3.18	-19.71	0.00	0.00	0.00
	1013	0.50	27.11	-2.39	0.00	0.00	0.00
	1113	0.67	27.17	-2.77	0.00	0.00	0.00
	1213	0.46	20.73	-1.69	0.00	0.00	0.00
	1014	0.75	15.99	-2.33	0.00	0.00	0.00
	1114	0.97	14.20	-2.69	0.00	0.00	0.00
	1214	0.75	7.75	-1.61	0.00	0.00	0.00
	1015	-5.22	21.98	-3.42	0.00	0.00	0.00
	1115	-6.00	21.19	-3.97	0.00	0.00	0.00
	1215	-6.22	14.74	-2.89	0.00	0.00	0.00
	1016	-3.16	17.10	-3.69	0.00	0.00	0.00
	1116	-3.59	15.50	-4.29	0.00	0.00	0.00
	1216	-3.81	9.05	-3.21	0.00	0.00	0.00
	1021	-10.46	42.06	-3.25	0.00	0.00	0.00
	1022	0.34	12.92	-13.44	0.00	0.00	0.00

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    STAAD SPACE                                         -- PAGE NO.    9
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SUPPORT REACTIONS -UNIT KN   METE   STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	1121	11.70	-0.20	-3.25	0.00	0.00	0.00
	1122	0.90	28.94	6.94	0.00	0.00	0.00
	1221	-10.59	35.63	-2.43	0.00	0.00	0.00
	1222	0.21	6.49	-12.62	0.00	0.00	0.00
	1321	11.57	-6.62	-2.43	0.00	0.00	0.00
	1322	0.77	22.52	7.75	0.00	0.00	0.00
401	1001	-0.46	34.25	3.47	0.00	0.00	0.00
	1011	-1.17	35.13	2.42	0.00	0.00	0.00
	1111	-1.51	33.98	2.79	0.00	0.00	0.00
	1211	-1.33	26.48	1.81	0.00	0.00	0.00
	1012	-1.24	31.87	0.55	0.00	0.00	0.00
	1112	-1.60	30.17	0.61	0.00	0.00	0.00
	1212	-1.42	22.68	-0.37	0.00	0.00	0.00
	1013	-0.28	35.40	1.69	0.00	0.00	0.00
	1113	-0.48	34.28	1.94	0.00	0.00	0.00
	1213	-0.30	26.79	0.96	0.00	0.00	0.00
	1014	-0.71	16.03	1.81	0.00	0.00	0.00
	1114	-0.97	11.68	2.08	0.00	0.00	0.00
	1214	-0.79	4.19	1.10	0.00	0.00	0.00
	1015	-0.64	17.15	3.34	0.00	0.00	0.00
	1115	-0.90	12.99	3.87	0.00	0.00	0.00
	1215	-0.72	5.50	2.89	0.00	0.00	0.00
	1016	3.87	11.60	3.53	0.00	0.00	0.00
	1116	4.36	6.52	4.09	0.00	0.00	0.00
	1216	4.55	-0.98	3.11	0.00	0.00	0.00
	1021	-10.53	11.55	3.07	0.00	0.00	0.00
	1022	-1.09	35.61	3.12	0.00	0.00	0.00
	1121	9.55	38.44	2.84	0.00	0.00	0.00
	1122	0.11	14.38	2.78	0.00	0.00	0.00
	1221	-10.45	3.43	2.32	0.00	0.00	0.00
	1222	-1.00	27.49	2.38	0.00	0.00	0.00
	1321	9.64	30.31	2.09	0.00	0.00	0.00
	1322	0.19	6.25	2.04	0.00	0.00	0.00
414	1001	0.66	27.65	3.81	0.00	0.00	0.00
	1011	1.15	28.84	2.91	0.00	0.00	0.00
	1111	1.44	29.20	3.38	0.00	0.00	0.00
	1211	1.22	22.75	2.30	0.00	0.00	0.00
	1012	1.21	26.97	1.58	0.00	0.00	0.00
	1112	1.51	27.01	1.83	0.00	0.00	0.00
	1212	1.29	20.57	0.74	0.00	0.00	0.00
	1013	0.50	27.11	2.39	0.00	0.00	0.00
	1113	0.67	27.17	2.77	0.00	0.00	0.00
	1213	0.46	20.73	1.69	0.00	0.00	0.00
	1014	0.75	15.99	2.33	0.00	0.00	0.00
	1114	0.97	14.20	2.69	0.00	0.00	0.00
	1214	0.75	7.75	1.61	0.00	0.00	0.00
	1015	-5.16	18.70	3.39	0.00	0.00	0.00
	1115	-5.92	17.36	3.94	0.00	0.00	0.00

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    STAAD SPACE                                         -- PAGE NO.   10
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SUPPORT REACTIONS -UNIT KN   METE   STRUCTURE TYPE = SPACE

JOINT LOAD FORCE-X FORCE-Y FORCE-Z MOM-X MOM-Y MOM Z

	1215	-6.14	10.91	2.86	0.00	0.00	0.00
	1016	-3.04	12.00	3.62	0.00	0.00	0.00
	1116	-3.45	9.55	4.21	0.00	0.00	0.00
	1216	-3.67	3.10	3.12	0.00	0.00	0.00
	1021	-10.46	42.06	3.25	0.00	0.00	0.00
	1022	0.93	28.96	3.17	0.00	0.00	0.00
	1121	11.70	-0.20	3.25	0.00	0.00	0.00
	1122	0.32	12.90	3.33	0.00	0.00	0.00
	1221	-10.59	35.63	2.43	0.00	0.00	0.00
	1222	0.80	22.53	2.35	0.00	0.00	0.00
	1321	11.57	-6.62	2.43	0.00	0.00	0.00
	1322	0.18	6.47	2.52	0.00	0.00	0.00
2006	1001	-0.17	48.88	1.17	0.00	0.00	0.00
	1011	-0.06	2.91	-37.67	0.00	0.00	0.00
	1111	-0.02	-13.01	-44.40	0.00	0.00	0.00
	1211	0.02	-21.62	-44.59	0.00	0.00	0.00
	1012	-0.01	-11.06	-30.99	0.00	0.00	0.00
	1112	0.04	-29.31	-36.60	0.00	0.00	0.00
	1212	0.07	-37.92	-36.79	0.00	0.00	0.00
	1013	-0.18	54.50	7.88	0.00	0.00	0.00
	1113	-0.17	47.17	8.74	0.00	0.00	0.00
	1213	-0.14	38.56	8.55	0.00	0.00	0.00
	1014	-0.04	13.50	11.57	0.00	0.00	0.00
	1114	0.00	-0.66	13.05	0.00	0.00	0.00
	1214	0.03	-9.27	12.86	0.00	0.00	0.00
	1015	-0.77	35.13	0.37	0.00	0.00	0.00
	1115	-0.85	24.58	-0.02	0.00	0.00	0.00
	1215	-0.82	15.97	-0.21	0.00	0.00	0.00
	1016	-0.08	27.46	-0.22	0.00	0.00	0.00
	1116	-0.04	15.63	-0.71	0.00	0.00	0.00
	1216	-0.01	7.02	-0.89	0.00	0.00	0.00
	1021	-4.96	42.80	0.68	0.00	0.00	0.00
	1022	-0.06	17.48	-15.07	0.00	0.00	0.00
	1121	4.74	20.58	0.76	0.00	0.00	0.00
	1122	-0.16	45.90	16.52	0.00	0.00	0.00
	1221	-4.92	30.48	0.38	0.00	0.00	0.00
	1222	-0.02	5.16	-15.38	0.00	0.00	0.00
	1321	4.78	8.27	0.46	0.00	0.00	0.00
	1322	-0.12	33.58	16.22	0.00	0.00	0.00
2008	1001	-0.05	47.67	1.92	0.00	0.00	0.00
	1011	-0.03	11.81	-20.05	0.00	0.00	0.00
	1111	-0.02	-4.89	-24.50	0.00	0.00	0.00
	1211	-0.01	-12.41	-24.69	0.00	0.00	0.00
	1012	-0.02	-1.06	-13.69	0.00	0.00	0.00
	1112	-0.01	-19.91	-17.08	0.00	0.00	0.00
	1212	0.00	-27.43	-17.27	0.00	0.00	0.00
	1013	-0.05	52.03	8.54	0.00	0.00	0.00
	1113	-0.05	[illegible]	8.86	0.00	0.00	0.00

STAAD SPACE

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	1213	-0.04	34.51	8.67	0.00	0.00	0.00
	1014	-0.02	14.33	14.04	0.00	0.00	0.00
	1114	-0.02	-1.94	15.27	0.00	0.00	0.00
	1214	0.00	-9.47	15.08	0.00	0.00	0.00
	1015	-0.60	31.80	1.12	0.00	0.00	0.00
	1115	-0.69	18.43	0.20	0.00	0.00	0.00
	1215	-0.68	10.91	0.01	0.00	0.00	0.00
	1016	0.02	25.33	0.81	0.00	0.00	0.00
	1116	0.04	10.88	-0.16	0.00	0.00	0.00
	1216	0.05	3.36	-0.35	0.00	0.00	0.00
	1021	-4.18	27.95	0.96	0.00	0.00	0.00
	1022	-0.01	22.11	-3.27	0.00	0.00	0.00
	1121	4.10	30.53	0.96	0.00	0.00	0.00
	1122	-0.07	36.37	5.19	0.00	0.00	0.00
	1221	-4.16	15.64	0.42	0.00	0.00	0.00
	1222	0.00	9.80	-3.81	0.00	0.00	0.00
	1321	4.11	18.22	0.42	0.00	0.00	0.00
	1322	-0.05	24.06	4.66	0.00	0.00	0.00
2009	1001	0.02	37.81	0.87	0.00	0.00	0.00
	1011	0.01	1.64	-30.41	0.00	0.00	0.00
	1111	0.01	-10.20	-35.81	0.00	0.00	0.00
	1211	0.00	-17.05	-35.95	0.00	0.00	0.00
	1012	0.01	-8.62	-25.35	0.00	0.00	0.00
	1112	0.00	-22.16	-29.90	0.00	0.00	0.00
	1212	0.00	-29.01	-30.04	0.00	0.00	0.00
	1013	0.02	41.65	5.89	0.00	0.00	0.00
	1113	0.02	36.49	6.54	0.00	0.00	0.00

	1213	0.02	29.64	6.40	0.00	0.00	0.00
	1014	0.01	11.60	8.72	0.00	0.00	0.00
	1114	0.01	1.42	9.84	0.00	0.00	0.00
	1214	0.01	-5.43	9.70	0.00	0.00	0.00
	1015	-0.59	25.99	0.23	0.00	0.00	0.00
	1115	-0.69	18.22	-0.06	0.00	0.00	0.00
	1215	-0.69	11.37	-0.20	0.00	0.00	0.00
	1016	0.13	23.19	-0.02	0.00	0.00	0.00
	1116	0.15	14.96	-0.35	0.00	0.00	0.00
	1216	0.15	8.10	-0.50	0.00	0.00	0.00
	1021	-4.97	7.37	-0.44	0.00	0.00	0.00
	1022	0.04	6.28	-20.50	0.00	0.00	0.00
	1121	5.00	42.38	1.53	0.00	0.00	0.00
	1122	0.00	43.47	21.58	0.00	0.00	0.00
	1221	-4.97	-2.09	-0.67	0.00	0.00	0.00
	1222	0.03	-3.18	-20.72	0.00	0.00	0.00
	1321	5.00	32.92	1.30	0.00	0.00	0.00
	1322	-0.01	34.01	21.36	0.00	0.00	0.00
4006	1001	-0.17	48.88	-1.17	0.00	0.00	0.00
	1011	-0.11	50.76	-2.86	0.00	0.00	0.00
	1111	-0.08	42.82	-2.89	0.00	0.00	0.00

STAAD SPACE

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	1211	-0.05	34.21	-2.70	0.00	0.00	0.00
	1012	-0.09	43.58	-9.41	0.00	0.00	0.00
	1112	-0.05	34.44	-10.53	0.00	0.00	0.00
	1212	-0.02	25.83	-10.35	0.00	0.00	0.00
	1013	-0.18	54.50	-7.88	0.00	0.00	0.00
	1113	-0.17	47.17	-8.74	0.00	0.00	0.00
	1213	-0.14	38.56	-8.55	0.00	0.00	0.00
	1014	-0.04	13.50	-11.57	0.00	0.00	0.00
	1114	0.00	-0.66	-13.05	0.00	0.00	0.00
	1214	0.03	-9.27	-12.86	0.00	0.00	0.00
	1015	-0.73	22.75	-0.35	0.00	0.00	0.00
	1115	-0.81	10.14	0.04	0.00	0.00	0.00
	1215	-0.78	1.53	0.23	0.00	0.00	0.00
	1016	-0.01	8.06	0.28	0.00	0.00	0.00
	1116	0.03	-7.00	0.78	0.00	0.00	0.00
	1216	0.06	-15.61	0.96	0.00	0.00	0.00
	1021	-4.96	42.80	-0.68	0.00	0.00	0.00
	1022	-0.16	45.86	-0.92	0.00	0.00	0.00
	1121	4.74	20.58	-0.76	0.00	0.00	0.00
	1122	-0.06	17.52	-0.53	0.00	0.00	0.00
	1221	-4.92	30.48	-0.38	0.00	0.00	0.00
	1222	-0.12	33.55	-0.62	0.00	0.00	0.00
	1321	4.78	8.27	-0.46	0.00	0.00	0.00
	1322	-0.02	5.20	-0.22	0.00	0.00	0.00
4008	1001	-0.05	47.67	-1.92	0.00	0.00	0.00
	1011	-0.04	41.72	-9.96	0.00	0.00	0.00
	1111	-0.03	30.01	-10.52	0.00	0.00	0.00
	1211	-0.02	22.48	-10.33	0.00	0.00	0.00
	1012	-0.03	35.15	-16.27	0.00	0.00	0.00
	1112	-0.02	22.34	-17.88	0.00	0.00	0.00
	1212	-0.01	14.82	-17.69	0.00	0.00	0.00
	1013	-0.05	52.03	-8.54	0.00	0.00	0.00
	1113	-0.05	42.03	-8.86	0.00	0.00	0.00
	1213	-0.04	34.51	-8.67	0.00	0.00	0.00
	1014	-0.02	14.33	-14.04	0.00	0.00	0.00
	1114	-0.02	-1.94	-15.27	0.00	0.00	0.00
	1214	0.00	-9.47	-15.08	0.00	0.00	0.00
	1015	-0.60	20.55	-1.19	0.00	0.00	0.00
	1115	-0.69	5.31	-0.28	0.00	0.00	0.00
	1215	-0.67	-2.22	-0.09	0.00	0.00	0.00
	1016	0.03	7.75	-0.92	0.00	0.00	0.00
	1116	0.04	-9.63	0.04	0.00	0.00	0.00
	1216	0.06	-17.15	0.22	0.00	0.00	0.00
	1021	-4.18	27.95	-0.96	0.00	0.00	0.00
	1022	-0.06	36.53	-5.19	0.00	0.00	0.00
	1121	4.10	30.53	-0.96	0.00	0.00	0.00
	1122	-0.02	21.96	3.27	0.00	0.00	0.00
	1221	-4.16	15.64	-0.42	0.00	0.00	0.00

STAAD SPACE

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

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JOINT  LOAD   FORCE-X    FORCE-Y    FORCE-Z     MOM-X      MOM-Y      MOM Z
0
       1222    -0.05      24.21      -4.66       0.00       0.00       0.00
       1321     4.11      18.22      -0.42       0.00       0.00       0.00
       1322    -0.01       9.65       3.81       0.00       0.00       0.00
  4009 1001     0.02      37.81      -0.87       0.00       0.00       0.00
       1011     0.02      40.99      -2.17       0.00       0.00       0.00
       1111     0.02      35.71      -2.20       0.00       0.00       0.00
       1211     0.02      28.86      -2.06       0.00       0.00       0.00
       1012     0.02      35.70      -7.10       0.00       0.00       0.00
       1112     0.02      29.55      -7.96       0.00       0.00       0.00
       1212     0.02      22.69      -7.82       0.00       0.00       0.00
       1013     0.02      41.65      -5.89       0.00       0.00       0.00
       1113     0.02      36.49      -6.54       0.00       0.00       0.00
       1213     0.02      29.64      -6.40       0.00       0.00       0.00
       1014     0.01      11.60      -8.72       0.00       0.00       0.00
       1114     0.01       1.42      -9.84       0.00       0.00       0.00
       1214     0.01      -5.43      -9.70       0.00       0.00       0.00
       1015    -0.60      16.80      -0.16       0.00       0.00       0.00
       1115    -0.70       7.49       0.14       0.00       0.00       0.00
       1215    -0.71       0.64       0.28       0.00       0.00       0.00
       1016     0.12       8.81       0.15       0.00       0.00       0.00
       1116     0.14      -1.83       0.51       0.00       0.00       0.00
       1216     0.13      -8.68       0.65       0.00       0.00       0.00
       1021    -4.97       7.37       0.44       0.00       0.00       0.00
       1022     0.00      43.36      -0.81       0.00       0.00       0.00
       1121     5.00      42.38      -1.53       0.00       0.00       0.00
       1122     0.03       6.40      -0.27       0.00       0.00       0.00
       1221    -4.97      -2.09       0.67       0.00       0.00       0.00
       1222     0.00      33.89      -0.59       0.00       0.00       0.00
       1321     5.00      32.92      -1.30       0.00       0.00       0.00
       1322     0.02      -3.06      -0.05       0.00       0.00       0.00

    ************** END OF LATEST ANALYSIS RESULT **************

    294. PRINT FORCE ENVELOPE NSECTION 5 ALL
0 FORCE    ENVELOPE NSECTION 5        0
 0    STAAD SPACE                                          -- PAGE NO.   14

        MEMBER FORCE ENVELOPE
        ---------------------

        ALL UNITS ARE KN    METE

    MEMB  DISTANCE         FY    LD         MZ    LD        FZ    LD         MY    LD

     21     0.00  MAX     15.94 1114       0.00 1322       4.78 1321       0.00 1322
                  MIN    -17.98 1211       0.00 1322      -4.96 1021       0.00 1322
            0.85  MAX      8.92 1114      11.91 1211       4.78 1321       4.07 1321
                  MIN     -9.99 1211     -10.59 1114      -4.96 1021      -4.23 1021
            1.70  MAX      2.25 1013      17.01 1211       4.78 1321       8.15 1321
                  MIN     -2.00 1211     -15.19 1114      -4.96 1021      -8.46 1021
            2.55  MAX      6.17 1111      15.32 1211       4.78 1321      12.22 1321
                  MIN     -5.31 1214     -13.82 1114      -4.96 1021     -12.68 1021
            3.41  MAX     14.16 1111       6.82 1211       4.78 1321      16.29 1321
                  MIN    -12.33 1214      -7.66 1013      -4.96 1021     -16.91 1021
            4.26  MAX     22.15 1111       7.66 1214       4.78 1321      20.36 1321
                  MIN    -19.35 1214      -9.60 1011      -4.96 1021     -21.14 1021

  |---------------------------------------------------------------------------
  | MAX/MIN FORCE VALUES FOR MEMB       21, AMONGST ALL SECT LOCATIONS
  |              FY/   DIST   LD          MZ/   DIST   LD
  |              FZ    DIST   LD          MY    DIST   LD         FX      DIST   LD
  |
  | MAX.      22.15   4.26 1111        17.01   1.70 1211
  |            4.78   0.00 1321        20.36   4.26 1321       54.52 C   0.00 1013
  | MIN.     -19.35   4.26 1214       -15.19   1.70 1114
  |           -4.96   4.26 1021       -21.14   4.26 1021       17.26 T   4.26 1212
  |---------------------------------------------------------------------------

     22     0.00  MAX     15.27 1114       0.00 1322       4.11 1321       0.00 1322
                  MIN    -24.69 1211       0.00 1322      -4.18 1021       0.00 1322
            0.86  MAX      8.55 1114      17.94 1211       4.11 1321       3.54 1321
                  MIN    -17.04 1211     -10.24 1114      -4.18 1021      -3.59 1021
            1.72  MAX      5.19 1122      29.31 1211       4.11 1321       7.07 1321
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                  MIN     -9.40 1211     -14.71 1114      -4.18 1021      -7.18 1021
            2.58  MAX      5.19 1122      34.11 1211       4.11 1321      10.61 1321
                  MIN     -5.07 1214     -13.93 1014      -4.18 1021     -10.78 1021
            3.44  MAX      6.16 1011      32.32 1211       4.11 1321      14.14 1321
                  MIN    -11.79 1214     -17.86 1122      -4.18 1021     -14.37 1021
            4.30  MAX     13.73 1111      28.17 1212       4.11 1321      17.68 1321
                  MIN    -18.51 1214     -22.33 1122      -4.18 1021     -17.96 1021
0     STAAD SPACE                                                 -- PAGE NO.   15

  |---------------------------------------------------------------------------
  | MAX/MIN FORCE VALUES FOR MEMB       22, AMONGST ALL SECT LOCATIONS
  |              FY/   DIST   LD          MZ/   DIST   LD
  |              FZ    DIST   LD          MY    DIST   LD         FX      DIST   LD
  |
  | MAX.      15.27   0.00 1114        34.11   2.58 1211
  |            4.11   0.00 1321        17.68   4.30 1321       52.03 C   0.00 1013
  | MIN.     -24.69   0.00 1211       -22.33   4.30 1122
  |           -4.18   4.30 1021       -17.96   4.30 1021       29.21 T   4.30 1212
  |---------------------------------------------------------------------------

     23     0.00  MAX     11.99 1114       0.00 1322       5.00 1121       0.00 1322
                  MIN    -13.58 1211       0.00 1322      -4.97 1221       0.00 1322
            0.84  MAX      6.72 1114       8.93 1211       5.00 1121       4.22 1121
                  MIN     -7.58 1211      -7.90 1114      -4.97 1221      -4.20 1221
            1.69  MAX      1.68 1013      12.79 1211       5.00 1121       8.44 1121
                  MIN     -1.57 1211     -11.35 1114      -4.97 1221      -8.39 1221
            2.53  MAX      4.57 1111      11.58 1211       5.00 1121      12.67 1121
                  MIN     -3.96 1214     -10.35 1114      -4.97 1221     -12.59 1221
            3.38  MAX     10.57 1111       5.31 1211       5.00 1121      16.89 1121
                  MIN     -9.23 1214      -5.66 1013      -4.97 1221     -16.79 1221
            4.22  MAX     16.58 1111       5.58 1214       5.00 1121      21.11 1121
                  MIN    -14.50 1214      -6.87 1011      -4.97 1221     -20.99 1221

  |---------------------------------------------------------------------------
  | MAX/MIN FORCE VALUES FOR MEMB       23, AMONGST ALL SECT LOCATIONS
  |              FY/   DIST   LD          MZ/   DIST   LD
  |              FZ    DIST   LD          MY    DIST   LD         FX      DIST   LD
  |
  | MAX.      16.58   4.22 1111        12.79   1.69 1211
  |            5.00   0.00 1121        21.11   4.22 1121       41.59 C   0.00 1013
  | MIN.     -14.50   4.22 1214       -11.35   1.69 1114
  |           -4.97   4.22 1221       -20.99   4.22 1221       12.15 T   4.22 1212
  |---------------------------------------------------------------------------

     41     0.00  MAX      0.15 1216       0.00 1322       4.78 1321       0.00 1322
                  MIN    -15.94 1114       0.00 1322      -4.96 1021       0.00 1322
            0.85  MAX      0.15 1216      10.59 1114       4.78 1321       4.07 1321
                  MIN     -8.92 1114      -0.12 1216      -4.96 1021      -4.23 1021
            1.70  MAX      0.15 1216      15.19 1114       4.78 1321       8.15 1321
                  MIN     -2.25 1013      -0.25 1216      -4.96 1021      -8.46 1021
            2.55  MAX      5.31 1214      13.82 1114       4.78 1321      12.22 1321
                  MIN     -1.17 1001      -0.37 1216      -4.96 1021     -12.68 1021
            3.41  MAX     12.33 1214       7.66 1013       4.78 1321      16.29 1321
                  MIN     -1.17 1001      -0.49 1216      -4.96 1021     -16.91 1021
            4.26  MAX     19.35 1214       4.98 1001       4.78 1321      20.36 1321
                  MIN     -1.17 1001      -8.36 1212      -4.96 1021     -21.14 1021
0     STAAD SPACE                                                 -- PAGE NO.   16

  |---------------------------------------------------------------------------
  | MAX/MIN FORCE VALUES FOR MEMB       41, AMONGST ALL SECT LOCATIONS
  |              FY/   DIST   LD          MZ/   DIST   LD
  |              FZ    DIST   LD          MY    DIST   LD         FX      DIST   LD
  |
  | MAX.      19.35   4.26 1214        15.19   1.70 1114
  |            4.78   0.00 1321        20.36   4.26 1321       54.52 C   0.00 1013
  | MIN.     -15.94   0.00 1114        -8.36   4.26 1212
  |           -4.96   4.26 1021       -21.14   4.26 1021       16.95 T   4.26 1216
  |---------------------------------------------------------------------------

     42     0.00  MAX      3.81 1322       0.00 1322       4.11 1321       0.00 1322
                  MIN    -17.88 1112       0.00 1322      -4.18 1021       0.00 1322
            0.86  MAX      3.81 1322      13.35 1112       4.11 1321       3.54 1321
                  MIN    -13.16 1112      -3.28 1322      -4.18 1021      -3.59 1021
            1.72  MAX      3.81 1322      22.63 1112       4.11 1321       7.07 1321
                  MIN     -8.44 1112      -6.55 1322      -4.18 1021      -7.18 1021
            2.58  MAX      5.07 1214      27.86 1112       4.11 1321      10.61 1321
                  MIN     -6.46 1011      -9.83 1322      -4.18 1021     -10.78 1021
            3.44  MAX     11.79 1214      29.03 1112       4.11 1321      14.14 1321
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           MIN    -5.30 1011    -13.11 1322      -4.18 1021     -14.37 1021
     4.30  MAX    18.51 1214     30.60 1111       4.11 1321      17.68 1321
           MIN    -5.19 1022    -16.39 1322      -4.18 1021     -17.96 1021


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      42, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.      18.51    4.30 1214       30.60    4.30 1111
 |            4.11    0.00 1321       17.68    4.30 1321     52.03 C   0.00 1013
 | MIN.     -17.88    0.00 1112      -16.39    4.30 1322
 |           -4.18    4.30 1021      -17.96    4.30 1021     18.94 T   4.30 1216
 |-------------------------------------------------------------------------


  43   0.00  MAX     0.09 1216      0.00 1322       5.00 1121       0.00 1322
             MIN   -11.99 1114      0.00 1322      -4.97 1221       0.00 1322
       0.84  MAX     0.09 1216      7.90 1114       5.00 1121       4.22 1121
             MIN    -6.72 1114     -0.08 1216      -4.97 1221      -4.20 1221
       1.69  MAX     0.09 1216     11.35 1114       5.00 1121       8.44 1121
             MIN    -1.68 1013     -0.16 1216      -4.97 1221      -8.39 1221
       2.53  MAX     3.96 1214     10.35 1114       5.00 1121      12.67 1121
             MIN    -0.87 1001     -0.24 1216      -4.97 1221     -12.59 1221
       3.38  MAX     9.23 1214      5.66 1013       5.00 1121      16.89 1121
             MIN    -0.87 1001     -0.32 1216      -4.97 1221     -16.79 1221
       4.22  MAX    14.50 1214      3.68 1001       5.00 1121      21.11 1121
             MIN    -0.87 1001     -6.02 1212      -4.97 1221     -20.99 1221
0    STAAD SPACE                                                -- PAGE NO.    17


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB      43, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.      14.50    4.22 1214       11.35    1.69 1114
 |            5.00    0.00 1121       21.11    4.22 1121     43.04 C   0.00 1022
 | MIN.     -11.99    0.00 1114       -6.02    4.22 1212
 |           -4.97    4.22 1221      -20.99    4.22 1221     10.23 T   4.22 1216
 |-------------------------------------------------------------------------


 201   0.00  MAX    12.85 1121      0.00 1322       4.63 1211       0.00 1322
             MIN   -10.13 1221      0.00 1322      -4.69 1114       0.00 1322
       0.53  MAX    12.83 1121      5.37 1221       2.28 1211       1.83 1211
             MIN   -10.14 1221     -6.80 1121      -2.62 1114      -1.94 1114
       1.06  MAX    12.81 1121     10.75 1221      -0.08 1211       2.41 1211
             MIN   -10.16 1221    -13.59 1121      -0.55 1114      -2.77 1114
       1.59  MAX    12.79 1121     16.13 1221       1.62 1214       1.75 1211
             MIN   -10.17 1221    -20.38 1121      -2.52 1111      -2.52 1114
       2.12  MAX    12.78 1121     21.53 1221       3.69 1214      -0.16 1211
             MIN   -10.18 1221    -27.15 1121      -4.87 1111      -1.16 1114
       2.65  MAX    12.76 1121     26.93 1221       5.76 1214       1.54 1214
             MIN   -10.20 1221    -33.91 1121      -7.22 1111      -3.57 1111


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     201, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.      12.85    0.00 1121       26.93    2.65 1221
 |            5.76    2.65 1214        2.41    1.06 1211     38.84 C   0.00 1121
 | MIN.     -10.20    2.65 1221      -33.91    2.65 1121
 |           -7.22    2.65 1111       -3.57    2.65 1111      0.22 T   2.65 1212
 |-------------------------------------------------------------------------


 202   0.00  MAX     9.68 1021     33.93 1121       4.98 1114       2.64 1211
             MIN    -5.63 1321    -26.91 1221      -5.24 1211      -2.70 1114
       0.15  MAX     9.67 1021     34.63 1121       4.40 1114       1.90 1211
             MIN    -5.64 1321    -28.22 1221      -4.58 1211      -2.00 1114
       0.30  MAX     9.66 1021     35.34 1121       3.82 1114       1.27 1211
             MIN    -5.64 1321    -29.52 1221      -3.92 1211      -1.39 1114
       0.45  MAX     9.65 1021     36.04 1121       3.23 1114       0.74 1211
             MIN    -5.65 1321    -30.83 1221      -3.25 1211      -0.86 1114
       0.60  MAX     9.64 1021     36.75 1121       2.65 1114       0.30 1211
             MIN    -5.66 1321    -32.13 1221      -2.59 1211      -0.43 1114
       0.75  MAX     9.62 1021     37.46 1121       2.07 1114      -0.01 1013
             MIN    -5.67 1321    -33.43 1221      -1.93 1211      -0.07 1114
0    STAAD SPACE                                                -- PAGE NO.    18
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 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     202, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.       9.68    0.00 1021       37.46    0.75 1121
 |            4.98    0.00 1114        2.64    0.00 1211     28.67 C   0.00 1121
 | MIN.      -5.67    0.75 1321      -33.43    0.75 1221
 |           -5.24    0.00 1211       -2.70    0.00 1114      8.14 T   0.75 1212
 |-------------------------------------------------------------------------


 203   0.00  MAX    16.23 1013     37.46 1121       2.07 1114      -0.01 1013
             MIN    -5.76 1212    -33.43 1221      -1.93 1211      -0.07 1114
       0.15  MAX    16.19 1013     35.89 1121       1.64 1122       0.23 1013
             MIN    -5.79 1212    -34.12 1221      -1.27 1211      -0.28 1211
       0.30  MAX    16.15 1013     34.45 1321       1.64 1122       0.46 1122
             MIN    -5.82 1212    -34.94 1021      -0.61 1211      -0.42 1211
       0.45  MAX    16.12 1013     33.42 1321       1.64 1122       0.70 1122
             MIN    -5.85 1212    -36.15 1021      -0.28 1222      -0.46 1211
       0.60  MAX    16.08 1013     32.40 1321       1.64 1122       0.95 1122
             MIN    -5.88 1212    -37.36 1021      -0.51 1214      -0.40 1211
       0.75  MAX    16.04 1013     31.38 1321       1.64 1122       1.19 1122
             MIN    -5.92 1212    -38.56 1021      -1.09 1214      -0.27 1212


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     203, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.      16.23    0.00 1013       37.46    0.00 1121
 |            2.07    0.00 1114        1.19    0.75 1122     26.80 C   0.00 1121
 | MIN.      -5.92    0.75 1212      -38.56    0.75 1021
 |           -1.93    0.00 1211       -0.46    0.45 1211      9.32 T   0.75 1221
 |-------------------------------------------------------------------------


 204   0.00  MAX    12.12 1321     31.39 1321       0.57 1212       0.44 1122
             MIN   -13.93 1021    -38.55 1021      -0.40 1122      -0.45 1211
       0.15  MAX    12.08 1321     29.58 1321       0.57 1212       0.38 1122
             MIN   -13.98 1021    -36.47 1021      -0.40 1122      -0.37 1211
       0.30  MAX    12.04 1321     27.79 1321       0.57 1212       0.33 1122
             MIN   -14.03 1021    -34.38 1021      -0.40 1122      -0.29 1211
       0.45  MAX    12.00 1321     26.00 1321       0.57 1212       0.27 1122
             MIN   -14.09 1021    -32.29 1021      -0.40 1122      -0.20 1211
       0.60  MAX    11.96 1321     24.21 1321       0.57 1212       0.21 1122
             MIN   -14.14 1021    -30.18 1021      -0.40 1122      -0.12 1211
       0.75  MAX    11.92 1321     22.43 1321       0.57 1212       0.15 1122
             MIN   -14.20 1021    -28.07 1021      -0.40 1122      -0.04 1211
0    STAAD SPACE                                                -- PAGE NO.    19


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     204, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
 | MAX.      12.12    0.00 1321       31.39    0.00 1321
 |            0.57    0.00 1212        0.44    0.00 1122      4.90 C   0.00 1012
 | MIN.     -14.20    0.75 1021      -38.55    0.00 1021
 |           -0.40    0.75 1122       -0.45    0.00 1211      6.95 T   0.75 1216
 |-------------------------------------------------------------------------


 205   0.00  MAX    12.40 1321     22.43 1321       0.57 1212       0.15 1122
             MIN   -14.38 1021    -28.07 1021      -0.40 1122      -0.04 1211
       0.63  MAX    12.22 1321     14.63 1321       0.57 1212       0.34 1112
             MIN   -14.61 1021    -18.89 1021      -0.40 1122      -0.11 1322
       1.27  MAX    12.04 1321      6.95 1321       0.57 1212       0.69 1112
             MIN   -14.85 1021     -9.56 1021      -0.40 1122      -0.35 1322
       1.90  MAX    11.86 1321      0.25 1212       0.57 1212       1.04 1212
             MIN   -15.09 1021     -0.76 1121      -0.40 1122      -0.60 1122
       2.53  MAX    11.68 1321      9.56 1021       0.57 1212       1.40 1212
             MIN   -15.33 1021     -8.07 1321      -0.40 1122      -0.85 1122
       3.17  MAX    11.50 1321     19.35 1021       0.57 1212       1.76 1212
             MIN   -15.57 1021    -15.41 1321      -0.40 1122      -1.10 1122


 |-------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB     205, AMONGST ALL SECT LOCATIONS
 |             FY/    DIST   LD          MZ/   DIST   LD
 |             FZ     DIST   LD          MY    DIST   LD        FX       DIST   LD
 |
```

```
| MAX.     12.40   0.00 1321     22.43   0.00 1321
|           0.57   0.00 1212      1.76   3.17 1212        4.94 C    0.00 1011
| MIN.    -15.57   3.17 1021    -28.07   0.00 1021
|          -0.40   3.17 1122     -1.10   3.17 1122        6.93 T    3.17 1216
|-----------------------------------------------------------------------------


206    0.00  MAX     3.60 1121      5.56 1121      1.06 1122      4.25 1212
             MIN    -1.40 1221     -2.40 1221     -1.93 1212     -2.36 1122
       0.77  MAX     3.31 1121      2.90 1121      1.06 1122      2.76 1212
             MIN    -1.62 1221     -1.24 1221     -1.93 1212     -1.54 1122
       1.54  MAX     3.02 1121      0.77 1013      1.06 1122      1.27 1212
             MIN    -1.84 1221     -0.73 1212     -1.93 1212     -0.73 1122
       2.31  MAX     2.73 1121      1.63 1021      1.06 1122      0.09 1322
             MIN    -2.06 1221     -1.78 1321     -1.93 1212     -0.22 1111
       3.08  MAX     2.44 1121      3.27 1221      1.06 1122      0.90 1122
             MIN    -2.28 1221     -3.74 1121     -1.93 1212     -1.70 1212
       3.85  MAX     2.15 1321      5.11 1221      1.06 1122      1.72 1122
             MIN    -2.50 1021     -5.51 1121     -1.93 1212     -3.18 1212
□   STAAD SPACE                                            -- PAGE NO.   20


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      206, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.      3.60   0.00 1121      5.56   0.00 1121
|           1.06   0.00 1122      4.25   0.00 1212        4.53 C    0.00 1211
| MIN.     -2.50   3.85 1021     -5.51   3.85 1121
|          -1.93   3.85 1212     -3.18   3.85 1212        8.06 T    3.85 1116
|-----------------------------------------------------------------------------


207    0.00  MAX     2.16 1321      5.11 1221      1.06 1122      1.72 1122
             MIN    -2.53 1021     -5.51 1121     -1.93 1212     -3.18 1212
       0.23  MAX     2.09 1321      5.69 1221      1.06 1122      1.96 1122
             MIN    -2.61 1021     -5.99 1121     -1.93 1212     -3.63 1212
       0.46  MAX     2.03 1321      6.30 1221      1.06 1122      2.21 1122
             MIN    -2.70 1021     -6.46 1121     -1.93 1212     -4.07 1212
       0.69  MAX     1.96 1321      6.91 1221      1.06 1122      2.45 1122
             MIN    -2.79 1021     -6.90 1121     -1.93 1212     -4.51 1212
       0.92  MAX     1.90 1321      7.54 1221      1.06 1122      2.69 1122
             MIN    -2.87 1021     -7.33 1121     -1.93 1212     -4.96 1212
       1.15  MAX     1.83 1321      8.19 1221      1.06 1122      2.94 1122
             MIN    -2.96 1021     -7.73 1121     -1.93 1212     -5.40 1212


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      207, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.      2.16   0.00 1321      8.19   1.15 1221
|           1.06   0.00 1122      2.94   1.15 1122        4.52 C    1.15 1211
| MIN.     -2.96   1.15 1021     -7.73   1.15 1121
|          -1.93   1.15 1212     -5.40   1.15 1212        8.06 T    0.00 1116
|-----------------------------------------------------------------------------


208    0.00  MAX     3.98 1121      9.96 1321      3.08 1212      3.55 1122
             MIN    -3.23 1221     -9.78 1021     -1.79 1122     -6.53 1212
       0.80  MAX     3.70 1321      6.91 1321      3.08 1212      2.12 1122
             MIN    -3.48 1021     -7.12 1021     -1.79 1122     -4.06 1212
       1.60  MAX     3.48 1321      4.03 1321      3.08 1212      0.69 1322
             MIN    -3.78 1021     -4.21 1021     -1.79 1122     -1.60 1112
       2.40  MAX     3.25 1321      1.43 1121      3.08 1212      0.87 1212
             MIN    -4.09 1021     -1.15 1221     -1.79 1122     -0.74 1122
       3.20  MAX     3.02 1321      2.33 1021      3.08 1212      3.33 1212
             MIN    -4.39 1021     -1.16 1321     -1.79 1122     -2.17 1122
       4.00  MAX     2.79 1321      5.96 1021      3.08 1212      5.80 1212
             MIN    -4.69 1021     -3.49 1321     -1.79 1122     -3.60 1122
□   STAAD SPACE                                            -- PAGE NO.   21


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      208, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.      3.98   0.00 1121      9.96   0.00 1321
|           3.08   0.00 1212      5.80   4.00 1212        4.07 C    4.00 1211
| MIN.     -4.69   4.00 1021     -9.78   0.00 1021
|          -1.79   4.00 1122     -6.53   0.00 1212        8.03 T    0.00 1116
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|-----------------------------------------------------------------------------

209    0.00  MAX    23.30 1121     18.11 1121      2.01 1122      2.93 1212
             MIN   -19.41 1221    -15.51 1221     -1.85 1222     -2.97 1122
       0.37  MAX    23.16 1121      9.44 1121      2.01 1122      2.36 1212
             MIN   -19.51 1221     -8.25 1221     -1.85 1222     -2.22 1122
       0.75  MAX    23.02 1121      0.85 1321      2.01 1122      1.79 1212
             MIN   -19.62 1221     -0.98 1021     -1.85 1222     -1.47 1122
       1.12  MAX    22.88 1121      6.39 1221      2.01 1122      1.23 1112
             MIN   -19.72 1221     -7.75 1121     -1.85 1222     -0.72 1322
       1.49  MAX    22.74 1121     13.77 1221      2.01 1122      0.67 1112
             MIN   -19.83 1221    -16.26 1121     -1.85 1222      0.01 1216
       1.87  MAX    22.60 1121     21.20 1221      2.01 1122      0.78 1122
             MIN   -19.94 1221    -24.72 1121     -1.85 1222     -0.65 1222


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      209, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.     23.30   0.00 1121     21.20   1.87 1221
|           2.01   0.00 1122      2.93   0.00 1212        4.77 C    1.87 1021
| MIN.    -19.94   1.87 1221    -24.72   1.87 1121
|          -1.85   1.87 1222     -2.97   0.00 1122        7.45 T    0.00 1116
|-----------------------------------------------------------------------------


210    0.00  MAX    18.38 1121     21.20 1221      5.20 1022      0.77 1122
             MIN   -15.87 1221    -24.72 1121     -5.03 1322     -0.64 1222
       0.15  MAX    18.33 1121     23.56 1221      5.20 1022      0.15 1022
             MIN   -15.91 1221    -27.46 1121     -5.03 1322     -0.14 1211
       0.30  MAX    18.28 1121     25.94 1221      5.20 1022      0.93 1022
             MIN   -15.95 1221    -30.19 1121     -5.03 1322     -0.75 1322
       0.45  MAX    18.22 1121     28.32 1221      5.20 1022      1.70 1022
             MIN   -15.99 1221    -32.91 1121     -5.03 1322     -1.50 1322
       0.60  MAX    18.17 1121     30.71 1221      5.20 1022      2.48 1022
             MIN   -16.03 1221    -35.62 1121     -5.03 1322     -2.25 1322
       0.75  MAX    18.11 1121     33.10 1221      5.20 1022      3.25 1022
             MIN   -16.07 1221    -38.32 1121     -5.03 1322     -3.00 1322
□   STAAD SPACE                                            -- PAGE NO.   22


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      210, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.     18.38   0.00 1121     33.10   0.75 1221
|           5.20   0.00 1022      3.25   0.75 1022       16.55 C    0.75 1021
| MIN.    -16.07   0.75 1221    -38.32   0.75 1121
|          -5.03   0.75 1322     -3.00   0.75 1322       17.37 T    0.00 1121
|-----------------------------------------------------------------------------


211    0.00  MAX     2.53 1212     33.09 1221      0.56 1214      1.39 1022
             MIN   -12.15 1021    -38.33 1121     -1.81 1022     -0.48 1322
       0.15  MAX     2.50 1212     34.55 1221      0.51 1322      1.12 1022
             MIN   -12.19 1021    -38.22 1121     -1.81 1022     -0.47 1211
       0.30  MAX     2.47 1212     36.01 1221      0.51 1322      0.85 1022
             MIN   -12.22 1021    -38.09 1121     -1.81 1022     -0.47 1211
       0.45  MAX     2.43 1212     37.48 1221      0.70 1211      0.58 1022
             MIN   -12.26 1021    -37.96 1121     -1.81 1022     -0.40 1211
       0.60  MAX     2.40 1212     39.21 1021      1.17 1211      0.31 1022
             MIN   -12.30 1021    -38.09 1321     -1.81 1022     -0.26 1211
       0.75  MAX     2.37 1212     41.05 1021      1.65 1211      0.05 1222
             MIN   -12.34 1021    -38.31 1321     -1.81 1022     -0.11 1122


|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB      211, AMONGST ALL SECT LOCATIONS
|            FY/   DIST   LD          MZ/   DIST   LD
|            FZ    DIST   LD          MY    DIST   LD          FX      DIST   LD
|
| MAX.      2.53   0.00 1212     41.05   0.75 1021
|           1.65   0.75 1211      1.39   0.00 1022       30.79 C    0.75 1021
| MIN.    -12.34   0.75 1021    -38.33   0.00 1121
|          -1.81   0.75 1022     -0.48   0.00 1322       18.54 T    0.00 1321
|-----------------------------------------------------------------------------


212    0.00  MAX     5.89 1221     41.05 1021      1.65 1211      0.04 1222
             MIN    -8.82 1121    -38.31 1321     -1.81 1022     -0.10 1122
       0.15  MAX     5.89 1221     40.26 1021      2.12 1211      0.23 1211
```

```
             MIN     -8.83 1121     -37.09 1321      -2.16 1114     -0.36 1114
       0.30  MAX      5.88 1221      39.48 1021       2.59 1211      0.58 1211
             MIN     -8.84 1121     -35.87 1321      -2.58 1114     -0.71 1114
       0.45  MAX      5.87 1221      38.70 1021       3.06 1211      1.00 1211
             MIN     -8.85 1121     -34.64 1321      -2.99 1114     -1.13 1114
       0.60  MAX      5.86 1221      37.92 1021       3.53 1211      1.49 1211
             MIN     -8.86 1121     -33.42 1321      -3.40 1114     -1.60 1114
       0.75  MAX      5.85 1221      37.15 1021       4.00 1211      2.06 1211
             MIN     -8.87 1121     -32.19 1321      -3.82 1114     -2.14 1114
```
STAAD SPACE


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 MAX/MIN FORCE VALUES FOR MEMB     212, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.       5.89    0.00 1221       41.05    0.00 1021
            4.00    0.75 1211        2.06    0.75 1211       33.03 C     0.75 1021
 MIN.      -8.87    0.75 1121      -38.31    0.00 1321
           -3.82    0.75 1114       -2.14    0.75 1114       16.52 T     0.00 1321

213    0.00  MAX     12.19 1321      32.20 1321       5.31 1111      0.91 1214
             MIN    -13.97 1021     -37.13 1021      -4.02 1214     -2.96 1111
       0.53  MAX     12.18 1321      25.75 1321       3.63 1111      0.14 1222
             MIN    -13.99 1021     -29.72 1021      -2.55 1214     -1.23 1122
       1.06  MAX     12.16 1321      19.30 1321       1.95 1111      1.05 1211
             MIN    -14.01 1021     -22.31 1021      -1.08 1214     -1.95 1114
       1.59  MAX     12.15 1321      12.86 1321       0.58 1122      1.59 1211
             MIN    -14.03 1021     -14.88 1021      -0.07 1222     -2.07 1114
       2.12  MAX     12.14 1321       6.43 1321       1.96 1114      1.24 1211
             MIN    -14.04 1021      -7.44 1021      -1.50 1211     -1.43 1114
       2.65  MAX     12.12 1321       0.00 1322       3.43 1114      0.00 1322
             MIN    -14.06 1021       0.00 1322      -3.17 1211      0.00 1322

 MAX/MIN FORCE VALUES FOR MEMB     213, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.      12.19    0.00 1321       32.20    0.00 1321
            5.31    0.00 1111        1.59    1.59 1211       42.58 C     2.65 1021
 MIN.     -14.06    2.65 1021      -37.13    0.00 1021
           -4.02    0.00 1214       -2.96    0.00 1111        5.14 T     0.00 1321

401    0.00  MAX     12.85 1121       0.00 1322       4.69 1114      0.00 1322
             MIN    -10.13 1221       0.00 1322       0.05 1222      0.00 1322
       0.53  MAX     12.83 1121       5.37 1221       2.62 1114      1.94 1114
             MIN    -10.14 1221      -6.80 1121       0.05 1222      0.02 1222
       1.06  MAX     12.81 1121      10.75 1221       0.55 1114      2.77 1114
             MIN    -10.16 1221     -13.59 1121       0.05 1222      0.05 1222
       1.59  MAX     12.79 1121      16.13 1221       0.44 1122      2.52 1114
             MIN    -10.17 1221     -20.38 1121      -1.62 1214      0.07 1222
       2.12  MAX     12.78 1121      21.53 1221       0.44 1122      1.16 1114
             MIN    -10.18 1221     -27.15 1121      -3.69 1214      0.10 1222
       2.65  MAX     12.76 1121      26.93 1221       0.44 1122      1.18 1122
             MIN    -10.20 1221     -33.91 1121      -5.76 1214     -1.54 1214
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STAAD SPACE


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 MAX/MIN FORCE VALUES FOR MEMB     401, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.      12.85    0.00 1121       26.93    2.65 1221
            4.69    0.00 1114        2.77    1.06 1114       38.84 C     0.00 1121
 MIN.     -10.20    2.65 1221      -33.91    2.65 1121
           -5.76    2.65 1214       -1.54    2.65 1214        0.07 C     2.65 1216

402    0.00  MAX      9.68 1021      33.93 1121      -0.13 1322      2.70 1114
             MIN     -5.63 1321     -26.91 1221      -4.98 1114      0.15 1322
       0.15  MAX      9.67 1021      34.63 1121      -0.13 1322      2.00 1114
             MIN     -5.64 1321     -28.22 1221      -4.40 1114      0.13 1322
       0.30  MAX      9.66 1021      35.34 1121      -0.13 1322      1.39 1114
             MIN     -5.64 1321     -29.52 1221      -3.82 1114      0.11 1322
       0.45  MAX      9.65 1021      36.04 1121      -0.13 1322      0.86 1114
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             MIN     -5.65 1321     -30.83 1221      -3.23 1114      0.09 1322
       0.60  MAX      9.64 1021      36.75 1121      -0.13 1322      0.43 1114
             MIN     -5.66 1321     -32.13 1221      -2.65 1114      0.07 1322
       0.75  MAX      9.62 1021      37.46 1121      -0.13 1322      0.07 1114
             MIN     -5.67 1321     -33.43 1221      -2.07 1114      0.00 1222

 MAX/MIN FORCE VALUES FOR MEMB     402, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.       9.68    0.00 1021       37.46    0.75 1121
           -0.13    0.00 1322        2.70    0.00 1114       28.67 C     0.00 1121
 MIN.      -5.67    0.75 1321      -33.43    0.75 1221
           -4.98    0.00 1114        0.00    0.75 1222       10.64 T     0.75 1216

403    0.00  MAX     16.23 1013      37.46 1121      -0.13 1322      0.07 1114
             MIN     -2.79 1214     -33.43 1221      -2.07 1114      0.00 1222
       0.15  MAX     16.19 1013      35.89 1121      -0.13 1322      0.03 1322
             MIN     -2.82 1214     -34.12 1221      -1.56 1113     -0.23 1013
       0.30  MAX     16.15 1013      34.45 1321      -0.13 1322      0.01 1322
             MIN     -2.86 1214     -34.94 1021      -1.27 1113     -0.44 1113
       0.45  MAX     16.12 1013      33.42 1321      -0.07 1214      0.00 1322
             MIN     -2.89 1214     -36.15 1021      -1.23 1022     -0.60 1113
       0.60  MAX     16.08 1013      32.40 1321       0.51 1214     -0.02 1322
             MIN     -2.92 1214     -37.36 1021      -1.23 1022     -0.73 1022
       0.75  MAX     16.04 1013      31.38 1321       1.09 1214     -0.04 1322
             MIN     -2.95 1214     -38.56 1021      -1.23 1022     -0.92 1022
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STAAD SPACE


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 MAX/MIN FORCE VALUES FOR MEMB     403, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.      16.23    0.00 1013       37.46    0.00 1121
            1.09    0.75 1214        0.07    0.00 1114       26.80 C     0.00 1121
 MIN.      -2.95    0.75 1214      -38.56    0.75 1021
           -2.07    0.00 1114       -0.92    0.75 1022       12.54 T     0.75 1216

404    0.00  MAX     12.12 1321      31.39 1321       0.61 1112      0.18 1322
             MIN    -13.93 1021     -38.55 1021      -0.28 1322     -0.53 1112
       0.15  MAX     12.08 1321      29.58 1321       0.61 1112      0.14 1322
             MIN    -13.98 1021     -36.47 1021      -0.28 1322     -0.44 1112
       0.30  MAX     12.04 1321      27.79 1321       0.61 1112      0.09 1322
             MIN    -14.03 1021     -34.38 1021      -0.28 1322     -0.35 1112
       0.45  MAX     12.00 1321      26.00 1321       0.61 1112      0.05 1322
             MIN    -14.09 1021     -32.29 1021      -0.28 1322     -0.26 1112
       0.60  MAX     11.96 1321      24.21 1321       0.61 1112      0.01 1322
             MIN    -14.14 1021     -30.18 1021      -0.28 1322     -0.17 1112
       0.75  MAX     11.92 1321      22.43 1321       0.61 1112     -0.03 1322
             MIN    -14.20 1021     -28.07 1021      -0.28 1322     -0.10 1113

 MAX/MIN FORCE VALUES FOR MEMB     404, AMONGST ALL SECT LOCATIONS
             FY/    DIST   LD         MZ/    DIST   LD
             FZ     DIST   LD         MY     DIST   LD          FX         DIST   LD

 MAX.      12.12    0.00 1321       31.39    0.00 1321
            0.61    0.00 1112        0.18    0.00 1322        3.89 C     0.00 1121
 MIN.     -14.20    0.75 1021      -38.55    0.00 1021
           -0.28    0.75 1322       -0.53    0.00 1112        6.89 T     0.75 1216

405    0.00  MAX     12.40 1321      22.43 1321       0.61 1112     -0.03 1322
             MIN    -14.38 1021     -28.07 1021      -0.28 1322     -0.10 1113
       0.63  MAX     12.22 1321      14.63 1321       0.61 1112      0.32 1211
             MIN    -14.61 1021     -18.89 1021      -0.28 1322     -0.23 1122
       1.27  MAX     12.04 1321       6.95 1321       0.61 1112      0.69 1212
             MIN    -14.85 1021      -9.56 1021      -0.28 1322     -0.40 1122
       1.90  MAX     11.86 1321       0.07 1221       0.61 1112      1.07 1112
             MIN    -15.09 1021      -0.76 1121      -0.28 1322     -0.57 1322
       2.53  MAX     11.68 1321       9.56 1021       0.61 1112      1.46 1112
             MIN    -15.33 1021      -8.07 1321      -0.28 1322     -0.75 1322
       3.17  MAX     11.50 1321      19.35 1021       0.61 1112      1.84 1112
             MIN    -15.57 1021     -15.41 1321      -0.28 1322     -0.93 1322
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STAAD SPACE

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 | MAX/MIN FORCE VALUES FOR MEMB     405, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.     12.40  0.00 1321        22.43   0.00 1321
 |           0.61  0.00 1112         1.84   3.17 1112       3.17 C   0.00 1013
 | MIN.    -15.57  3.17 1021       -28.07   0.00 1021
 |          -0.28  3.17 1322        -0.93   3.17 1322       7.17 T   3.17 1216
 -----------------------------------------------------------------------------

 406     0.00  MAX     3.60 1121      5.56 1121      1.02 1322      4.31 1112
               MIN    -1.40 1221     -2.40 1221     -1.95 1112     -2.24 1322
         0.77  MAX     3.31 1121      2.90 1121      1.02 1322      2.81 1112
               MIN    -1.62 1221     -1.24 1221     -1.95 1112     -1.45 1322
         1.54  MAX     3.02 1121      0.77 1013      1.02 1322      1.30 1112
               MIN    -1.84 1221     -0.40 1214     -1.95 1112     -0.67 1322
         2.31  MAX     2.73 1121      1.63 1021      1.02 1322      0.12 1122
               MIN    -2.06 1221     -1.78 1321     -1.95 1112     -0.20 1212
         3.08  MAX     2.44 1121      3.27 1221      1.02 1322      0.91 1322
               MIN    -2.28 1221     -3.74 1121     -1.95 1112     -1.70 1112
         3.85  MAX     2.15 1321      5.11 1221      1.02 1322      1.69 1322
               MIN    -2.50 1021     -5.51 1121     -1.95 1112     -3.21 1112

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 | MAX/MIN FORCE VALUES FOR MEMB     406, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.      3.60  0.00 1121         5.56   0.00 1121
 |           1.02  0.00 1322         4.31   0.00 1112       2.32 C   0.00 1322
 | MIN.     -2.50  3.85 1021        -5.51   3.85 1121
 |          -1.95  3.85 1112        -3.21   3.85 1112       8.23 T   3.85 1116
 -----------------------------------------------------------------------------

 407     0.00  MAX     2.16 1321      5.11 1221      1.02 1322      1.69 1322
               MIN    -2.53 1021     -5.51 1121     -1.95 1112     -3.21 1112
         0.23  MAX     2.09 1321      5.69 1221      1.02 1322      1.93 1322
               MIN    -2.61 1021     -5.99 1121     -1.95 1112     -3.66 1112
         0.46  MAX     2.03 1321      6.30 1221      1.02 1322      2.16 1322
               MIN    -2.70 1021     -6.46 1121     -1.95 1112     -4.11 1112
         0.69  MAX     1.96 1321      6.91 1221      1.02 1322      2.40 1322
               MIN    -2.79 1021     -6.90 1121     -1.95 1112     -4.56 1112
         0.92  MAX     1.90 1321      7.54 1221      1.02 1322      2.63 1322
               MIN    -2.87 1021     -7.33 1121     -1.95 1112     -5.01 1112
         1.15  MAX     1.83 1321      8.19 1221      1.02 1322      2.87 1322
               MIN    -2.96 1021     -7.73 1121     -1.95 1112     -5.46 1112
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 | MAX/MIN FORCE VALUES FOR MEMB     407, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.      2.16  0.00 1321         8.19   1.15 1221
 |           1.02  0.00 1322         2.87   1.15 1322       2.31 C   1.15 1322
 | MIN.     -2.96  1.15 1021        -7.73   1.15 1121
 |          -1.95  1.15 1112        -5.46   1.15 1112       8.23 T   0.00 1116
 -----------------------------------------------------------------------------

 408     0.00  MAX     3.98 1121      9.96 1321      3.13 1112      3.47 1322
               MIN    -3.23 1221     -9.78 1021     -1.72 1322     -6.59 1112
         0.80  MAX     3.70 1321      6.91 1321      3.13 1112      2.09 1322
               MIN    -3.48 1021     -7.12 1021     -1.72 1322     -4.09 1112
         1.60  MAX     3.48 1321      4.03 1321      3.13 1112      0.72 1122
               MIN    -3.78 1021     -4.21 1021     -1.72 1322     -1.59 1212
         2.40  MAX     3.25 1321      1.43 1121      3.13 1112      0.91 1112
               MIN    -4.09 1021     -1.15 1221     -1.72 1322     -0.66 1322
         3.20  MAX     3.02 1321      2.33 1021      3.13 1112      3.41 1112
               MIN    -4.39 1021     -1.16 1321     -1.72 1322     -2.03 1322
         4.00  MAX     2.79 1321      5.96 1021      3.13 1112      5.92 1112
               MIN    -4.69 1021     -3.49 1321     -1.72 1322     -3.41 1322

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 | MAX/MIN FORCE VALUES FOR MEMB     408, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
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```
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.      3.98  0.00 1121         9.96   0.00 1321
 |           3.13  0.00 1112         5.92   4.00 1112       2.08 C   4.00 1322
 | MIN.     -4.69  4.00 1021        -9.78   0.00 1021
 |          -1.72  4.00 1322        -6.59   0.00 1112       8.20 T   0.00 1116
 -----------------------------------------------------------------------------

 409     0.00  MAX    23.30 1121     18.11 1121      1.78 1322      3.08 1112
               MIN   -19.41 1221    -15.51 1221     -1.95 1022     -2.73 1322
         0.37  MAX    23.16 1121      9.44 1121      1.78 1322      2.47 1112
               MIN   -19.51 1221     -8.25 1221     -1.95 1022     -2.06 1322
         0.75  MAX    23.02 1121      0.85 1321      1.78 1322      1.85 1112
               MIN   -19.62 1221     -0.98 1021     -1.95 1022     -1.40 1322
         1.12  MAX    22.88 1121      6.39 1221      1.78 1322      1.25 1212
               MIN   -19.72 1221     -7.75 1121     -1.95 1022     -0.73 1122
         1.49  MAX    22.74 1121     13.77 1221      1.78 1322      0.65 1211
               MIN   -19.83 1221    -16.26 1121     -1.95 1022     -0.08 1122
         1.87  MAX    22.60 1121     21.20 1221      1.78 1322      0.60 1322
               MIN   -19.94 1221    -24.72 1121     -1.95 1022     -0.73 1022
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 | MAX/MIN FORCE VALUES FOR MEMB     409, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.     23.30  0.00 1121        21.20   1.87 1221
 |           1.78  0.00 1322         3.08   0.00 1112       4.77 C   1.87 1021
 | MIN.    -19.94  1.87 1221       -24.72   1.87 1121
 |          -1.95  1.87 1022        -2.73   0.00 1322       7.59 T   0.00 1116
 -----------------------------------------------------------------------------

 410     0.00  MAX    18.38 1121     21.20 1221      5.10 1222      0.59 1322
               MIN   -15.87 1221    -24.72 1121     -5.26 1122     -0.72 1022
         0.15  MAX    18.33 1121     23.56 1221      5.10 1222      0.06 1222
               MIN   -15.91 1221    -27.46 1121     -5.26 1122     -0.22 1112
         0.30  MAX    18.28 1121     25.94 1221      5.10 1222      0.82 1222
               MIN   -15.95 1221    -30.19 1121     -5.26 1122     -1.00 1122
         0.45  MAX    18.22 1121     28.32 1221      5.10 1222      1.58 1222
               MIN   -15.99 1221    -32.91 1121     -5.26 1122     -1.79 1122
         0.60  MAX    18.17 1121     30.71 1221      5.10 1222      2.34 1222
               MIN   -16.03 1221    -35.62 1121     -5.26 1122     -2.57 1122
         0.75  MAX    18.11 1121     33.10 1221      5.10 1222      3.10 1222
               MIN   -16.07 1221    -38.32 1121     -5.26 1122     -3.36 1122

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 | MAX/MIN FORCE VALUES FOR MEMB     410, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.     18.38  0.00 1121        33.10   0.75 1221
 |           5.10  0.00 1222         3.10   0.75 1222      16.55 C   0.75 1021
 | MIN.    -16.07  0.75 1221       -38.32   0.75 1121
 |          -5.26  0.75 1122        -3.36   0.75 1122      17.37 T   0.00 1121
 -----------------------------------------------------------------------------

 411     0.00  MAX     1.62 1321     33.09 1221      2.20 1122      0.75 1222
               MIN   -12.15 1021    -38.33 1121     -0.90 1222     -1.66 1122
         0.15  MAX     1.59 1321     34.55 1221      2.20 1122      0.61 1222
               MIN   -12.19 1021    -38.22 1121     -0.90 1222     -1.33 1122
         0.30  MAX     1.56 1321     36.01 1221      2.20 1122      0.48 1222
               MIN   -12.22 1021    -38.09 1121     -0.90 1222     -1.00 1122
         0.45  MAX     1.53 1321     37.48 1221      2.20 1122      0.35 1222
               MIN   -12.26 1021    -37.96 1121     -0.90 1222     -0.67 1122
         0.60  MAX     1.51 1321     39.21 1021      2.20 1122      0.21 1222
               MIN   -12.30 1021    -38.09 1321     -0.90 1222     -0.34 1122
         0.75  MAX     1.48 1321     41.05 1021      2.20 1122      0.08 1022
               MIN   -12.34 1021    -38.31 1321     -0.90 1222     -0.02 1322
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 | MAX/MIN FORCE VALUES FOR MEMB     411, AMONGST ALL SECT LOCATIONS
 |           FY/   DIST   LD         MZ/    DIST   LD
 |           FZ    DIST   LD         MY     DIST   LD         FX      DIST   LD
 |
 | MAX.      1.62  0.00 1321        41.05   0.75 1021
 |           2.20  0.00 1122         0.75   0.00 1222      30.79 C   0.75 1021
```

```
| MIN.    -12.34    0.75 1021      -38.33    0.00 1121
|          -0.90    0.75 1222       -1.66    0.00 1122       18.54 T    0.00 1321
|-------------------------------------------------------------------------------

412    0.00  MAX     5.89 1221     41.05 1021      2.20 1122      0.08 1022
             MIN    -8.82 1121    -38.31 1321     -0.90 1222     -0.02 1322
       0.15  MAX     5.89 1221     40.26 1021      2.20 1122      0.36 1114
             MIN    -8.83 1121    -37.09 1321     -0.90 1222     -0.06 1222
       0.30  MAX     5.88 1221     39.48 1021      2.58 1114      0.71 1114
             MIN    -8.84 1121    -35.87 1321     -0.90 1222     -0.20 1222
       0.45  MAX     5.87 1221     38.70 1021      2.99 1114      1.13 1114
             MIN    -8.85 1121    -34.64 1321     -0.90 1222     -0.33 1222
       0.60  MAX     5.86 1221     37.92 1021      3.40 1114      1.60 1114
             MIN    -8.86 1121    -33.42 1321     -0.90 1222     -0.46 1222
       0.75  MAX     5.85 1221     37.15 1021      3.82 1114      2.14 1114
             MIN    -8.87 1121    -32.19 1321     -0.90 1222     -0.60 1222

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB     412, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.      5.89    0.00 1221       41.05    0.00 1021
|           3.82    0.75 1114        2.14    0.75 1114       33.03 C    0.75 1021
| MIN.     -8.87    0.75 1121      -38.31    0.00 1321
|          -0.90    0.75 1222       -0.60    0.75 1222       16.52 T    0.00 1321
|-------------------------------------------------------------------------------

413    0.00  MAX    12.19 1321     32.20 1321      4.02 1214      0.99 1116
             MIN   -13.97 1021    -37.13 1021     -0.38 1116     -0.99 1212
       0.53  MAX    12.18 1321     25.75 1321      2.55 1214      1.04 1114
             MIN   -13.99 1021    -29.72 1021     -0.38 1116      0.13 1211
       1.06  MAX    12.16 1321     19.30 1321      1.08 1214      1.95 1114
             MIN   -14.01 1021    -22.31 1021     -0.38 1116      0.22 1322
       1.59  MAX    12.15 1321     12.86 1321     -0.06 1211      2.07 1114
             MIN   -14.03 1021    -14.88 1021     -0.49 1114      0.15 1322
       2.12  MAX    12.14 1321      6.43 1321     -0.14 1322      1.43 1114
             MIN   -14.04 1021     -7.44 1021     -1.96 1114      0.07 1322
       2.65  MAX    12.12 1321      0.00 1322     -0.14 1322      0.00 1322
             MIN   -14.06 1021      0.00 1322     -3.43 1114      0.00 1322
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|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB     413, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.     12.19    0.00 1321       32.20    0.00 1321
|           4.02    0.00 1214        2.07    1.59 1114       42.58 C    2.65 1021
| MIN.    -14.06    2.65 1021      -37.13    0.00 1021
|          -3.43    2.65 1114       -0.99    0.00 1212        5.14 T    0.00 1321
|-------------------------------------------------------------------------------

1001   0.00  MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
             MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
       0.32  MAX     0.00 1322      0.00 1322      0.90 1116      0.14 1116
             MIN     0.00 1322      0.00 1322      0.58 1322      0.09 1322
       0.64  MAX     0.00 1322      0.00 1322      1.81 1116      0.58 1116
             MIN     0.00 1322      0.00 1322      1.16 1322      0.37 1322
       0.96  MAX     0.00 1322      0.00 1322      2.71 1116      1.30 1116
             MIN     0.00 1322      0.00 1322      1.74 1322      0.84 1322
       1.28  MAX     0.00 1322      0.00 1322      3.61 1116      2.31 1116
             MIN     0.00 1322      0.00 1322      2.32 1322      1.49 1322
       1.60  MAX     0.00 1322      0.00 1322      4.51 1116      3.61 1116
             MIN     0.00 1322      0.00 1322      2.90 1322      2.32 1322

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    1001, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.      0.00    0.00 1001        0.00    0.00 1001
|           4.51    1.60 1001        3.61    1.60 1001        0.00      0.00 1001
| MIN.      0.00    1.60 1322        0.00    1.60 1322
|           0.00    0.00 1322        0.00    0.00 1322        0.16 T    1.60 1116
|-------------------------------------------------------------------------------

1002   0.00  MAX     0.13 1212      0.76 1211     -3.69 1222      5.29 1122
```



```
             MIN    -0.04 1021     -0.62 1114     -6.91 1114      2.47 1222
       0.50  MAX     0.13 1212      0.70 1211     -2.78 1222      2.40 1122
             MIN    -0.04 1021     -0.62 1114     -5.50 1114      0.85 1222
       1.00  MAX     0.13 1212      0.64 1211     -1.87 1222      0.13 1322
             MIN    -0.04 1021     -0.61 1114     -4.09 1114     -0.52 1114
       1.50  MAX     0.13 1212      0.59 1211     -0.97 1222     -0.77 1211
             MIN    -0.04 1021     -0.60 1114     -2.75 1122     -2.21 1114
       2.00  MAX     0.13 1212      0.53 1211     -0.06 1222     -1.12 1211
             MIN    -0.04 1021     -0.60 1114     -1.55 1122     -3.20 1114
       2.50  MAX     0.13 1212      0.47 1111      0.92 1022     -0.96 1211
             MIN    -0.04 1021     -0.60 1214     -0.41 1322     -3.48 1114
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|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    1002, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.      0.13    0.00 1212        0.76    0.00 1211
|           0.92    2.50 1022        5.29    0.00 1122       27.12 C    0.00 1111
| MIN.     -0.04    2.50 1021       -0.62    0.00 1114
|          -6.91    0.00 1114       -3.48    2.50 1114        7.55 T    2.50 1322
|-------------------------------------------------------------------------------

1003   0.00  MAX     0.28 1111      0.47 1111      0.13 1322     -0.96 1211
             MIN    -0.06 1322     -0.60 1214     -1.02 1111     -3.48 1114
       0.50  MAX     0.28 1111      0.34 1211      1.27 1122     -1.14 1222
             MIN    -0.06 1322     -0.60 1114      0.09 1211     -3.20 1114
       1.00  MAX     0.28 1111      0.20 1211      2.68 1114     -0.74 1222
             MIN    -0.06 1322     -0.60 1114      1.10 1211     -2.21 1114
       1.50  MAX     0.28 1111      0.07 1211      4.09 1114      0.10 1222
             MIN    -0.06 1322     -0.61 1114      2.10 1211     -0.52 1114
       2.00  MAX     0.28 1111      0.02 1216      5.50 1114      2.19 1001
             MIN    -0.06 1322     -0.62 1114      3.06 1222      1.12 1212
       2.50  MAX     0.28 1111      0.05 1322      6.91 1114      5.20 1001
             MIN    -0.06 1322     -0.62 1114      3.96 1222      3.01 1212

|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    1003, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.      0.28    0.00 1111        0.47    0.00 1111
|           6.91    2.50 1114        5.20    2.50 1001       27.11 C    2.50 1111
| MIN.     -0.06    2.50 1322       -0.62    2.50 1114
|          -1.02    0.00 1111       -3.48    0.00 1114        7.55 T    0.00 1322
|-------------------------------------------------------------------------------

1004   0.00  MAX     0.00 1322      0.00 1322     -2.90 1322      3.61 1116
             MIN     0.00 1322      0.00 1322     -4.51 1116      2.32 1322
       0.32  MAX     0.00 1322      0.00 1322     -2.32 1322      2.31 1116
             MIN     0.00 1322      0.00 1322     -3.61 1116      1.49 1322
       0.64  MAX     0.00 1322      0.00 1322     -1.74 1322      1.30 1116
             MIN     0.00 1322      0.00 1322     -2.71 1116      0.84 1322
       0.96  MAX     0.00 1322      0.00 1322     -1.16 1322      0.58 1116
             MIN     0.00 1322      0.00 1322     -1.81 1116      0.37 1322
       1.28  MAX     0.00 1322      0.00 1322     -0.58 1322      0.14 1116
             MIN     0.00 1322      0.00 1322     -0.90 1116      0.09 1322
       1.60  MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
             MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
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|-------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    1004, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD          MZ/    DIST   LD
|            FZ     DIST   LD          MY     DIST   LD         FX        DIST   LD
|
| MAX.      0.00    0.00 1001        0.00    0.00 1001
|           0.00    1.60 1001        3.61    0.00 1001        0.00      1.60 1001
| MIN.      0.00    1.60 1322        0.00    1.60 1322
|          -4.51    0.00 1116        0.00    1.60 1322        0.16 T    0.00 1116
|-------------------------------------------------------------------------------

2001   0.00  MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
             MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
       0.32  MAX     1.00 1212      0.31 1013      0.00 1322      0.00 1322
             MIN    -1.93 1013     -0.16 1212      0.00 1322      0.00 1322
       0.64  MAX     2.01 1212      1.23 1013      0.00 1322      0.00 1322
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               MIN    -3.85 1013      -0.64 1212       0.00 1322       0.00 1322
        0.96   MAX     3.01 1212       2.78 1013       0.00 1322       0.00 1322
               MIN    -5.78 1013      -1.45 1212       0.00 1322       0.00 1322
        1.28   MAX     4.01 1212       4.93 1013       0.00 1322       0.00 1322
               MIN    -7.70 1013      -2.57 1212       0.00 1322       0.00 1322
        1.60   MAX     5.02 1212       7.71 1013       0.00 1322       0.00 1322
               MIN    -9.63 1013      -4.02 1212       0.00 1322       0.00 1322


 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    2001, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.       5.02   1.60 1212        7.71   1.60 1013
 |            0.00   0.00 1001        0.00   0.00 1001        0.19 C     1.60 1212
 | MIN.      -9.63   1.60 1013       -4.02   1.60 1212
 |            0.00   1.60 1322        0.00   1.60 1322        0.36 T     1.60 1013
 |--------------------------------------------------------------------------------


 2002   0.00   MAX    14.59 1013       8.30 1013       0.01 1321       0.41 1013
               MIN    -6.99 1212      -4.10 1212      -0.13 1112       0.05 1216
        0.50   MAX    11.58 1013       1.75 1013       0.01 1321       0.39 1013
               MIN    -5.42 1212      -0.99 1212      -0.13 1112       0.04 1216
        1.00   MAX     8.58 1013       1.67 1214       0.01 1321       0.38 1013
               MIN    -3.85 1212      -3.30 1013      -0.13 1112       0.04 1216
        1.50   MAX     5.57 1013       2.92 1214       0.01 1321       0.36 1013
               MIN    -2.28 1212      -6.83 1013      -0.13 1112       0.03 1216
        2.00   MAX     2.56 1013       3.71 1214       0.01 1321       0.35 1013
               MIN    -1.34 1216      -8.87 1013      -0.13 1112       0.01 1212
        2.50   MAX     0.85 1212       4.06 1214       0.01 1321       0.33 1013
               MIN    -1.62 1116      -9.39 1013      -0.13 1112      -0.05 1212
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 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    2002, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.      14.59   0.00 1013        8.30   0.00 1013
 |            0.01   0.00 1321        0.41   0.00 1013       16.32 C     0.00 1021
 | MIN.      -6.99   0.00 1212       -9.39   2.50 1013
 |           -0.13   2.50 1112       -0.05   2.50 1212        2.09 T     2.50 1321
 |--------------------------------------------------------------------------------


 2003   0.00   MAX     1.75 1112       4.06 1214       0.05 1022       0.33 1013
               MIN    -1.07 1216      -9.39 1013      -0.07 1212      -0.05 1212
        0.50   MAX     1.47 1212       3.71 1214       0.05 1022       0.35 1013
               MIN    -2.56 1013      -8.87 1013      -0.07 1212      -0.09 1212
        1.00   MAX     2.06 1216       2.92 1214       0.05 1022       0.36 1013
               MIN    -5.57 1013      -6.83 1013      -0.07 1212      -0.12 1212
        1.50   MAX     3.63 1216       1.67 1214       0.05 1022       0.38 1013
               MIN    -8.58 1013      -3.30 1013      -0.07 1212      -0.16 1212
        2.00   MAX     5.20 1216       1.75 1013       0.05 1022       0.39 1013
               MIN   -11.58 1013      -0.92 1216      -0.07 1212      -0.20 1212
        2.50   MAX     6.77 1216       8.30 1013       0.05 1022       0.41 1013
               MIN   -14.59 1013      -3.91 1216      -0.07 1212      -0.24 1212


 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    2003, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.       6.77   2.50 1216        8.30   2.50 1013
 |            0.05   0.00 1022        0.41   2.50 1013       16.32 C     2.50 1021
 | MIN.     -14.59   2.50 1013       -9.39   0.00 1013
 |           -0.07   2.50 1212       -0.24   2.50 1212        2.09 T     0.00 1321
 |--------------------------------------------------------------------------------


 2004   0.00   MAX     9.63 1013       7.71 1013       0.00 1322       0.00 1322
               MIN    -5.02 1216      -4.02 1216       0.00 1322       0.00 1322
        0.32   MAX     7.70 1013       4.93 1013       0.00 1322       0.00 1322
               MIN    -4.01 1216      -2.57 1216       0.00 1322       0.00 1322
        0.64   MAX     5.78 1013       2.78 1013       0.00 1322       0.00 1322
               MIN    -3.01 1216      -1.45 1216       0.00 1322       0.00 1322
        0.96   MAX     3.85 1013       1.23 1013       0.00 1322       0.00 1322
               MIN    -2.01 1216      -0.64 1216       0.00 1322       0.00 1322
        1.28   MAX     1.93 1013       0.31 1013       0.00 1322       0.00 1322
               MIN    -1.00 1216      -0.16 1216       0.00 1322       0.00 1322
        1.60   MAX     0.00 1322       0.00 1322       0.00 1322       0.00 1322
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               MIN     0.00 1322       0.00 1322       0.00 1322       0.00 1322
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 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    2004, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.       9.63   0.00 1013        7.71   0.00 1013
 |            0.00   0.00 1001        0.00   0.00 1001        0.19 C     0.00 1216
 | MIN.      -5.02   0.00 1216       -4.02   0.00 1216
 |            0.00   1.60 1322        0.00   1.60 1322        0.36 T     0.00 1013
 |--------------------------------------------------------------------------------


 3001   0.00   MAX     0.00 1322       0.00 1322       0.00 1322       0.00 1322
               MIN     0.00 1322       0.00 1322       0.00 1322       0.00 1322
        0.32   MAX     1.77 1212       0.58 1013       0.00 1322       0.00 1322
               MIN    -3.61 1013      -0.28 1212       0.00 1322       0.00 1322
        0.64   MAX     3.54 1212       2.31 1013       0.00 1322       0.00 1322
               MIN    -7.22 1013      -1.13 1212       0.00 1322       0.00 1322
        0.96   MAX     5.31 1212       5.20 1013       0.00 1322       0.00 1322
               MIN   -10.83 1013      -2.55 1212       0.00 1322       0.00 1322
        1.28   MAX     7.08 1212       9.25 1013       0.00 1322       0.00 1322
               MIN   -14.45 1013      -4.54 1212       0.00 1322       0.00 1322
        1.60   MAX     8.86 1212      14.46 1013       0.00 1322       0.00 1322
               MIN   -18.06 1013      -7.09 1212       0.00 1322       0.00 1322


 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    3001, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.       8.86   1.60 1212       14.46   1.60 1013
 |            0.00   0.00 1001        0.00   0.00 1001        0.33 C     1.60 1212
 | MIN.     -18.06   1.60 1013       -7.09   1.60 1212
 |            0.00   1.60 1322        0.00   1.60 1322        0.68 T     1.60 1013
 |--------------------------------------------------------------------------------


 3002   0.00   MAX    28.29 1013      17.62 1013       0.99 1212       1.27 1322
               MIN    -9.56 1212     -11.68 1214      -0.51 1322      -2.49 1112
        0.50   MAX    22.64 1013       8.87 1011       0.99 1212       1.01 1322
               MIN    -6.80 1212      -8.43 1214      -0.51 1322      -2.00 1112
        1.00   MAX    17.00 1013       6.67 1211       0.99 1212       0.75 1322
               MIN    -4.14 1214      -6.27 1114      -0.51 1322      -1.50 1112
        1.50   MAX    11.36 1013       6.18 1211       0.99 1212       0.49 1322
               MIN    -2.57 1214     -12.13 1013      -0.51 1322      -1.00 1112
        2.00   MAX     5.72 1013       5.37 1212       0.99 1212       0.24 1322
               MIN    -2.27 1216     -16.41 1013      -0.51 1322      -0.51 1112
        2.50   MAX     4.27 1212       4.03 1211       0.99 1212       0.00 1222
               MIN    -2.68 1116     -17.85 1013      -0.51 1322      -0.03 1122
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 |--------------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    3002, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/   DIST   LD
 |            FZ     DIST   LD          MY    DIST   LD          FX         DIST   LD
 |
 | MAX.      28.29   0.00 1013       17.62   0.00 1013
 |            0.99   0.00 1112        1.27   0.00 1322       26.48 C     2.50 1111
 | MIN.      -9.56   0.00 1212      -17.85   2.50 1013
 |           -0.51   2.50 1322       -2.49   0.00 1112       15.54 T     0.00 1214
 |--------------------------------------------------------------------------------


 3003   0.00   MAX     5.50 1112       4.03 1211       0.99 1212       0.00 1222
               MIN    -2.54 1216     -17.85 1013      -0.51 1322      -0.03 1122
        0.50   MAX     5.09 1212       3.83 1216       0.99 1212       0.49 1212
               MIN    -5.72 1013     -16.41 1013      -0.51 1322      -0.28 1122
        1.00   MAX     4.71 1212       3.02 1216       0.99 1212       0.99 1212
               MIN   -11.36 1013     -12.13 1013      -0.51 1322      -0.54 1122
        1.50   MAX     5.76 1216       0.83 1216       0.99 1212       1.49 1212
               MIN   -17.00 1013      -6.27 1114      -0.51 1322      -0.80 1122
        2.00   MAX     8.53 1216       6.58 1001       0.99 1212       1.98 1212
               MIN   -22.64 1013      -8.43 1214      -0.51 1322      -1.05 1122
        2.50   MAX    11.30 1216      17.62 1013       0.99 1212       2.48 1212
               MIN   -28.29 1013     -11.68 1214      -0.51 1322      -1.31 1122


 |--------------------------------------------------------------------------------
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| MAX/MIN FORCE VALUES FOR MEMB    3003, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.      11.30   2.50 1216        17.62   2.50 1013
|            0.99   0.00 1112         2.48   2.50 1212       26.56 C     2.50 1111
| MIN.     -28.29   2.50 1013       -17.85   0.00 1013
|           -0.51   2.50 1322        -1.31   2.50 1122       15.54 T     2.50 1214
|------------------------------------------------------------------------------------

3004     0.00  MAX      18.06 1013      14.46 1013       0.00 1322       0.00 1322
               MIN      -8.86 1216      -7.09 1216       0.00 1322       0.00 1322
         0.32  MAX      14.45 1013       9.25 1013       0.00 1322       0.00 1322
               MIN      -7.08 1216      -4.54 1216       0.00 1322       0.00 1322
         0.64  MAX      10.83 1013       5.20 1013       0.00 1322       0.00 1322
               MIN      -5.31 1216      -2.55 1216       0.00 1322       0.00 1322
         0.96  MAX       7.22 1013       2.31 1013       0.00 1322       0.00 1322
               MIN      -3.54 1216      -1.13 1216       0.00 1322       0.00 1322
         1.28  MAX       3.61 1013       0.58 1013       0.00 1322       0.00 1322
               MIN      -1.77 1216      -0.28 1216       0.00 1322       0.00 1322
         1.60  MAX       0.00 1322       0.00 1322       0.00 1322       0.00 1322
               MIN       0.00 1322       0.00 1322       0.00 1322       0.00 1322
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|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    3004, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.      18.06   0.00 1013        14.46   0.00 1013
|            0.00   0.00 1001         0.00   0.00 1001        0.33 C     0.00 1216
| MIN.      -8.86   0.00 1216        -7.09   0.00 1216
|            0.00   1.60 1322         0.00   1.60 1322        0.68 T     0.00 1013
|------------------------------------------------------------------------------------

4001     0.00  MAX       0.00 1322       0.00 1322       0.00 1322       0.00 1322
               MIN       0.00 1322       0.00 1322       0.00 1322       0.00 1322
         0.32  MAX       1.77 1212       0.58 1013       0.00 1322       0.00 1322
               MIN      -3.61 1013      -0.28 1212       0.00 1322       0.00 1322
         0.64  MAX       3.54 1212       2.31 1013       0.00 1322       0.00 1322
               MIN      -7.22 1013      -1.13 1212       0.00 1322       0.00 1322
         0.96  MAX       5.31 1212       5.20 1013       0.00 1322       0.00 1322
               MIN     -10.83 1013      -2.55 1212       0.00 1322       0.00 1322
         1.28  MAX       7.08 1212       9.25 1013       0.00 1322       0.00 1322
               MIN     -14.45 1013      -4.54 1212       0.00 1322       0.00 1322
         1.60  MAX       8.86 1212      14.46 1013       0.00 1322       0.00 1322
               MIN     -18.06 1013      -7.09 1212       0.00 1322       0.00 1322

|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    4001, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.       8.86   1.60 1212        14.46   1.60 1013
|            0.00   0.00 1001         0.00   0.00 1001        0.34 C     1.60 1212
| MIN.     -18.06   1.60 1013        -7.09   1.60 1212
|            0.00   1.60 1322         0.00   1.60 1322        0.69 T     1.60 1013
|------------------------------------------------------------------------------------

4002     0.00  MAX      31.43 1013      30.15 1122       0.24 1322       1.13 1212
               MIN     -22.23 1212     -35.40 1212      -0.45 1212      -0.61 1122
         0.50  MAX      25.79 1013      19.08 1122       0.24 1322       0.90 1212
               MIN     -19.46 1212     -24.97 1212      -0.45 1212      -0.49 1122
         1.00  MAX      20.15 1013      10.27 1322       0.24 1322       0.68 1212
               MIN     -16.69 1212     -16.23 1112      -0.45 1212      -0.37 1122
         1.50  MAX      14.57 1122       4.42 1322       0.24 1322       0.45 1212
               MIN     -13.93 1212     -14.25 1013      -0.45 1212      -0.25 1122
         2.00  MAX      11.54 1122       2.03 1216       0.24 1322       0.22 1211
               MIN     -11.16 1212     -20.09 1013      -0.45 1212      -0.13 1122
         2.50  MAX       8.52 1122       3.15 1216       0.24 1322       0.00 1211
               MIN      -9.64 1111     -23.11 1013      -0.45 1212      -0.01 1014
0    STAAD SPACE                                                  -- PAGE NO.    37


|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    4002, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
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| MAX.      31.43   0.00 1013        30.15   0.00 1122
|            0.24   0.00 1122         1.13   0.00 1212        8.58 C     0.00 1011
| MIN.     -22.23   0.00 1212       -35.40   0.00 1212
|           -0.45   2.50 1212        -0.61   0.00 1122       18.76 T     0.00 1214
|------------------------------------------------------------------------------------

4003     0.00  MAX       7.33 1322       3.15 1216       0.24 1322       0.00 1211
               MIN     -10.66 1011     -23.11 1013      -0.45 1212      -0.01 1014
         0.50  MAX       5.56 1322       7.71 1211       0.24 1322       0.12 1322
               MIN     -14.00 1011     -20.09 1013      -0.45 1212      -0.23 1112
         1.00  MAX       3.80 1322      13.13 1211       0.24 1322       0.24 1322
               MIN     -17.35 1011     -14.25 1013      -0.45 1212      -0.46 1112
         1.50  MAX       5.86 1216      19.17 1211       0.24 1322       0.36 1322
               MIN     -20.69 1011     -12.41 1122      -0.45 1212      -0.68 1112
         2.00  MAX       8.63 1216      26.87 1111       0.24 1322       0.48 1322
               MIN     -25.79 1013     -12.25 1322      -0.45 1212      -0.91 1112
         2.50  MAX      11.40 1216      39.00 1011       0.24 1322       0.60 1322
               MIN     -31.43 1013     -11.95 1322      -0.45 1212      -1.13 1112


|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    4003, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.      11.40   2.50 1216        39.00   2.50 1011
|            0.24   0.00 1122         0.60   2.50 1322        9.51 C     2.50 1011
| MIN.     -31.43   2.50 1013       -23.11   0.00 1013
|           -0.45   2.50 1212        -1.13   2.50 1112       18.76 T     2.50 1214
|------------------------------------------------------------------------------------

4004     0.00  MAX      18.06 1013      14.46 1013       0.00 1322       0.00 1322
               MIN      -8.86 1216      -7.09 1216       0.00 1322       0.00 1322
         0.32  MAX      14.45 1013       9.25 1013       0.00 1322       0.00 1322
               MIN      -7.08 1216      -4.54 1216       0.00 1322       0.00 1322
         0.64  MAX      10.83 1013       5.20 1013       0.00 1322       0.00 1322
               MIN      -5.31 1216      -2.55 1216       0.00 1322       0.00 1322
         0.96  MAX       7.22 1013       2.31 1013       0.00 1322       0.00 1322
               MIN      -3.54 1216      -1.13 1216       0.00 1322       0.00 1322
         1.28  MAX       3.61 1013       0.58 1013       0.00 1322       0.00 1322
               MIN      -1.77 1216      -0.28 1216       0.00 1322       0.00 1322
         1.60  MAX       0.00 1322       0.00 1322       0.00 1322       0.00 1322
               MIN       0.00 1322       0.00 1322       0.00 1322       0.00 1322
0    STAAD SPACE                                                  -- PAGE NO.    38


|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    4004, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.      18.06   0.00 1013        14.46   0.00 1013
|            0.00   0.00 1001         0.00   0.00 1001        0.34 C     0.00 1216
| MIN.      -8.86   0.00 1216        -7.09   0.00 1216
|            0.00   1.60 1322         0.00   1.60 1322        0.69 T     0.00 1013
|------------------------------------------------------------------------------------

5001     0.00  MAX       0.00 1322       0.00 1322       0.00 1322       0.00 1322
               MIN       0.00 1322       0.00 1322       0.00 1322       0.00 1322
         0.32  MAX       1.25 1212       0.42 1013       0.00 1322       0.00 1322
               MIN      -2.64 1013      -0.20 1212       0.00 1322       0.00 1322
         0.64  MAX       2.51 1212       1.69 1013       0.00 1322       0.00 1322
               MIN      -5.29 1013      -0.80 1212       0.00 1322       0.00 1322
         0.96  MAX       3.76 1212       3.81 1013       0.00 1322       0.00 1322
               MIN      -7.93 1013      -1.81 1212       0.00 1322       0.00 1322
         1.28  MAX       5.01 1212       6.77 1013       0.00 1322       0.00 1322
               MIN     -10.58 1013      -3.21 1212       0.00 1322       0.00 1322
         1.60  MAX       6.27 1212      10.58 1013       0.00 1322       0.00 1322
               MIN     -13.22 1013      -5.02 1212       0.00 1322       0.00 1322


|------------------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB    5001, AMONGST ALL SECT LOCATIONS
|            FY/    DIST   LD         MZ/    DIST   LD
|            FZ     DIST   LD         MY     DIST   LD         FX          DIST   LD
|
| MAX.       6.27   1.60 1212        10.58   1.60 1013
|            0.00   0.00 1001         0.00   0.00 1001        0.24 C     1.60 1212
| MIN.     -13.22   1.60 1013        -5.02   1.60 1212
|            0.00   1.60 1322         0.00   1.60 1322        0.50 T     1.60 1013
|------------------------------------------------------------------------------------
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5002     0.00  MAX    20.71 1013    12.90 1001     0.26 1322     2.88 1112
               MIN    -6.86 1212    -8.37 1214    -1.14 1212    -0.63 1322
         0.50  MAX    16.58 1013     6.43 1011     0.26 1322     2.31 1112
               MIN    -4.90 1212    -6.13 1214    -1.14 1212    -0.49 1322
         1.00  MAX    12.45 1013     4.81 1211     0.26 1322     1.74 1112
               MIN    -2.94 1212    -4.67 1114    -1.14 1212    -0.36 1322
         1.50  MAX     8.31 1013     4.49 1211     0.26 1322     1.16 1112
               MIN    -1.74 1214    -8.88 1013    -1.14 1212    -0.23 1322
         2.00  MAX     4.18 1013     3.91 1211     0.26 1322     0.59 1111
               MIN    -1.59 1216   -12.01 1013    -1.14 1212    -0.10 1322
         2.50  MAX     2.93 1212     3.06 1211     0.26 1322     0.03 1122
               MIN    -1.92 1116   -13.07 1013    -1.14 1212     0.01 1222
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 MAX/MIN FORCE VALUES FOR MEMB     5002, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.     20.71   0.00 1013       12.90   0.00 1001
           0.26   0.00 1122        2.88   0.00 1112       22.29 C    0.00 1111
 MIN.     -6.86   0.00 1212      -13.07   2.50 1013
          -1.14   2.50 1212       -0.63   0.00 1322       11.70 T    0.00 1214
 ---------------------------------------------------------------------------

5003     0.00  MAX     4.05 1112     3.06 1211     0.26 1322     0.03 1122
               MIN    -1.84 1216   -13.07 1013    -1.14 1212     0.01 1222
         0.50  MAX     3.72 1212     2.66 1216     0.26 1322     0.16 1122
               MIN    -4.18 1013   -12.01 1013    -1.14 1212    -0.56 1212
         1.00  MAX     3.40 1212     2.11 1216     0.26 1322     0.30 1122
               MIN    -8.31 1013    -8.88 1013    -1.14 1212    -1.13 1212
         1.50  MAX     4.03 1216     0.58 1216     0.26 1322     0.43 1122
               MIN   -12.45 1013    -4.67 1114    -1.14 1212    -1.71 1212
         2.00  MAX     5.99 1216     4.85 1001     0.26 1322     0.56 1122
               MIN   -16.58 1013    -6.13 1214    -1.14 1212    -2.28 1212
         2.50  MAX     7.95 1216    12.90 1001     0.26 1322     0.69 1122
               MIN   -20.71 1013    -8.37 1214    -1.14 1212    -2.85 1212

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 MAX/MIN FORCE VALUES FOR MEMB     5003, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.      7.95   2.50 1216       12.90   2.50 1001
           0.26   0.00 1122        0.69   2.50 1122       22.36 C    2.50 1111
 MIN.    -20.71   2.50 1013      -13.07   0.00 1013
          -1.14   2.50 1212       -2.85   2.50 1212       11.70 T    2.50 1214
 ---------------------------------------------------------------------------

5004     0.00  MAX    13.22 1013    10.58 1013     0.00 1322     0.00 1322
               MIN    -6.27 1216    -5.02 1216     0.00 1322     0.00 1322
         0.32  MAX    10.58 1013     6.77 1013     0.00 1322     0.00 1322
               MIN    -5.01 1216    -3.21 1216     0.00 1322     0.00 1322
         0.64  MAX     7.93 1013     3.81 1013     0.00 1322     0.00 1322
               MIN    -3.76 1216    -1.81 1216     0.00 1322     0.00 1322
         0.96  MAX     5.29 1013     1.69 1013     0.00 1322     0.00 1322
               MIN    -2.51 1216    -0.80 1216     0.00 1322     0.00 1322
         1.28  MAX     2.64 1013     0.42 1013     0.00 1322     0.00 1322
               MIN    -1.25 1216    -0.20 1216     0.00 1322     0.00 1322
         1.60  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
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 MAX/MIN FORCE VALUES FOR MEMB     5004, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.     13.22   0.00 1013       10.58   0.00 1013
           0.00   0.00 1001        0.00   0.00 1001        0.24 C    0.00 1216
 MIN.     -6.27   0.00 1216       -5.02   0.00 1216
           0.00   1.60 1322        0.00   1.60 1322        0.50 T    0.00 1013
 ---------------------------------------------------------------------------

6001     0.00  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         0.32  MAX     0.52 1212     0.21 1013     0.00 1322     0.00 1322
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               MIN    -1.28 1013    -0.08 1212     0.00 1322     0.00 1322
         0.64  MAX     1.05 1212     0.82 1013     0.00 1322     0.00 1322
               MIN    -2.57 1013    -0.34 1212     0.00 1322     0.00 1322
         0.96  MAX     1.57 1212     1.85 1013     0.00 1322     0.00 1322
               MIN    -3.85 1013    -0.75 1212     0.00 1322     0.00 1322
         1.28  MAX     2.09 1212     3.29 1013     0.00 1322     0.00 1322
               MIN    -5.14 1013    -1.34 1212     0.00 1322     0.00 1322
         1.60  MAX     2.62 1212     5.14 1013     0.00 1322     0.00 1322
               MIN    -6.42 1013    -2.09 1212     0.00 1322     0.00 1322

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 MAX/MIN FORCE VALUES FOR MEMB     6001, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.      2.62   1.60 1212        5.14   1.60 1013
           0.00   0.00 1001        0.00   0.00 1001        0.10 C    1.60 1212
 MIN.     -6.42   1.60 1013       -2.09   1.60 1212
           0.00   1.60 1322        0.00   1.60 1322        0.24 T    1.60 1013
 ---------------------------------------------------------------------------

6002     0.00  MAX     9.64 1013     5.63 1013     0.93 1122     2.22 1222
               MIN    -3.74 1212    -2.07 1212    -0.90 1222    -2.58 1122
         0.50  MAX     7.63 1013     1.31 1013     0.93 1122     1.76 1222
               MIN    -2.92 1212    -0.41 1212    -0.90 1222    -2.11 1122
         1.00  MAX     5.62 1013     1.03 1214     0.93 1122     1.31 1222
               MIN    -2.11 1212    -2.01 1013    -0.90 1222    -1.65 1122
         1.50  MAX     3.62 1013     1.70 1212     0.93 1122     0.86 1222
               MIN    -1.29 1212    -4.32 1013    -0.90 1222    -1.18 1122
         2.00  MAX     1.61 1013     2.14 1212     0.93 1122     0.41 1222
               MIN    -0.77 1216    -5.63 1013    -0.90 1222    -0.71 1122
         2.50  MAX     0.35 1212     2.32 1214     0.93 1122     0.02 1212
               MIN    -1.09 1116    -5.93 1013    -0.90 1222    -0.25 1122
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 MAX/MIN FORCE VALUES FOR MEMB     6002, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.      9.64   0.00 1013        5.63   0.00 1013
           0.93   0.00 1122        2.22   0.00 1222       13.96 C    0.00 1022
 MIN.     -3.74   0.00 1212       -5.93   2.50 1013
          -0.90   2.50 1222       -2.58   0.00 1122        0.79 T    2.50 1322
 ---------------------------------------------------------------------------

6003     0.00  MAX     1.24 1112     2.32 1214     0.93 1322     0.02 1212
               MIN    -0.57 1216    -5.93 1013    -0.96 1022    -0.25 1122
         0.50  MAX     0.92 1212     2.09 1214     0.93 1322     0.27 1322
               MIN    -1.61 1013    -5.63 1013    -0.96 1022    -0.58 1022
         1.00  MAX     1.07 1214     1.66 1214     0.93 1322     0.74 1322
               MIN    -3.62 1013    -4.32 1013    -0.96 1022    -1.06 1022
         1.50  MAX     1.89 1216     1.03 1214     0.93 1322     1.21 1322
               MIN    -5.62 1013    -2.01 1013    -0.96 1022    -1.54 1022
         2.00  MAX     2.70 1216     1.31 1013     0.93 1322     1.67 1322
               MIN    -7.63 1013    -0.36 1216    -0.96 1022    -2.02 1022
         2.50  MAX     3.52 1216     5.63 1013     0.93 1322     2.14 1322
               MIN    -9.64 1013    -1.92 1216    -0.96 1022    -2.50 1022

 ---------------------------------------------------------------------------
 MAX/MIN FORCE VALUES FOR MEMB     6003, AMONGST ALL SECT LOCATIONS
           FY/    DIST   LD          MZ/    DIST   LD
           FZ     DIST   LD          MY     DIST   LD          FX        DIST   LD

 MAX.      3.52   2.50 1216        5.63   2.50 1013
           0.93   0.00 1322        2.14   2.50 1322       13.96 C    2.50 1022
 MIN.     -9.64   2.50 1013       -5.93   0.00 1013
          -0.96   2.50 1022       -2.50   2.50 1022        0.80 T    0.00 1322
 ---------------------------------------------------------------------------

6004     0.00  MAX     6.42 1013     5.14 1013     0.00 1322     0.00 1322
               MIN    -2.62 1216    -2.09 1216     0.00 1322     0.00 1322
         0.32  MAX     5.14 1013     3.29 1013     0.00 1322     0.00 1322
               MIN    -2.09 1216    -1.34 1216     0.00 1322     0.00 1322
         0.64  MAX     3.85 1013     1.85 1013     0.00 1322     0.00 1322
               MIN    -1.57 1216    -0.75 1216     0.00 1322     0.00 1322
         0.96  MAX     2.57 1013     0.82 1013     0.00 1322     0.00 1322
               MIN    -1.05 1216    -0.34 1216     0.00 1322     0.00 1322
         1.28  MAX     1.28 1013     0.21 1013     0.00 1322     0.00 1322
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          MIN    -0.52 1216     -0.08 1216      0.00 1322      0.00 1322
     1.60 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
          MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
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 | MAX/MIN FORCE VALUES FOR MEMB   6004, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       6.42    0.00 1013       5.14    0.00 1013
 |            0.00    0.00 1001       0.00    0.00 1001       0.10 C     0.00 1216
 | MIN.      -2.62    0.00 1216      -2.09    0.00 1216
 |            0.00    1.60 1322       0.00    1.60 1322       0.24 T     0.00 1013
 -----------------------------------------------------------------------

7001     0.00 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         0.32 MAX     0.00 1322      0.00 1322      0.92 1116      0.15 1116
              MIN     0.00 1322      0.00 1322      0.59 1322      0.09 1322
         0.64 MAX     0.00 1322      0.00 1322      1.84 1116      0.59 1116
              MIN     0.00 1322      0.00 1322      1.19 1322      0.38 1322
         0.96 MAX     0.00 1322      0.00 1322      2.77 1116      1.33 1116
              MIN     0.00 1322      0.00 1322      1.78 1322      0.85 1322
         1.28 MAX     0.00 1322      0.00 1322      3.69 1116      2.36 1116
              MIN     0.00 1322      0.00 1322      2.37 1322      1.52 1322
         1.60 MAX     0.00 1322      0.00 1322      4.61 1116      3.69 1116
              MIN     0.00 1322      0.00 1322      2.96 1322      2.37 1322

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 | MAX/MIN FORCE VALUES FOR MEMB   7001, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       0.00    0.00 1001       0.00    0.00 1001
 |            4.61    1.60 1001       3.69    1.60 1001       0.00       0.00 1001
 | MIN.       0.00    1.60 1322       0.00    1.60 1322
 |            0.00    0.00 1322       0.00    0.00 1322       0.17 T     1.60 1116
 -----------------------------------------------------------------------

7002     0.00 MAX     0.14 1022      0.46 1114     -4.14 1222      5.28 1001
              MIN    -0.11 1212     -0.59 1211     -6.99 1114      3.29 1211
         0.50 MAX     0.14 1022      0.45 1114     -3.21 1222      2.22 1001
              MIN    -0.11 1212     -0.55 1211     -5.55 1114      1.30 1212
         1.00 MAX     0.14 1022      0.45 1114     -2.29 1222      0.11 1222
              MIN    -0.11 1212     -0.50 1211     -4.11 1114     -0.41 1114
         1.50 MAX     0.14 1022      0.44 1114     -1.31 1211     -0.80 1222
              MIN    -0.11 1212     -0.45 1211     -2.67 1114     -2.11 1114
         2.00 MAX     0.14 1022      0.43 1214     -0.28 1211     -1.25 1222
              MIN    -0.11 1212     -0.40 1111     -1.23 1114     -3.08 1114
         2.50 MAX     0.14 1022      0.42 1214      0.85 1111     -1.24 1222
              MIN    -0.11 1212     -0.36 1111     -0.03 1322     -3.34 1114
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 | MAX/MIN FORCE VALUES FOR MEMB   7002, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       0.14    0.00 1022       0.46    0.00 1114
 |            0.85    2.50 1111       5.28    0.00 1001      23.17 C     0.00 1111
 | MIN.      -0.11    2.50 1212      -0.59    0.00 1211
 |           -6.99    0.00 1114      -3.34    2.50 1114       5.04 T     2.50 1322
 -----------------------------------------------------------------------

7003     0.00 MAX     0.02 1222      0.42 1214      0.33 1322     -1.24 1222
              MIN    -0.23 1111     -0.36 1111     -0.86 1022     -3.34 1114
         0.50 MAX     0.02 1222      0.43 1214      1.49 1122     -1.36 1211
              MIN    -0.23 1111     -0.24 1111      0.14 1222     -3.08 1114
         1.00 MAX     0.02 1222      0.44 1114      2.72 1122     -0.94 1211
              MIN    -0.23 1111     -0.13 1222      1.07 1222     -2.11 1114
         1.50 MAX     0.02 1222      0.45 1114      4.11 1114      0.14 1322
              MIN    -0.23 1111     -0.14 1222      1.99 1222     -0.41 1114
         2.00 MAX     0.02 1222      0.45 1114      5.55 1114      2.40 1122
              MIN    -0.23 1111     -0.15 1222      2.92 1222      0.90 1222
         2.50 MAX     0.02 1222      0.46 1114      6.99 1114      5.31 1122
              MIN    -0.23 1111     -0.15 1222      3.85 1222      2.59 1222
```



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 -----------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB   7003, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       0.02    0.00 1222       0.46    2.50 1114
 |            6.99    2.50 1114       5.31    2.50 1122      23.17 C     2.50 1111
 | MIN.      -0.23    2.50 1111      -0.36    0.00 1111
 |           -0.86    0.00 1022      -3.34    0.00 1114       5.03 T     0.00 1322
 -----------------------------------------------------------------------

7004     0.00 MAX     0.00 1322      0.00 1322     -2.96 1322      3.69 1116
              MIN     0.00 1322      0.00 1322     -4.61 1116      2.37 1322
         0.32 MAX     0.00 1322      0.00 1322     -2.37 1322      2.36 1116
              MIN     0.00 1322      0.00 1322     -3.69 1116      1.52 1322
         0.64 MAX     0.00 1322      0.00 1322     -1.78 1322      1.33 1116
              MIN     0.00 1322      0.00 1322     -2.77 1116      0.85 1322
         0.96 MAX     0.00 1322      0.00 1322     -1.19 1322      0.59 1116
              MIN     0.00 1322      0.00 1322     -1.84 1116      0.38 1322
         1.28 MAX     0.00 1322      0.00 1322     -0.59 1322      0.15 1116
              MIN     0.00 1322      0.00 1322     -0.92 1116      0.09 1322
         1.60 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
0   STAAD SPACE                                        -- PAGE NO.   44


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 | MAX/MIN FORCE VALUES FOR MEMB   7004, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       0.00    0.00 1001       0.00    0.00 1001
 |            0.00    1.60 1001       3.69    0.00 1001       0.00       1.60 1001
 | MIN.       0.00    1.60 1322       0.00    1.60 1322
 |           -4.61    0.00 1116       0.00    1.60 1322       0.17 T     0.00 1116
 -----------------------------------------------------------------------

8001     0.00 MAX     0.29 1116      0.00 1322      0.00 1322      0.00 1322
              MIN     0.18 1322      0.00 1322      0.00 1322      0.00 1322
         1.13 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         2.26 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         3.40 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         4.53 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         5.66 MAX    -0.18 1322      0.00 1322      0.00 1322      0.00 1322
              MIN    -0.29 1116      0.00 1322      0.00 1322      0.00 1322

 -----------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB   8001, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
 |              FZ     DIST   LD        MY     DIST   LD        FX         DIST   LD
 |
 | MAX.       0.29    0.00 1001       0.00    0.00 1001
 |            0.00    0.00 1001       0.00    0.00 1001      13.48 C     0.00 1322
 | MIN.      -0.29    5.66 1116       0.00    5.66 1322
 |            0.00    5.66 1322       0.00    5.66 1322      26.86 T     5.66 1112
 -----------------------------------------------------------------------

8002     0.00 MAX     0.29 1116      0.00 1322      0.00 1322      0.00 1322
              MIN     0.18 1322      0.00 1322      0.00 1322      0.00 1322
         1.13 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         2.26 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         3.40 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         4.53 MAX     0.00 1322      0.00 1322      0.00 1322      0.00 1322
              MIN     0.00 1322      0.00 1322      0.00 1322      0.00 1322
         5.66 MAX    -0.18 1322      0.00 1322      0.00 1322      0.00 1322
              MIN    -0.29 1116      0.00 1322      0.00 1322      0.00 1322
0   STAAD SPACE                                        -- PAGE NO.   45


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 | MAX/MIN FORCE VALUES FOR MEMB   8002, AMONGST ALL SECT LOCATIONS
 |              FY/    DIST   LD        MZ/    DIST   LD
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```
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.29   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       2.54 T   0.00 1321
| MIN.   -0.29   5.66 1116        0.00    5.66 1322
|         0.00   5.66 1322        0.00    5.66 1322       7.81 T   5.66 1114
|-----------------------------------------------------------------------------

8003     0.00  MAX     0.29 1116     0.00 1322     0.00 1322     0.00 1322
               MIN     0.18 1322     0.00 1322     0.00 1322     0.00 1322
         1.04  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.08  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.12  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         4.16  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.20  MAX    -0.18 1322     0.00 1322     0.00 1322     0.00 1322
               MIN    -0.29 1116     0.00 1322     0.00 1322     0.00 1322

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8003, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.29   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       9.29 C   0.00 1322
| MIN.   -0.29   5.20 1116        0.00    5.20 1322
|         0.00   5.20 1322        0.00    5.20 1322      19.14 T   5.20 1022
|-----------------------------------------------------------------------------

8004     0.00  MAX     0.29 1116     0.00 1322     0.00 1322     0.00 1322
               MIN     0.18 1322     0.00 1322     0.00 1322     0.00 1322
         1.04  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.08  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.12  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         4.16  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.20  MAX    -0.18 1322     0.00 1322     0.00 1322     0.00 1322
               MIN    -0.29 1116     0.00 1322     0.00 1322     0.00 1322
0    STAAD SPACE                                             -- PAGE NO.   46

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8004, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.29   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       0.99 T   0.00 1321
| MIN.   -0.29   5.20 1116        0.00    5.20 1322
|         0.00   5.20 1322        0.00    5.20 1322       8.85 T   5.20 1021
|-----------------------------------------------------------------------------

8005     0.00  MAX     0.36 1116     0.00 1322     0.00 1322     0.00 1322
               MIN     0.23 1322     0.00 1322     0.00 1322     0.00 1322
         1.27  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.53  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.80  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.07  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         6.33  MAX    -0.23 1322     0.00 1322     0.00 1322     0.00 1322
               MIN    -0.36 1116     0.00 1322     0.00 1322     0.00 1322

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8005, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.36   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       4.79 C   0.00 1321
| MIN.   -0.36   6.33 1116        0.00    6.33 1322
|         0.00   6.33 1322        0.00    6.33 1322       8.66 T   6.33 1021
```



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|-----------------------------------------------------------------------------

8006     0.00  MAX     0.36 1116     0.00 1322     0.00 1322     0.00 1322
               MIN     0.23 1322     0.00 1322     0.00 1322     0.00 1322
         1.27  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.53  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.80  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.07  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         6.33  MAX    -0.23 1322     0.00 1322     0.00 1322     0.00 1322
               MIN    -0.36 1116     0.00 1322     0.00 1322     0.00 1322
0    STAAD SPACE                                             -- PAGE NO.   47

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8006, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.36   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       4.79 C   6.33 1321
| MIN.   -0.36   6.33 1116        0.00    6.33 1322
|         0.00   6.33 1322        0.00    6.33 1322       8.66 T   0.00 1021
|-----------------------------------------------------------------------------

8007     0.00  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         1.31  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.63  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.94  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.25  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         6.57  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8007, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.00   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001       0.00     0.00 1001
| MIN.    0.00   6.57 1322        0.00    6.57 1322
|         0.00   6.57 1322        0.00    6.57 1322       3.80 T   6.57 1214
|-----------------------------------------------------------------------------

8008     0.00  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         1.31  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         2.63  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         3.94  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         5.25  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
         6.57  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
               MIN     0.00 1322     0.00 1322     0.00 1322     0.00 1322
0    STAAD SPACE                                             -- PAGE NO.   48

|-----------------------------------------------------------------------------
| MAX/MIN FORCE VALUES FOR MEMB   8008, AMONGST ALL SECT LOCATIONS
|         FY/    DIST   LD         MZ/    DIST   LD
|         FZ     DIST   LD         MY     DIST   LD         FX      DIST   LD
|
| MAX.    0.00   0.00 1001        0.00    0.00 1001
|         0.00   0.00 1001        0.00    0.00 1001      20.33 C   0.00 1122
| MIN.    0.00   6.57 1322        0.00    6.57 1322
|         0.00   6.57 1322        0.00    6.57 1322      35.00 T   6.57 1212
|-----------------------------------------------------------------------------

8009     0.00  MAX     0.00 1322     0.00 1322     0.00 1322     0.00 1322
```

```
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      1.31 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.62 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.93 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.23 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      6.54 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8009, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.00   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001        1.28 C     0.00 1121
 | MIN.       0.00   6.54 1322        0.00    6.54 1322
 |            0.00   6.54 1322        0.00    6.54 1322        2.82 T     6.54 1214
 |-----------------------------------------------------------------------------


8010  0.00 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      1.31 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.62 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.93 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.23 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      6.54 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
0    STAAD SPACE                                                -- PAGE NO.    49


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8010, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.00   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001       26.97 C     0.00 1122
 | MIN.       0.00   6.54 1322        0.00    6.54 1322
 |            0.00   6.54 1322        0.00    6.54 1322       29.27 T     6.54 1212
 |-----------------------------------------------------------------------------


8011  0.00 MAX      0.32 1116      0.00 1322      0.00 1322      0.00 1322
           MIN      0.21 1322      0.00 1322      0.00 1322      0.00 1322
      1.13 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.25 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.38 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      4.51 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.63 MAX     -0.21 1322      0.00 1322      0.00 1322      0.00 1322
           MIN     -0.32 1116      0.00 1322      0.00 1322      0.00 1322


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8011, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.32   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001        0.78 T     5.63 1221
 | MIN.      -0.32   5.63 1116        0.00    5.63 1322
 |            0.00   5.63 1322        0.00    5.63 1322        4.02 T     0.00 1013
 |-----------------------------------------------------------------------------


8012  0.00 MAX      0.32 1116      0.00 1322      0.00 1322      0.00 1322
           MIN      0.21 1322      0.00 1322      0.00 1322      0.00 1322
      1.13 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.25 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.38 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
```



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           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      4.51 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.63 MAX     -0.21 1322      0.00 1322      0.00 1322      0.00 1322
           MIN     -0.32 1116      0.00 1322      0.00 1322      0.00 1322
0    STAAD SPACE                                                -- PAGE NO.    50


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8012, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.32   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001        0.78 T     0.00 1221
 | MIN.      -0.32   5.63 1116        0.00    5.63 1322
 |            0.00   5.63 1322        0.00    5.63 1322        8.82 T     5.63 1112
 |-----------------------------------------------------------------------------


8013  0.00 MAX      0.29 1116      0.00 1322      0.00 1322      0.00 1322
           MIN      0.18 1322      0.00 1322      0.00 1322      0.00 1322
      1.04 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.08 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.12 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      4.16 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.20 MAX     -0.18 1322      0.00 1322      0.00 1322      0.00 1322
           MIN     -0.29 1116      0.00 1322      0.00 1322      0.00 1322


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8013, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.29   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001        4.17 T     5.20 1221
 | MIN.      -0.29   5.20 1116        0.00    5.20 1322
 |            0.00   5.20 1322        0.00    5.20 1322        7.90 T     0.00 1114
 |-----------------------------------------------------------------------------


8014  0.00 MAX      0.29 1116      0.00 1322      0.00 1322      0.00 1322
           MIN      0.18 1322      0.00 1322      0.00 1322      0.00 1322
      1.04 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.08 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.12 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      4.16 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.20 MAX     -0.18 1322      0.00 1322      0.00 1322      0.00 1322
           MIN     -0.29 1116      0.00 1322      0.00 1322      0.00 1322
0    STAAD SPACE                                                -- PAGE NO.    51


 |-----------------------------------------------------------------------------
 | MAX/MIN FORCE VALUES FOR MEMB    8014, AMONGST ALL SECT LOCATIONS
 |            FY/    DIST   LD          MZ/    DIST   LD
 |            FZ     DIST   LD          MY     DIST   LD          FX          DIST   LD
 |
 | MAX.       0.29   0.00 1001        0.00    0.00 1001
 |            0.00   0.00 1001        0.00    0.00 1001        9.11 C     5.20 1322
 | MIN.      -0.29   5.20 1116        0.00    5.20 1322
 |            0.00   5.20 1322        0.00    5.20 1322       19.02 T     0.00 1022
 |-----------------------------------------------------------------------------


8015  0.00 MAX      0.29 1116      0.00 1322      0.00 1322      0.00 1322
           MIN      0.18 1322      0.00 1322      0.00 1322      0.00 1322
      1.13 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      2.26 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      3.40 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      4.53 MAX      0.00 1322      0.00 1322      0.00 1322      0.00 1322
           MIN      0.00 1322      0.00 1322      0.00 1322      0.00 1322
      5.66 MAX     -0.18 1322      0.00 1322      0.00 1322      0.00 1322
```

MIN	-0.29 1116	0.00 1322	0.00 1322	0.00 1322

MAX/MIN FORCE VALUES FOR MEMB 8015, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST DIST	LD LD	MZ/ MY	DIST DIST	LD LD	FX	DIST	LD
MAX.	0.29	0.00	1001	0.00	0.00	1001			
	0.00	0.00	1001	0.00	0.00	1001	2.91 T	5.66	1221
MIN.	-0.29	5.66	1116	0.00	5.66	1322			
	0.00	5.66	1322	0.00	5.66	1322	7.07 T	0.00	1114

8016	0.00	MAX	0.29 1116	0.00 1322	0.00 1322	0.00 1322
		MIN	0.18 1322	0.00 1322	0.00 1322	0.00 1322
	1.13	MAX	0.00 1322	0.00 1322	0.00 1322	0.00 1322
		MIN	0.00 1322	0.00 1322	0.00 1322	0.00 1322
	2.26	MAX	0.00 1322	0.00 1322	0.00 1322	0.00 1322
		MIN	0.00 1322	0.00 1322	0.00 1322	0.00 1322
	3.40	MAX	0.00 1322	0.00 1322	0.00 1322	0.00 1322
		MIN	0.00 1322	0.00 1322	0.00 1322	0.00 1322
	4.53	MAX	0.00 1322	0.00 1322	0.00 1322	0.00 1322
		MIN	0.00 1322	0.00 1322	0.00 1322	0.00 1322
	5.66	MAX	-0.18 1322	0.00 1322	0.00 1322	0.00 1322
		MIN	-0.29 1116	0.00 1322	0.00 1322	0.00 1322

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MAX/MIN FORCE VALUES FOR MEMB 8016, AMONGST ALL SECT LOCATIONS

	FY/ FZ	DIST DIST	LD LD	MZ/ MY	DIST DIST	LD LD	FX	DIST	LD
MAX.	0.29	0.00	1001	0.00	0.00	1001			
	0.00	0.00	1001	0.00	0.00	1001	8.29 C	0.00	1322
MIN.	-0.29	5.66	1116	0.00	5.66	1322			
	0.00	5.66	1322	0.00	5.66	1322	21.97 T	5.66	1112

********** END OF FORCE ENVELOPE FROM INTERNAL STORAGE **********

295. LOAD LIST 2001 2002 2011 TO 2016
296. PRINT JOINT DISPLACEMENTS ALL
JOINT DISPLACE ALL
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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
102	2001	-0.0178	-0.1096	-0.0083	-0.0010	-0.0001	-0.0002
	2002	-0.0112	-0.1036	-0.0086	-0.0009	-0.0001	-0.0002
	2011	0.0491	-0.1940	0.0595	-0.0015	-0.0004	-0.0001
	2012	0.0394	-0.1255	0.0495	-0.0011	-0.0002	0.0000
	2013	-0.0601	-0.0669	-0.0126	-0.0007	0.0001	-0.0003
	2014	-0.0441	0.0242	-0.0188	-0.0002	0.0002	-0.0001
	2015	0.2227	-0.0795	-0.0090	-0.0009	-0.0001	-0.0009
	2016	-0.0682	-0.0978	-0.0095	-0.0009	-0.0001	-0.0002
104	2001	-0.0744	-0.0011	-0.0033	0.0002	0.0006	-0.0003
	2002	-0.0706	-0.0052	-0.0023	0.0002	0.0005	-0.0003
	2011	-0.0542	0.0133	0.0805	0.0002	0.0004	-0.0001
	2012	-0.0597	0.0161	0.0775	0.0001	0.0004	0.0000
	2013	-0.0895	0.0061	-0.0052	0.0003	0.0007	-0.0005
	2014	-0.0697	-0.0232	0.0021	-0.0001	0.0005	-0.0001
	2015	0.2550	-0.0415	-0.0032	0.0000	0.0005	-0.0006
	2016	-0.0899	-0.0660	-0.0028	-0.0002	0.0004	-0.0004
106	2001	0.0284	-0.0038	-0.0047	0.0000	0.0000	0.0001
	2002	0.0257	-0.0033	-0.0038	0.0000	0.0000	0.0001
	2011	0.2723	-0.1024	0.1205	-0.0006	-0.0016	0.0000
	2012	0.2664	-0.0166	0.1100	-0.0001	-0.0016	0.0000
	2013	0.0335	0.0701	-0.0135	0.0005	0.0000	0.0002
	2014	0.0103	0.1470	-0.0142	0.0009	0.0000	0.0000
	2015	0.3452	-0.0226	-0.0041	-0.0001	0.0000	0.0001
	2016	-0.0172	-0.0326	-0.0032	-0.0002	0.0000	0.0001
108	2001	0.0237	0.0320	-0.0088	0.0002	0.0000	0.0000
	2002	0.0204	0.0154	-0.0060	0.0001	0.0000	0.0000

Waste Stock Yard Rev1 - Jebel Ali

	2011	-0.1068	0.1994	1.8884	0.0012	0.0007	0.0000
	2012	-0.1128	0.2843	1.8898	0.0018	0.0007	0.0000
	2013	0.0312	0.1052	-0.0182	0.0007	0.0000	0.0000
	2014	0.0109	0.1793	-0.0186	0.0011	0.0000	0.0000
	2015	0.3446	0.0118	-0.0222	0.0001	0.0000	-0.0001
	2016	-0.0148	0.0005	-0.0291	0.0000	0.0000	0.0000
109	2001	0.0115	-0.0034	-0.0036	0.0000	0.0001	-0.0001
	2002	0.0081	-0.0029	-0.0029	0.0000	0.0001	-0.0001
	2011	-0.2975	-0.0757	0.1000	-0.0005	0.0019	0.0000
	2012	-0.3037	-0.0123	0.0921	-0.0001	0.0019	0.0000
	2013	0.0207	0.0511	-0.0100	0.0003	0.0000	-0.0001
	2014	0.0016	0.1084	-0.0107	0.0007	0.0000	0.0000
	2015	0.3356	-0.0164	-0.0032	-0.0001	0.0000	-0.0001
	2016	-0.0229	-0.0244	-0.0027	-0.0001	0.0000	-0.0001
111	2001	0.0945	-0.0024	-0.0015	0.0001	-0.0004	0.0001
	2002	0.0876	-0.0057	-0.0009	0.0001	-0.0004	0.0001
	2011	0.1119	0.0027	0.0618	0.0001	-0.0006	0.0001
	2012	0.1082	0.0026	0.0595	0.0001	-0.0007	0.0000
	2013	0.1137	0.0025	-0.0025	0.0002	-0.0005	0.0002
	2014	0.0734	-0.0189	0.0018	0.0000	-0.0004	0.0001
	2015	0.4055	-0.0315	-0.0010	0.0000	-0.0004	0.0000
	2016	0.0421	-0.0440	-0.0005	-0.0001	-0.0003	0.0002

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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
113	2001	0.0333	-0.1070	-0.0090	-0.0010	0.0001	0.0001
	2002	0.0261	-0.1027	-0.0092	-0.0010	0.0001	0.0000
	2011	-0.0200	-0.1518	0.0412	-0.0013	0.0002	0.0000
	2012	-0.0202	-0.1035	0.0344	-0.0010	0.0002	0.0000
	2013	0.0633	-0.0759	-0.0120	-0.0008	0.0000	0.0001
	2014	0.0340	-0.0134	-0.0162	-0.0004	-0.0001	0.0000
	2015	0.3104	-0.1254	-0.0094	-0.0009	0.0001	-0.0005
	2016	0.0224	-0.0950	-0.0096	-0.0009	0.0001	0.0002
201	2001	0.0000	0.0000	0.0000	-0.0003	-0.0002	0.0003
	2002	0.0000	0.0000	0.0000	-0.0003	-0.0002	0.0002
	2011	0.0000	0.0000	0.0000	0.0017	-0.0002	0.0001
	2012	0.0000	0.0000	0.0000	0.0009	-0.0002	0.0000
	2013	0.0000	0.0000	0.0000	-0.0011	-0.0001	0.0004
	2014	0.0000	0.0000	0.0000	-0.0019	-0.0001	0.0001
	2015	0.0000	0.0000	0.0000	-0.0003	-0.0002	-0.0007
	2016	0.0000	0.0000	0.0000	-0.0004	-0.0002	0.0006
202	2001	-0.0311	-0.0105	-0.0119	0.0005	-0.0001	-0.0002
	2002	-0.0258	-0.0092	-0.0121	0.0005	-0.0001	-0.0002
	2011	-0.0085	-0.0059	0.0525	0.0000	-0.0004	-0.0001
	2012	0.0044	-0.0030	0.0450	0.0004	-0.0002	0.0000
	2013	-0.0429	-0.0134	-0.0144	0.0008	0.0001	-0.0003
	2014	-0.0093	-0.0056	-0.0176	0.0013	0.0002	-0.0001
	2015	0.2149	0.0124	-0.0123	0.0006	-0.0001	-0.0009
	2016	-0.0806	-0.0128	-0.0126	0.0006	-0.0001	-0.0002
203	2001	-0.0074	-0.0162	0.0148	0.0005	0.0009	-0.0004
	2002	-0.0059	-0.0140	0.0178	0.0005	0.0011	-0.0003
	2011	0.0019	-0.0085	0.0842	0.0007	0.0008	-0.0002
	2012	0.0065	-0.0036	0.0905	0.0009	0.0012	0.0000
	2013	-0.0107	-0.0212	0.0138	0.0004	0.0009	-0.0005
	2014	-0.0017	-0.0075	0.0377	0.0009	0.0019	-0.0001
	2015	0.2846	-0.0014	0.0197	0.0006	0.0012	-0.0009
	2016	-0.0547	-0.0183	0.0239	0.0007	0.0014	-0.0004
204	2001	0.0184	-0.0403	-0.0016	0.0004	0.0006	-0.0003
	2002	0.0157	-0.0342	-0.0010	0.0003	0.0005	-0.0003
	2011	0.0130	-0.0190	0.0819	0.0003	0.0004	-0.0001
	2012	0.0083	-0.0053	0.0785	0.0001	0.0004	0.0000
	2013	0.0245	-0.0543	-0.0026	0.0006	0.0007	-0.0005
	2014	0.0065	-0.0152	0.0019	0.0000	0.0005	-0.0001
	2015	0.3320	-0.0441	-0.0029	0.0001	0.0005	-0.0006
	2016	-0.0278	-0.0425	-0.0035	0.0000	0.0004	-0.0004
205	2001	0.0229	-0.0550	-0.0303	-0.0015	-0.0001	-0.0001
	2002	0.0196	-0.0467	-0.0292	-0.0014	-0.0001	-0.0001
	2011	0.0144	-0.0245	0.0522	0.0001	0.0005	0.0000
	2012	0.0080	-0.0056	0.0450	0.0000	0.0005	0.0000
	2013	0.0305	-0.0738	-0.0354	-0.0016	-0.0001	-0.0001
	2014	0.0080	-0.0201	-0.0297	-0.0015	-0.0001	0.0000
	2015	0.3410	-0.0720	-0.0299	-0.0014	-0.0001	-0.0002
	2016	-0.0221	-0.0593	-0.0291	-0.0014	-0.0001	-0.0001

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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
206	2001	0.0211	-0.0123	-0.0044	0.0004	0.0000	0.0001
	2002	0.0181	-0.0101	-0.0035	0.0003	0.0000	0.0001
	2011	0.0122	-0.0067	0.1169	-0.0005	-0.0016	0.0000
	2012	0.0063	-0.0022	0.1095	-0.0001	-0.0016	0.0000
	2013	0.0277	-0.0171	-0.0102	0.0010	0.0000	0.0002
	2014	0.0071	-0.0046	-0.0085	0.0010	0.0000	0.0000
	2015	0.3396	-0.0108	-0.0045	0.0002	0.0000	0.0001
	2016	-0.0223	-0.0085	-0.0041	0.0000	0.0000	0.0001
207	2001	0.0213	-0.0126	-0.0063	0.0005	0.0000	0.0000
	2002	0.0183	-0.0105	-0.0044	0.0003	0.0000	0.0000
	2011	0.0108	-0.0061	1.7439	0.0072	-0.0031	0.0000
	2012	0.0049	-0.0030	1.7474	0.0076	-0.0031	0.0000
	2013	0.0278	-0.0159	-0.0126	0.0011	0.0000	0.0000
	2014	0.0074	-0.0070	-0.0109	0.0011	0.0000	0.0000
	2015	0.3412	-0.0056	-0.0195	0.0002	0.0000	0.0000
	2016	-0.0197	-0.0089	-0.0265	0.0001	0.0001	0.0000
208	2001	0.0213	-0.0121	-0.0071	0.0006	0.0000	0.0000
	2002	0.0184	-0.0095	-0.0051	0.0004	0.0000	0.0000
	2011	0.0104	-0.0035	1.8961	0.0014	0.0007	0.0000
	2012	0.0045	0.0007	1.9006	0.0018	0.0007	0.0000
	2013	0.0279	-0.0165	-0.0135	0.0012	0.0000	0.0000
	2014	0.0075	-0.0043	-0.0116	0.0012	0.0000	0.0000
	2015	0.3417	-0.0099	-0.0213	0.0004	0.0000	-0.0001
	2016	-0.0189	-0.0078	-0.0288	0.0003	0.0000	0.0000
209	2001	0.0216	-0.0094	-0.0033	0.0003	0.0001	-0.0001
	2002	0.0187	-0.0077	-0.0027	0.0002	0.0001	-0.0001
	2011	0.0090	-0.0052	0.0974	-0.0003	0.0019	0.0000
	2012	0.0031	-0.0020	0.0917	-0.0001	0.0019	0.0000
	2013	0.0281	-0.0129	-0.0076	0.0007	0.0000	-0.0001
	2014	0.0078	-0.0037	-0.0065	0.0008	0.0000	0.0000
	2015	0.3437	-0.0079	-0.0035	0.0001	0.0000	-0.0001
	2016	-0.0162	-0.0070	-0.0033	0.0000	0.0000	-0.0001
210	2001	0.0209	-0.0284	-0.0234	-0.0014	0.0001	0.0000
	2002	0.0181	-0.0245	-0.0229	-0.0014	0.0001	0.0000
	2011	0.0081	-0.0152	0.0064	-0.0022	-0.0002	0.0000
	2012	0.0023	-0.0071	0.0008	-0.0023	-0.0002	0.0000
	2013	0.0271	-0.0365	-0.0264	-0.0014	0.0001	0.0000
	2014	0.0074	-0.0139	-0.0242	-0.0014	0.0001	0.0000
	2015	0.3438	-0.0312	-0.0228	-0.0013	0.0001	-0.0001
	2016	-0.0161	-0.0363	-0.0222	-0.0013	0.0002	0.0000
211	2001	0.0220	-0.0251	-0.0005	0.0003	-0.0004	0.0001
	2002	0.0190	-0.0218	-0.0001	0.0002	-0.0004	0.0001
	2011	0.0081	-0.0143	0.0627	0.0002	-0.0006	0.0001
	2012	0.0020	-0.0075	0.0601	0.0001	-0.0007	0.0000
	2013	0.0284	-0.0322	-0.0009	0.0004	-0.0005	0.0002
	2014	0.0078	-0.0127	0.0017	0.0000	-0.0004	0.0001
	2015	0.3436	-0.0329	-0.0008	0.0001	-0.0004	0.0000
	2016	-0.0136	-0.0298	-0.0009	0.0000	-0.0003	0.0002

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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
212	2001	0.0346	-0.0130	0.0175	0.0005	-0.0010	0.0002
	2002	0.0297	-0.0115	0.0195	0.0006	-0.0011	0.0002
	2011	0.0145	-0.0081	0.0804	0.0010	-0.0014	0.0001
	2012	0.0044	-0.0050	0.0847	0.0011	-0.0017	0.0000
	2013	0.0451	-0.0162	0.0177	0.0005	-0.0011	0.0002
	2014	0.0132	-0.0075	0.0329	0.0008	-0.0017	0.0001
	2015	0.3449	-0.0311	0.0209	0.0006	-0.0012	-0.0001
	2016	0.0061	-0.0123	0.0235	0.0007	-0.0013	0.0003
213	2001	0.0445	-0.0103	-0.0125	0.0005	0.0001	0.0000
	2002	0.0380	-0.0092	-0.0126	0.0006	0.0001	0.0000
	2011	0.0196	-0.0066	0.0358	0.0002	0.0002	0.0000
	2012	0.0064	-0.0044	0.0307	0.0005	0.0002	0.0000
	2013	0.0580	-0.0125	-0.0142	0.0008	0.0000	0.0001
	2014	0.0176	-0.0063	-0.0164	0.0011	-0.0001	0.0000
	2015	0.3233	-0.0343	-0.0127	0.0006	0.0001	-0.0005
	2016	0.0299	-0.0075	-0.0128	0.0006	0.0001	0.0002
214	2001	0.0000	0.0000	0.0000	-0.0003	0.0001	-0.0003
	2002	0.0000	0.0000	0.0000	-0.0003	0.0002	-0.0002
	2011	0.0000	0.0000	0.0000	0.0011	0.0002	-0.0001
	2012	0.0000	0.0000	0.0000	0.0005	0.0002	0.0000
	2013	0.0000	0.0000	0.0000	-0.0009	0.0001	-0.0003

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	2014	0.0000	0.0000	0.0000	-0.0015	0.0001	-0.0001
	2015	0.0000	0.0000	0.0000	-0.0003	0.0001	-0.0016
	2016	0.0000	0.0000	0.0000	-0.0004	0.0001	-0.0003
302	2001	-0.0529	-0.4010	0.0000	0.0000	0.0000	0.0025
	2002	-0.0496	-0.4037	0.0000	0.0000	0.0000	0.0027
	2011	-0.0875	-0.2762	0.0562	0.0001	0.0000	0.0067
	2012	-0.0445	-0.3440	0.0524	0.0001	-0.0001	0.0035
	2013	-0.0232	-0.4501	0.0000	0.0000	0.0000	-0.0014
	2014	0.0340	-0.5145	0.0000	0.0000	0.0000	-0.0036
	2015	0.2020	-0.3799	-0.0005	0.0000	0.0000	0.0022
	2016	-0.1001	-0.4094	-0.0007	0.0000	0.0000	0.0032
304	2001	0.0801	-0.1396	0.0000	0.0000	0.0000	0.0017
	2002	0.0763	-0.1116	0.0000	0.0000	0.0000	0.0010
	2011	0.0374	-0.0787	0.0809	0.0000	-0.0001	0.0011
	2012	0.0393	-0.0305	0.0764	0.0001	-0.0001	-0.0005
	2013	0.0949	-0.2013	0.0000	0.0000	0.0000	0.0033
	2014	0.0725	-0.0188	0.0000	0.0000	0.0000	-0.0019
	2015	0.3905	-0.0908	-0.0033	0.0000	0.0000	-0.0001
	2016	0.0235	-0.0486	-0.0054	-0.0001	0.0000	-0.0012
306	2001	0.0138	-0.1540	0.0000	0.0000	0.0000	0.0003
	2002	0.0107	-0.1228	0.0000	0.0000	0.0000	0.0003
	2011	0.0083	-0.0325	0.1118	0.0002	0.0008	0.0002
	2012	0.0042	-0.0473	0.1073	0.0002	0.0008	0.0001
	2013	0.0217	-0.2797	0.0000	0.0000	0.0000	0.0003
	2014	0.0040	-0.1594	0.0000	0.0000	0.0000	0.0001
	2015	0.3314	-0.0967	-0.0022	0.0000	0.0000	0.0003
	2016	-0.0321	-0.0492	-0.0035	-0.0001	0.0000	0.0003

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JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
308	2001	0.0191	-0.2120	0.0000	0.0000	0.0000	0.0001
	2002	0.0165	-0.1519	0.0000	0.0000	0.0000	0.0001
	2011	0.0121	-0.0980	1.8976	-0.0005	-0.0003	0.0001
	2012	0.0061	-0.1113	1.9048	-0.0005	-0.0003	0.0001
	2013	0.0247	-0.3372	0.0000	0.0000	0.0000	0.0001
	2014	0.0043	-0.2154	0.0000	0.0000	0.0000	0.0001
	2015	0.3390	-0.1542	-0.0162	0.0000	0.0000	0.0000
	2016	-0.0238	-0.1073	-0.0253	0.0000	0.0000	0.0001
309	2001	0.0313	-0.1134	0.0000	0.0000	0.0000	-0.0004
	2002	0.0287	-0.0910	0.0000	0.0000	0.0000	-0.0004
	2011	0.0218	-0.0231	0.0930	0.0001	-0.0009	-0.0003
	2012	0.0149	-0.0350	0.0895	0.0001	-0.0009	-0.0002
	2013	0.0354	-0.2051	0.0000	0.0000	0.0000	-0.0003
	2014	0.0137	-0.1196	0.0000	0.0000	0.0000	-0.0002
	2015	0.3526	-0.0716	-0.0017	0.0000	0.0000	-0.0004
	2016	-0.0090	-0.0382	-0.0027	0.0000	0.0000	-0.0004
311	2001	-0.0280	-0.0899	0.0000	0.0000	0.0000	-0.0009
	2002	-0.0307	-0.0721	0.0000	0.0000	0.0000	-0.0004
	2011	-0.0099	-0.0532	0.0610	0.0000	0.0002	-0.0006
	2012	-0.0212	-0.0252	0.0578	0.0000	0.0002	0.0004
	2013	-0.0266	-0.1248	0.0000	0.0000	0.0000	-0.0018
	2014	-0.0478	-0.0196	0.0000	0.0000	0.0000	0.0014
	2015	0.2957	-0.0669	-0.0014	0.0000	0.0000	-0.0001
	2016	-0.0632	-0.0396	-0.0024	0.0000	0.0000	0.0009
313	2001	0.0628	-0.4112	0.0000	0.0000	0.0000	-0.0021
	2002	0.0573	-0.4127	0.0000	0.0000	0.0000	-0.0022
	2011	0.0734	-0.3187	0.0425	0.0000	0.0000	-0.0043
	2012	0.0435	-0.3665	0.0400	0.0000	0.0000	-0.0026
	2013	0.0545	-0.4475	0.0000	0.0000	0.0000	-0.0001
	2014	-0.0002	-0.4906	0.0000	0.0000	0.0000	0.0012
	2015	0.3439	-0.4379	-0.0002	0.0000	0.0000	-0.0028
	2016	0.0419	-0.4113	-0.0003	0.0000	0.0000	-0.0022
401	2001	0.0000	0.0000	0.0000	0.0003	0.0002	0.0003
	2002	0.0000	0.0000	0.0000	0.0003	0.0002	0.0002
	2011	0.0000	0.0000	0.0000	0.0006	0.0003	0.0003
	2012	0.0000	0.0000	0.0000	0.0014	0.0002	0.0002
	2013	0.0000	0.0000	0.0000	0.0011	0.0001	0.0004
	2014	0.0000	0.0000	0.0000	0.0019	0.0001	0.0001
	2015	0.0000	0.0000	0.0000	0.0003	0.0002	-0.0008
	2016	0.0000	0.0000	0.0000	0.0003	0.0002	0.0005
402	2001	-0.0311	-0.0105	0.0119	-0.0005	0.0001	-0.0002
	2002	-0.0258	-0.0092	0.0121	-0.0005	0.0001	-0.0002
	2011	-0.0320	-0.0123	0.0598	-0.0001	0.0002	-0.0002
	2012	-0.0274	-0.0110	0.0597	-0.0005	0.0000	-0.0001
	2013	-0.0429	-0.0134	0.0144	-0.0008	-0.0001	-0.0003
	2014	-0.0093	-0.0056	0.0176	-0.0013	-0.0002	-0.0001
	2015	0.2281	0.0151	0.0115	-0.0006	0.0001	-0.0009
	2016	-0.0613	-0.0087	0.0114	-0.0006	0.0002	-0.0001

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
403	2001	-0.0074	-0.0162	-0.0148	-0.0005	-0.0009	-0.0004
	2002	-0.0059	-0.0140	-0.0178	-0.0005	-0.0011	-0.0003
	2011	-0.0114	-0.0173	0.0674	0.0002	-0.0002	-0.0004
	2012	-0.0113	-0.0150	0.0510	-0.0001	-0.0007	-0.0003
	2013	-0.0107	-0.0212	-0.0138	-0.0004	-0.0009	-0.0005
	2014	-0.0017	-0.0075	-0.0377	-0.0009	-0.0019	-0.0001
	2015	0.2908	0.0030	-0.0215	-0.0006	-0.0011	-0.0008
	2016	-0.0466	-0.0115	-0.0270	-0.0007	-0.0013	-0.0002
404	2001	0.0184	-0.0403	0.0016	-0.0004	-0.0006	-0.0003
	2002	0.0157	-0.0342	0.0010	-0.0003	-0.0005	-0.0003
	2011	0.0115	-0.0387	0.0791	-0.0001	-0.0001	-0.0003
	2012	0.0069	-0.0319	0.0732	0.0001	-0.0001	-0.0002
	2013	0.0245	-0.0543	0.0026	-0.0006	-0.0007	-0.0005
	2014	0.0065	-0.0152	-0.0019	0.0000	-0.0005	-0.0001
	2015	0.3302	-0.0321	-0.0032	-0.0003	-0.0005	-0.0005
	2016	-0.0323	-0.0238	-0.0065	-0.0001	-0.0004	-0.0002
405	2001	0.0229	-0.0550	0.0303	0.0015	0.0001	-0.0001
	2002	0.0196	-0.0467	0.0292	0.0014	0.0001	-0.0001
	2011	0.0158	-0.0523	0.0763	0.0017	0.0006	-0.0001
	2012	0.0103	-0.0428	0.0733	0.0018	0.0006	-0.0001
	2013	0.0305	-0.0738	0.0354	0.0016	0.0001	-0.0001
	2014	0.0080	-0.0201	0.0297	0.0015	0.0001	0.0000
	2015	0.3378	-0.0555	0.0232	0.0013	0.0001	-0.0002
	2016	-0.0289	-0.0336	0.0183	0.0012	0.0001	-0.0001
406	2001	0.0211	-0.0123	0.0044	-0.0004	0.0000	0.0001
	2002	0.0181	-0.0101	0.0035	-0.0003	0.0000	0.0001
	2011	0.0143	-0.0158	0.1065	0.0000	-0.0016	0.0001
	2012	0.0090	-0.0135	0.1046	-0.0003	-0.0016	0.0001
	2013	0.0277	-0.0171	0.0102	-0.0010	0.0000	0.0002
	2014	0.0071	-0.0046	0.0085	-0.0010	0.0000	0.0000
	2015	0.3368	-0.0069	0.0003	-0.0003	0.0000	0.0001
	2016	-0.0283	-0.0023	-0.0025	-0.0002	0.0001	0.0001
407	2001	0.0213	-0.0126	0.0063	-0.0005	0.0000	0.0000
	2002	0.0183	-0.0105	0.0044	-0.0003	0.0000	0.0000
	2011	0.0162	-0.0145	1.7449	0.0074	-0.0031	0.0000
	2012	0.0110	-0.0130	1.7537	0.0071	-0.0031	0.0000
	2013	0.0278	-0.0159	0.0126	-0.0011	0.0000	0.0000
	2014	0.0074	-0.0070	0.0109	-0.0011	0.0000	0.0000
	2015	0.3384	-0.0029	-0.0105	-0.0004	0.0000	0.0000
	2016	-0.0256	-0.0047	-0.0205	-0.0004	0.0000	0.0000
408	2001	0.0213	-0.0121	0.0071	-0.0006	0.0000	0.0000
	2002	0.0184	-0.0095	0.0051	-0.0004	0.0000	0.0000
	2011	0.0168	-0.0131	1.8986	0.0014	0.0008	0.0000
	2012	0.0116	-0.0110	1.9084	0.0010	0.0008	0.0000
	2013	0.0279	-0.0165	0.0135	-0.0012	0.0000	0.0000
	2014	0.0075	-0.0043	0.0116	-0.0012	0.0000	0.0000
	2015	0.3389	-0.0063	-0.0110	-0.0005	0.0000	-0.0001
	2016	-0.0248	-0.0022	-0.0216	-0.0004	0.0000	0.0000

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
409	2001	0.0216	-0.0094	0.0033	-0.0003	-0.0001	-0.0001
	2002	0.0187	-0.0077	0.0027	-0.0002	-0.0001	-0.0001
	2011	0.0187	-0.0126	0.0884	0.0000	0.0018	-0.0001
	2012	0.0136	-0.0109	0.0870	-0.0002	0.0018	-0.0001
	2013	0.0281	-0.0129	0.0076	-0.0007	0.0000	-0.0001
	2014	0.0078	-0.0037	0.0065	-0.0008	0.0000	0.0000
	2015	0.3409	-0.0050	0.0002	-0.0002	-0.0001	-0.0001
	2016	-0.0221	-0.0025	-0.0019	-0.0001	-0.0001	-0.0001
410	2001	0.0209	-0.0284	0.0234	0.0014	-0.0001	0.0000
	2002	0.0181	-0.0245	0.0229	0.0014	-0.0001	0.0000
	2011	0.0181	-0.0294	0.0276	-0.0005	-0.0004	0.0000
	2012	0.0131	-0.0255	0.0256	-0.0004	-0.0004	0.0000
	2013	0.0271	-0.0365	0.0264	0.0014	-0.0001	0.0000
	2014	0.0074	-0.0139	0.0242	0.0014	-0.0001	0.0000
	2015	0.3412	-0.0241	0.0202	0.0013	-0.0001	-0.0001
	2016	-0.0218	-0.0252	0.0175	0.0013	-0.0001	0.0000

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
411	2001	0.0220	-0.0251	0.0005	-0.0003	0.0004	0.0001
	2002	0.0190	-0.0218	0.0001	-0.0002	0.0004	0.0001
	2011	0.0193	-0.0256	0.0590	-0.0001	-0.0002	0.0001
	2012	0.0141	-0.0222	0.0549	0.0000	-0.0002	0.0001
	2013	0.0284	-0.0322	0.0009	-0.0004	0.0005	0.0002
	2014	0.0078	-0.0127	-0.0017	0.0000	0.0004	0.0001
	2015	0.3406	-0.0269	-0.0018	-0.0002	0.0004	0.0000
	2016	-0.0198	-0.0205	-0.0036	-0.0001	0.0004	0.0002
412	2001	0.0346	-0.0130	-0.0175	-0.0005	0.0010	0.0002
	2002	0.0297	-0.0115	-0.0195	-0.0006	0.0011	0.0002
	2011	0.0315	-0.0141	0.0559	0.0002	0.0000	0.0002
	2012	0.0244	-0.0125	0.0447	0.0000	0.0003	0.0002
	2013	0.0451	-0.0162	-0.0177	-0.0005	0.0011	0.0002
	2014	0.0132	-0.0075	-0.0329	-0.0008	0.0017	0.0001
	2015	0.3383	-0.0286	-0.0214	-0.0006	0.0011	-0.0002
	2016	-0.0057	-0.0083	-0.0244	-0.0007	0.0012	0.0003
413	2001	0.0445	-0.0103	0.0125	-0.0005	-0.0001	0.0000
	2002	0.0380	-0.0092	0.0126	-0.0006	-0.0001	0.0000
	2011	0.0410	-0.0115	0.0492	-0.0001	-0.0001	0.0000
	2012	0.0327	-0.0103	0.0491	-0.0004	0.0000	0.0000
	2013	0.0580	-0.0125	0.0142	-0.0008	0.0000	0.0001
	2014	0.0176	-0.0063	0.0164	-0.0011	0.0001	0.0000
	2015	0.3137	-0.0326	0.0123	-0.0006	-0.0001	-0.0005
	2016	0.0136	-0.0047	0.0122	-0.0006	-0.0001	0.0002
414	2001	0.0000	0.0000	0.0000	0.0003	-0.0001	-0.0003
	2002	0.0000	0.0000	0.0000	0.0003	-0.0002	-0.0002
	2011	0.0000	0.0000	0.0000	0.0005	-0.0002	-0.0003
	2012	0.0000	0.0000	0.0000	0.0011	-0.0002	-0.0002
	2013	0.0000	0.0000	0.0000	0.0009	-0.0001	-0.0003
	2014	0.0000	0.0000	0.0000	0.0015	-0.0001	-0.0001
	2015	0.0000	0.0000	0.0000	0.0003	-0.0002	-0.0016
	2016	0.0000	0.0000	0.0000	0.0003	-0.0002	-0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
502	2001	-0.0178	-0.1096	0.0083	0.0010	0.0001	-0.0002
	2002	-0.0112	-0.1036	0.0086	0.0009	0.0001	-0.0002
	2011	0.0031	-0.1792	0.0537	0.0014	0.0002	-0.0002
	2012	-0.0202	-0.1078	0.0561	0.0010	0.0000	-0.0001
	2013	-0.0601	-0.0669	0.0126	0.0007	-0.0001	-0.0003
	2014	-0.0441	0.0242	0.0188	0.0002	-0.0002	-0.0001
	2015	0.2443	-0.0759	0.0081	0.0009	0.0001	-0.0009
	2016	-0.0366	-0.0918	0.0083	0.0009	0.0002	-0.0001
504	2001	-0.0744	-0.0011	0.0033	-0.0002	-0.0006	-0.0003
	2002	-0.0706	-0.0052	0.0023	-0.0002	-0.0005	-0.0003
	2011	-0.0091	-0.0419	0.0791	0.0000	-0.0001	-0.0003
	2012	-0.0102	-0.0617	0.0721	0.0002	-0.0001	-0.0002
	2013	-0.0895	0.0061	0.0052	-0.0003	-0.0007	-0.0005
	2014	-0.0697	-0.0232	-0.0021	0.0001	-0.0005	-0.0001
	2015	0.2501	-0.0012	-0.0018	-0.0002	-0.0005	-0.0005
	2016	-0.0995	-0.0028	-0.0055	-0.0001	-0.0004	-0.0002
506	2001	0.0284	-0.0038	0.0047	0.0000	0.0000	0.0001
	2002	0.0257	-0.0033	0.0038	0.0000	0.0000	0.0001
	2011	-0.2356	-0.0615	0.1048	0.0003	-0.0016	0.0001
	2012	-0.2439	0.0048	0.1053	-0.0001	-0.0016	0.0001
	2013	0.0335	0.0701	0.0135	-0.0005	0.0000	0.0002
	2014	0.0103	0.1470	0.0142	-0.0009	0.0000	0.0000
	2015	0.3450	0.0146	0.0012	-0.0001	0.0000	0.0001
	2016	-0.0197	0.0256	-0.0015	-0.0002	0.0001	0.0001
508	2001	0.0237	0.0320	0.0088	-0.0002	0.0000	0.0000
	2002	0.0204	0.0154	0.0060	-0.0001	0.0000	0.0000
	2011	0.1371	-0.2727	1.8887	0.0017	0.0008	0.0000
	2012	0.1331	-0.2102	1.9008	0.0013	0.0008	0.0000
	2013	0.0312	0.1052	0.0182	-0.0007	0.0000	0.0000
	2014	0.0109	0.1793	0.0186	-0.0011	0.0000	0.0000
	2015	0.3394	0.0528	-0.0087	-0.0003	0.0000	-0.0001
	2016	-0.0248	0.0644	-0.0191	-0.0004	0.0000	0.0000
509	2001	0.0115	-0.0034	0.0036	0.0000	-0.0001	-0.0001
	2002	0.0081	-0.0029	0.0029	0.0000	-0.0001	-0.0001
	2011	0.3086	-0.0489	0.0870	0.0003	0.0018	-0.0001
	2012	0.3066	0.0006	0.0874	0.0000	0.0018	-0.0001
	2013	0.0207	0.0511	0.0100	-0.0003	0.0000	-0.0001
	2014	0.0016	0.1084	0.0107	-0.0007	0.0000	0.0000
	2015	0.3281	0.0106	0.0008	-0.0001	-0.0001	-0.0001
	2016	-0.0363	0.0178	-0.0011	-0.0001	-0.0001	-0.0001
511	2001	0.0945	-0.0024	0.0015	-0.0001	0.0004	0.0001
	2002	0.0876	-0.0057	0.0009	-0.0001	0.0004	0.0001
	2011	-0.0072	-0.0298	0.0588	0.0000	-0.0002	0.0001

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	2012	-0.0133	-0.0423	0.0541	0.0001	-0.0002	0.0001
	2013	0.1137	0.0025	0.0025	-0.0002	0.0005	0.0002
	2014	0.0734	-0.0189	-0.0018	0.0000	0.0004	0.0001
	2015	0.4045	-0.0094	-0.0010	-0.0001	0.0004	0.0000
	2016	0.0382	-0.0094	-0.0030	-0.0001	0.0004	0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
513	2001	0.0333	-0.1070	0.0090	0.0010	-0.0001	0.0001
	2002	0.0261	-0.1027	0.0092	0.0010	-0.0001	0.0000
	2011	0.0209	-0.1733	0.0433	0.0014	-0.0001	0.0001
	2012	0.0287	-0.1233	0.0450	0.0011	0.0000	0.0001
	2013	0.0633	-0.0759	0.0120	0.0008	0.0000	0.0001
	2014	0.0340	-0.0134	0.0162	0.0004	0.0001	0.0000
	2015	0.2947	-0.1238	0.0090	0.0009	-0.0001	-0.0005
	2016	-0.0042	-0.0924	0.0090	0.0009	-0.0001	0.0002
2006	2001	0.0000	0.0000	0.0000	-0.0002	0.0000	-0.0001
	2002	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0001
	2011	0.0000	0.0000	0.0000	0.0015	-0.0016	-0.0001
	2012	0.0000	0.0000	0.0000	0.0009	-0.0016	0.0000
	2013	0.0000	0.0000	0.0000	-0.0009	0.0000	-0.0002
	2014	0.0000	0.0000	0.0000	-0.0013	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0012
	2016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2008	2001	0.0000	0.0000	0.0000	-0.0003	0.0000	-0.0001
	2002	0.0000	0.0000	0.0000	-0.0002	0.0000	-0.0001
	2011	0.0000	0.0000	0.0000	0.0067	0.0007	0.0000
	2012	0.0000	0.0000	0.0000	0.0061	0.0007	0.0000
	2013	0.0000	0.0000	0.0000	-0.0010	0.0000	-0.0001
	2014	0.0000	0.0000	0.0000	-0.0013	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	-0.0003	0.0000	-0.0012
	2016	0.0000	0.0000	0.0000	-0.0002	0.0000	0.0001
2009	2001	0.0000	0.0000	0.0000	-0.0001	0.0001	0.0000
	2002	0.0000	0.0000	0.0000	-0.0001	0.0001	0.0000
	2011	0.0000	0.0000	0.0000	0.0011	0.0019	0.0000
	2012	0.0000	0.0000	0.0000	0.0007	0.0019	0.0000
	2013	0.0000	0.0000	0.0000	-0.0006	0.0000	-0.0001
	2014	0.0000	0.0000	0.0000	-0.0009	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	-0.0001	0.0000	-0.0012
	2016	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001
4006	2001	0.0000	0.0000	0.0000	0.0002	0.0000	-0.0001
	2002	0.0000	0.0000	0.0000	0.0001	0.0000	-0.0001
	2011	0.0000	0.0000	0.0000	0.0005	-0.0016	-0.0001
	2012	0.0000	0.0000	0.0000	0.0010	-0.0016	-0.0001
	2013	0.0000	0.0000	0.0000	0.0009	0.0000	-0.0002
	2014	0.0000	0.0000	0.0000	0.0013	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	0.0001	0.0000	-0.0012
	2016	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001
4008	2001	0.0000	0.0000	0.0000	0.0003	0.0000	-0.0001
	2002	0.0000	0.0000	0.0000	0.0002	0.0000	-0.0001
	2011	0.0000	0.0000	0.0000	0.0060	0.0008	-0.0001
	2012	0.0000	0.0000	0.0000	0.0066	0.0008	0.0000
	2013	0.0000	0.0000	0.0000	0.0010	0.0000	-0.0001
	2014	0.0000	0.0000	0.0000	0.0013	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	0.0002	0.0000	-0.0011
	2016	0.0000	0.0000	0.0000	0.0001	0.0000	0.0001

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
4009	2001	0.0000	0.0000	0.0000	0.0001	-0.0001	0.0000
	2002	0.0000	0.0000	0.0000	0.0001	-0.0001	0.0000
	2011	0.0000	0.0000	0.0000	0.0004	0.0018	0.0000
	2012	0.0000	0.0000	0.0000	0.0008	0.0018	0.0000
	2013	0.0000	0.0000	0.0000	0.0006	0.0000	-0.0001
	2014	0.0000	0.0000	0.0000	0.0009	0.0000	0.0000
	2015	0.0000	0.0000	0.0000	0.0001	-0.0001	-0.0012
	2016	0.0000	0.0000	0.0000	0.0001	-0.0001	0.0001

************** END OF LATEST ANALYSIS RESULT **************


```
297. PARAMETER 1
298. CODE BS5950
299. PY 345000 ALL
300. CHECK CODE ALL
STEEL DESIGN
```

STAAD.Pro CODE CHECKING - (BSI)
****

PROGRAM CODE REVISION V2.9_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
21	TAPERED	PASS 44.40 C	BS-4.7 (C) -0.38	0.091 -8.04	2013 2.48
22	TAPERED	PASS 8.80 C	BS-4.7 (C) -0.06	0.146 21.94	2011 2.51
23	TAPERED	PASS 19.74 C	BS-4.7 (C) 2.06	0.065 1.32	2015 4.22
41	TAPERED	PASS 44.40 C	BS-4.7 (C) -0.38	0.091 8.04	2013 2.48
42	TAPERED	PASS 32.98 C	BS-4.7 (C) 0.14	0.186 -25.25	2011 4.30
43	TAPERED	PASS 33.64 C	BS-4.7 (C) 0.05	0.063 5.95	2013 2.46
201	TAPERED	PASS 15.80 C	BS-4.7 (C) 2.49	0.110 1.75	2011 2.65
202	TAPERED	PASS 20.95 C	BS-4.7 (C) -1.48	0.102 6.06	2013 0.00
203	TAPERED	PASS 15.87 C	BS-4.7 (C) -0.68	0.075 6.48	2013 0.75
204	TAPERED	PASS 2.11 T	BS-4.8.2.2 0.08	0.054 -7.97	2015 0.00
205	TAPERED	PASS 4.09 C	BS-4.7 (C) -1.23	0.053 -1.21	2011 3.17
206	TAPERED	PASS 3.16 C	BS-4.7 (C) 2.99	0.110 0.95	2011 0.00
207	TAPERED	PASS 3.14 C	BS-4.7 (C) 3.84	0.133 -0.63	2012 1.15
208	TAPERED	PASS 2.79 C	BS-4.7 (C) -4.64	0.162 0.60	2012 0.00
209	TAPERED	PASS 2.72 C	BS-4.7 (C) 2.04	0.075 0.84	2011 0.00
210	TAPERED	PASS 2.54 C	BS-4.7 (C) 0.70	0.036 1.85	2011 0.75
211	TAPERED	PASS 10.32 C	BS-4.7 (C) 0.58	0.054 -4.32	2013 0.00
212	TAPERED	PASS 13.83 C	BS-4.7 (C) 1.17	0.074 -3.99	2013 0.75
213	TAPERED	PASS 14.90 C	BS-4.7 (C) -2.07	0.093 -1.44	2011 0.00
401	TAPERED	PASS 30.35 C	BS-4.7 (C) 1.00	0.087 -3.55	2013 1.55
402	TAPERED	PASS 20.95 C	BS-4.7 (C) 1.48	0.102 6.06	2013 0.00
403	TAPERED	PASS 15.87 C	BS-4.7 (C) 0.68	0.075 6.48	2013 0.75
404	TAPERED	PASS 2.11 C	BS-4.7 (C) -0.11	0.046 -6.46	2013 0.00
405	TAPERED	PASS 0.76 C	BS-4.7 (C) -1.32	0.057 -1.97	2011 3.17

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
406	TAPERED	PASS	BS-4.8.2.2	0.114	2011

```
                          4.23 T        3.07            1.64          0.00
  407     TAPERED          PASS     BS-4.8.2.2         0.134      2012
                          4.36 T        3.91           -0.31          1.15
  408     TAPERED          PASS     BS-4.8.2.2         0.159      2012
                          4.07 T       -4.72            0.17          0.00
  409     TAPERED          PASS     BS-4.8.2.2         0.080      2011
                          0.01 T        2.19            1.28          0.00
  410     TAPERED          PASS     BS-4.8.2.2         0.047      2011
                          0.30 T        0.86            2.96          0.75
  411     TAPERED          PASS     BS-4.7   (C)       0.054      2013
                         10.32 C       -0.58           -4.32          0.00
  412     TAPERED          PASS     BS-4.7   (C)       0.074      2013
                         13.83 C       -1.17           -3.99          0.75
  413     TAPERED          PASS     BS-4.7   (C)       0.083      2015
                         16.22 C        0.64           -7.17          0.00
 1001     TAPERED          PASS     BS-4.8.2.2         0.085      2001
                          0.12 T       -2.58            0.00          1.60
 1002     TAPERED          PASS     BS-4.7   (C)       0.136      2011
                         19.41 C        3.42            0.51          0.00
 1003     TAPERED          PASS     BS-4.7   (C)       0.130      2001
                          6.99 C       -3.71            0.06          2.50
 1004     TAPERED          PASS     BS-4.8.2.2         0.085      2001
                          0.12 T        2.58            0.00          0.00
 2001     TAPERED          PASS     BS-4.3.6           0.041      2013
                          0.30 T        0.00            6.42          1.60
 2002     TAPERED          PASS     BS-4.7   (C)       0.068      2013
                         10.05 C        0.28           -7.80          2.29
 2003     TAPERED          PASS     BS-4.7   (C)       0.068      2013
                         10.05 C        0.28           -7.80          0.21
 2004     TAPERED          PASS     BS-4.3.6           0.041      2013
                          0.30 T        0.00            6.42          0.00
 3001     TAPERED          PASS     BS-4.3.6           0.076      2013
                          0.56 T        0.00           12.05          1.60
 3002     TAPERED          PASS     BS-4.7   (C)       0.150      2011
                         20.35 C       -1.77           11.44          0.00
 3003     TAPERED          PASS     BS-4.7   (C)       0.119      2011
                         20.41 C       -1.75           -6.52          2.50
 3004     TAPERED          PASS     BS-4.3.6           0.076      2013
                          0.56 T        0.00           12.05          0.00
 4001     TAPERED          PASS     BS-4.3.6           0.076      2013
                          0.57 T        0.00           12.05          1.60
 4002     TAPERED          PASS     BS-4.7   (C)       0.129      2012
                          0.19 C        0.80          -16.42          0.00
 4003     TAPERED          PASS     BS-4.7   (C)       0.238      2011
                          7.93 C        0.81          -32.50          2.50
0   STAAD SPACE                                            -- PAGE NO.   65
0

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE        RESULT/   CRITICAL COND/     RATIO/     LOADING/
                           FX            MY             MZ       LOCATION
 =======================================================================

0
 4004     TAPERED          PASS     BS-4.3.6           0.076      2013
                          0.57 T        0.00           12.05          0.00
 5001     TAPERED          PASS     BS-4.3.6           0.056      2013
                          0.42 T        0.00            8.82          1.60
 5002     TAPERED          PASS     BS-4.7   (C)       0.137      2011
                         17.04 C        2.05            8.32          0.00
 5003     TAPERED          PASS     BS-4.7   (C)       0.114      2011
                         17.09 C        2.02           -4.89          2.50
 5004     TAPERED          PASS     BS-4.3.6           0.056      2013
                          0.42 T        0.00            8.82          0.00
 6001     TAPERED          PASS     BS-4.3.6           0.027      2013
                          0.20 T        0.00            4.28          1.60
 6002     TAPERED          PASS     BS-4.7   (C)       0.047      2013
                          9.14 C       -0.25            4.69          0.00
 6003     TAPERED          PASS     BS-4.7   (C)       0.047      2013
                          9.14 C        0.25           -4.69          2.50
 6004     TAPERED          PASS     BS-4.3.6           0.027      2013
                          0.20 T        0.00            4.28          0.00
 7001     TAPERED          PASS     BS-4.8.2.2         0.085      2001
                          0.12 T       -2.64            0.00          1.60
 7002     TAPERED          PASS     BS-4.7   (C)       0.129      2011
                         16.57 C        3.44           -0.40          0.00
 7003     TAPERED          PASS     BS-4.7   (C)       0.131      2011
                         16.57 C       -3.53           -0.17          2.50
 7004     TAPERED          PASS     BS-4.8.2.2         0.085      2001
                          0.12 T        2.64            0.00          0.00
 8001 ST UA90X90X6         PASS     BS-4.6   (T)       0.052      2012
                         19.18 T        0.00            0.00          5.66
 8002 ST UA90X90X6         PASS     BS-4.6   (T)       0.015      2014
```



```
                          5.58 T        0.00            0.00          5.66
 8003 ST UA90X90X6         PASS     BS-4.6   (T)       0.023      2012
                          8.50 T        0.00            0.00          5.20
 8004 ST UA90X90X6         PASS     BS-4.6   (T)       0.017      2014
                          6.13 T        0.00            0.00          5.20
 8005 ST UA90X90X6         PASS     BS-4.6   (T)       0.019      2012
                          7.10 T        0.00            0.00          6.33
 8006 ST UA90X90X6         PASS     BS-4.6   (T)       0.013      2013
                          4.81 T        0.00            0.00          0.00
 8007 ST UA90X90X6         PASS     BS-4.6   (T)       0.007      2014
                          2.72 T        0.00            0.00          0.00
 8008 ST UA90X90X6         PASS     BS-4.6   (T)       0.068      2012
                         25.00 T        0.00            0.00          0.00
 8009 ST UA90X90X6         PASS     BS-4.6   (T)       0.006      2014
                          2.01 T        0.00            0.00          0.00
 8010 ST UA90X90X6         PASS     BS-4.6   (T)       0.057      2012
                         20.91 T        0.00            0.00          0.00
0   STAAD SPACE                                            -- PAGE NO.   66
0

 ALL UNITS ARE - KN   METE (UNLESS OTHERWISE NOTED)

 MEMBER     TABLE        RESULT/   CRITICAL COND/     RATIO/     LOADING/
                           FX            MY             MZ       LOCATION
 =======================================================================

0
 8011 ST UA90X90X6         PASS     BS-4.6   (T)       0.009      2013
                          3.35 T        0.00            0.00          0.00
 8012 ST UA90X90X6         PASS     BS-4.6   (T)       0.019      2012
                          7.09 T        0.00            0.00          5.63
 8013 ST UA90X90X6         PASS     BS-4.6   (T)       0.016      2014
                          5.77 T        0.00            0.00          0.00
 8014 ST UA90X90X6         PASS     BS-4.6   (T)       0.021      2012
                          7.53 T        0.00            0.00          0.00
 8015 ST UA90X90X6         PASS     BS-4.6   (T)       0.014      2014
                          5.05 T        0.00            0.00          0.00
 8016 ST UA90X90X6         PASS     BS-4.6   (T)       0.043      2012
                         15.69 T        0.00            0.00          5.66

  ************** END OF TABULATED RESULT OF DESIGN **************

  301. STEEL TAKE OFF ALL
0    STAAD SPACE                                           -- PAGE NO.   67

0STEEL TAKE OFF 0
0
  STEEL TAKE-OFF
  --------------

     PROFILE                 LENGTH(METE)      WEIGHT(KN  )
0
  ISECT: TAPERED     21          25.56             10.609
  ISECT: TAPERED    201          96.85             30.533
  ISECT: TAPERED   7001           8.21              2.945
  ST  UA90X90X6                  93.60              7.622
                                           ----------------
                               TOTAL =             51.708

  ************ END OF DATA FROM INTERNAL STORAGE ************

  302. FINISH

           *********** END OF THE STAAD.Pro RUN ***********

             **** DATE= MAY 15,2007   TIME= 17: 2: 0 ****
```

2 FOUNDATION DESIGN CHECK

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 10 Footing Summary	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Soil Bearing Pressure	=	450	kN/m^2
Soil Density	=	20	kN/m^3
Concrete Density	=	24	kN/m^3
Footing Depth below slab soffit	=	1	m
Column Stump Area	=	0.25	m^2

Individual Footing

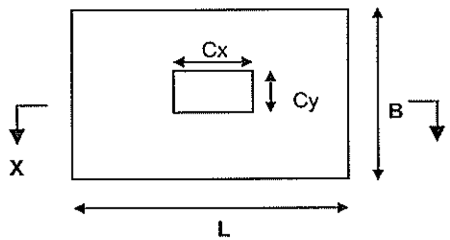
Column	Column Reaction (ULS)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footing size			Area of Footing (m^2)	Soil Pressure (kN/m^2)	Factor	Remarks
						L (m)	B (m)	H (m)				
401	55	34.2	28.224	108	142	1.400	1.400	0.600	1.96	55	8.18	OK

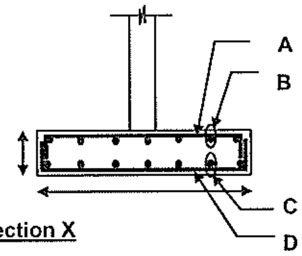
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 10 Footing Detail	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary





Section X

Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F3	1400	1400	600	450	10T12	143	10T12	143	10T16	143	10T16	143

Calculation/Sketch

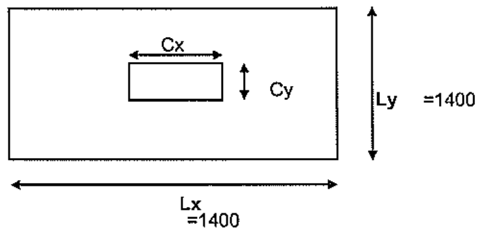
Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 10 Footing F3 For soil bearing pressure = 150kPa	Date	May '07	Job No.	068010.01
		Designed by	Rudy	Sheet No.	

REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. ***Material Properties***

Concrete grade	f_{cu}	=	35	N/mm^2
Steel grade	f_y	=	460	N/mm^2
Concrete density	γ_c	=	24	KN/m^3
Steel density	γ_s	=	78	KN/m^3



2. ***Design Data***

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	1400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect.depth (x-dir)	d_x	=	542	mm
Effect.depth (y-dir)	d_y	=	526	mm
Area of footing	A	=	1.96	m^2

3. ***Loadings***

Safe bearing pressure	p_b	=	$\boxed{450}$	KN/m
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	882.00	kN
Eccentricity	e	=	0	mm

4. ***Analysis***

Net pressure (SLS)	p_{max}	=	435.60	KN/m					
Net pressure (ULS)	p_{max}	=	653.4	KN/m					
Shear force (x-dir)	V_x	=	434.51	kN					
Shear force (y-dir)	V_y	=	434.51	kN					
Moment (x-dir)	M_x	=	103.20	kNm					
Moment (y-dir)	M_y	=	103.20	kNm					
Steel required (x-dir)	A_{sreq}	=	1092	mm^2		Nos	ϕ	Spacing	
Steel provided (x-dir)	A_{sprov}	=	2011	mm^2	O.K.	10	16	143	x-dir
Steel required (y-dir)	A_{sreq}	=	1092	mm^2		Nos	ϕ		
Steel provided (y-dir)	A_{sprov}	=	2011	mm^2	O.K.	10	16	143	y-dir
Top bars required (x-dir)	A'_{sreq}	=	1092	mm^2		Nos	ϕ		
Top bars provided (x-dir)	A'_{sprov}	=	1131	mm^2	O.K.	10	12	143	x-dir
Top bars required (y-dir)	A'_{sreq}	=	1092	mm^2		Nos	ϕ		
Top bars provided (y-dir)	A'_{sprov}	=	1131	mm^2	O.K.	10	12	143	y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Auxiliary Depot Jebel Ali - Building 10	Date	May '07	Job No.	068010.01
	Footing F3				
	For soil bearing pressure = 150kPa	Designed by	Rudy	Sheet No.	

5. ***Shear Check***

Checking for punching shear:

Shear force at face V_f = 1186.31 kN

Shear stress at face v_f = 1.22 N/mm^2 **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) V_c = 448.88 kN

Shear stress at face (x-dir) v_c = 0.59 N/mm^2 **O.K.**

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

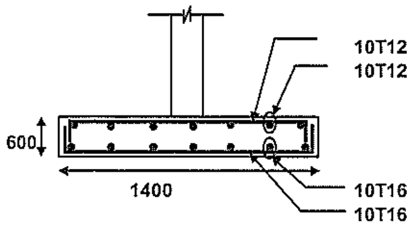
Checking for shear (y-dir):

Shear force at face (y-dir) V_c = 448.88 kN

Shear stress at face (y-dir) v_c = 0.61 N/mm^2 **O.K.**

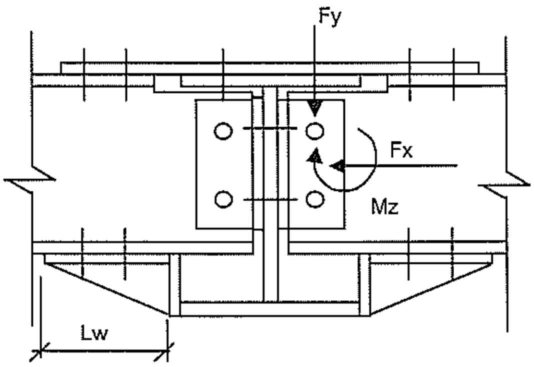
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm^2)	v_c (N/mm^2)	Enhanced v_c (N/mm^2)	Remarks
d	526	0.00	0.00	0.46	0.92	No Shear Links
2d	1052	0.00	0.00	0.46	0.46	No Shear Links

6. ***Detailing***



3 CONNECTION DESIGN

Secondary Beam Connection Design



From the analysis for the perimeter beam

Fy	=	16 kN
Fx	=	7 kN
Mz	=	28 kNm

Eccentricity of load, e = 60 mm

- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, Df	=	242 mm
Mmax bending moment (incl. moment due to ecc), Me	=	29.0 kNm
Shear on the flange, Vf	=	Me / Df
	=	119.9 kN
No of bolt provided	=	2
Shear per bolt	=	59.9 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, Vb	=	91.9 kN

Check weld length required

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = Vf / pw
= 114 mm
= 57 mm per face

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	8.5 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	100 mm
Tensile load per bolt	=	9.5 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN

From the analysis for the internal beam

Fy	=	31 kN
Fx	=	19 kN
Mz	=	39 kNm

Eccentricity of load, e = 60 mm

- Assume shear due to bending is resisted by bolt on the flange
- Assume vertical and horizontal shear is resisted by bolt on the web

Check bolts on the flanges

Beam depth - flange thickness, Df	=	242 mm
Mmax bending moment (incl. moment due to ecc), Me	=	40.9 kNm
Shear on the flange, Vf	=	Me / Df
	=	168.9 kN
No of bolt provided	=	2
Shear per bolt	=	84.5 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, Vb	=	91.9 kN

Check weld length required

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = Vf / pw
= 161 mm
= 80 mm per face

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	18.3 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	100 mm
Tensile load per bolt	=	21.5 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

Eccentricity of load, e = 60 mm

Mmax = 14 kNm
Vmax = 18 kN

Check shear on the web

- for the secondary beam web

No of bolt provided	=	2
Shear per bolt	=	9.0 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

No of bolt provided	=	2
Distance between bolt	=	100 mm
Tensile load per bolt	=	10.8 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN

Check bolts on the top flange

Beam depth - flange thickness, Df	=	242 mm
Mmax bending moment (incl. moment due to ecc), Me	=	15.5 kNm
Shear on the flange, Vf	=	Me / Df
	=	64.2 kN
No of bolt provided	=	2
Shear per bolt	=	32.1 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, Vb	=	91.9 kN

Calculation/Sketch

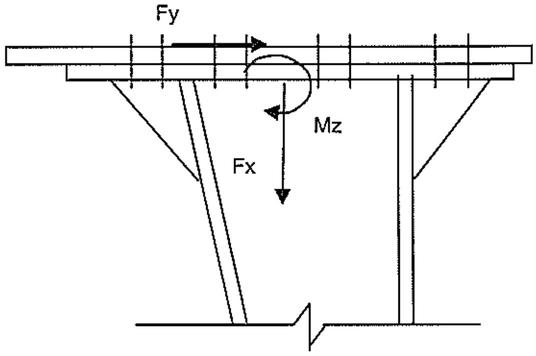
Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers

Site	Dubai Depot Waste Stock Yard (Jebel Ali)	Date	May '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Column Splice Connection Design (Supporting the Beam)

From the STAAD Pro analysis

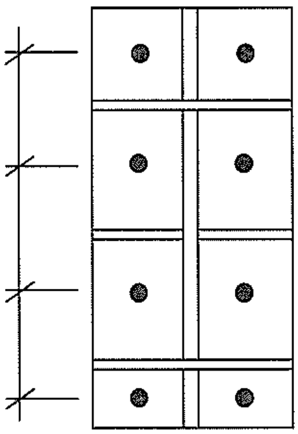
Fxc =	52	kN (compression)
Fxt =	19	kN (tension)
Fy =	248	kN (include shear due to beam moment)
Mz =	31	kNm



Total Tension
T = (n*Tb/d)*[(d-a)+(d-y1-a)+(d-y2-a) + ...]
where n = no of row of bolts and Tb is bolt tension capacity

Total Compression
= 0.5a*pc*b
where b = stiff bearing length
= web thickness + 2*plate thickness

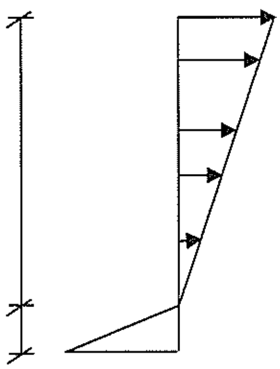
Moment Capacity
Mc = [nTb/(d-a)] * [$(d\text{-}a)^2$ + $(d\text{-}a\text{-}y1)^2$ + $(d\text{-}a\text{-}ys)^2$ + ...]



For Gr 8.8 bolt

Bolt size	=	16 mm
Tb	=	70.7 kN
Vb	=	58.9 kN
n	=	2
n total	=	8
web, tw	=	8 mm
plate, tp	=	15 mm
b	=	38 mm

y1	125	Trial a	=	43 mm
y2	100	T	=	263.2 kN
y3	125	C	=	263.2 kN
y4	0	T-C	=	0.00 kN
y5	0	Mc	=	61.9 kNm
y6	0			
Depth, d	350			



Tension force on the bolt, T1 = (Mz/Mc)*Tb
= 35 kN

[T1 + (Fxt/n total)] / Tb + Fy/(n total*Vb) = 1.05
< 1.4 Ok

pc = 325 N/mm2

Site	Dubai Depot	Date	May '07	Job No.	068010
	Waste Stock Yard (Jebel Ali)				
		Designed by	Rudy	Sheet No.	

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = max (T1 + 0.5*Fxt, Fy) / pw
= 236 mm
= 118 mm per face

From the plate analysis (refer attached) using
Prying Force, To = 10 kN

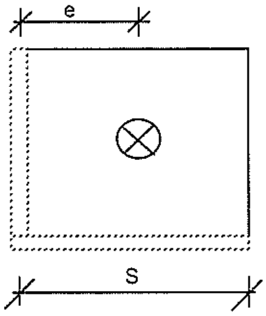


Plate width, So (mm)	Ave. support bending moment, Mo (kNm)
75	0.020
100	0.027
125	0.035
150	0.040

Actual - Prying force, T = 35 kN
Side, S = 100 mm
Bending Moment, M = (T/To)*Mo
= 0.094 kNm

Bolt Diameter, ϕ = 16 mm
Bolt distance, e = 50 mm
Thickness of plate, tp = 15 mm
py allow = 355 N/mm^2

Effective width of the plate resisting the bending moment, bp = min (2e+ϕ, S)
= 100 mm

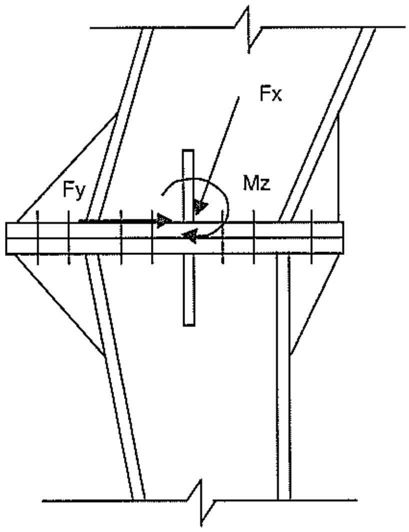
Section of modulus of the plate, Zp = bp*tp^2 / 6
= 3750 mm^2

M / Zp = 25 N/mm^2 Ok

Column Splice Connection Design

From the STAAD Pro analysis

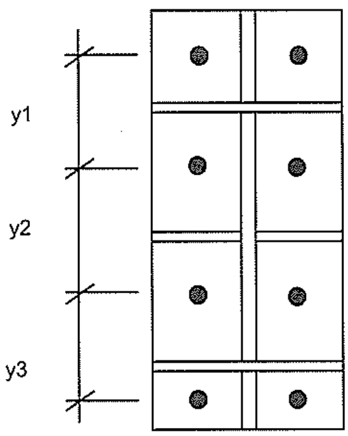
Fxc =	29	kN (compression)
Fxt =	8	kN (tension)
Fy =	10	kN
Mz =	34	kNm



Total Tension
T = (n*Tb/d)*[(d-a)+(d-y1-a)+(d-y2-a) + ...]
where n = no of row of bolts and Tb is bolt tension capacity

Total Compression
= 0.5a*pc*b
where b = stiff bearing length
= web thickness + 2*plate thickness

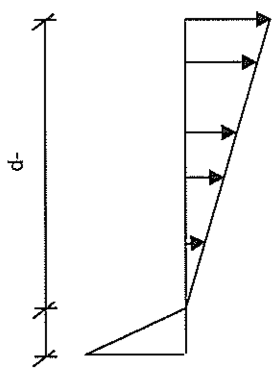
Moment Capacity
$Mc = [nTb/(d-a)] * [(d-a)^2 + (d-a-y1)^2 + (d-a-ys)^2 + ...]$



For Gr 8.8 bolt

Bolt size	=	16 mm
Tb	=	70.7 kN
Vb	=	58.9 kN
n	=	2
n total	=	12
web, tw	=	8 mm
plate, tp	=	15 mm
b	=	38 mm

y1	125	Trial a	=	43 mm
y2	100	T	=	263.2 kN
y3	125	C	=	263.2 kN
y4	0	T-C	=	0.00 kN
y5	0	Mc	=	61.9 kNm
y6	0			
Depth, d	350			



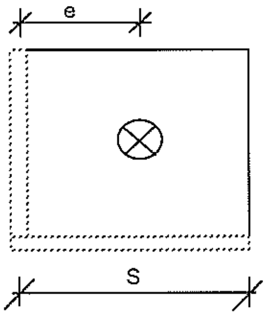
Tension force on the bolt, T1 = (Mz/Mc)*Tb
= 39 kN

[T1 + (Fxt/n total)] / Tb + Fy/(n total*Vb) = 0.57
< 1.4 Ok

pc = 325 N/mm2

<u>Check the effective weld length on the stiffener plate to resist bolt's tension</u>

Weld size, s =	6 mm	(up to 15mm thick)
Weld strength, pw =	1.05 kN/mm	(based on longitudinal capacity)
Weld length required =	(T1 + 0.5*Fxt) / pw	
=	41 mm	
=	20 mm per face	



From the plate analysis (refer attached) using
Prying Force, To = 10 kN

Plate width, So (mm)	Ave. support bending moment, Mo (kNm)
75	0.020
100	0.027
125	0.035
150	0.040

Actual - Prying force, T = 39 kN
Side, S = 75 mm
Bending Moment, M = (T/To)*Mo
= 0.079 kNm

Bolt Diameter, ϕ = 16 mm
Bolt distance, e = 36 mm
Thickness of plate, tp = 15 mm
py allow = 355 N/mm^2

Effective width of the plate resisting the bending moment, bp = min (2e+ϕ, S)
= 75 mm

Section of modulus of the plate, Zp = bp*tp^2 / 6
= 2812.5 mm^2

M / Zp = 28 N/mm^2 Ok